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What is a human but an ingenious machine designed to turn, with “infinite artfulness, the red wine of Shiraz into urine?” So said the storyteller in Isak Dinesen’s *Seven Gothic Tales*. More accurately but less poetically, human kidneys serve to convert more than 1700 L of blood per day into about 1 L of a highly concentrated fluid called *urine*. In so doing, the kidney excretes the waste products of metabolism; precisely regulates the body’s concentration of water, salt, calcium, phosphorus, and other anions and cations; and maintains the appropriate acid-base balance of plasma. The kidney also serves as an endocrine organ, secreting hormones such as erythropoietin and renin, and it regulates vitamin D metabolism.

Renal diseases are responsible for a great deal of morbidity and mortality. According to the 2020 US Renal Data System Annual Data Report, more than 808,000 Americans had end-stage kidney disease (ESKD), of whom two-thirds are maintained on dialysis, at a cost of approximately \$72,000 per person annually. The 1-year mortality rate of ESKD, when the enhanced risk for cardiovascular disease conferred by ESKD is considered, exceeds that of most newly diagnosed cancers. Acute kidney injury occurs in more than 2 million people worldwide, and it is a major risk factor for the development of chronic kidney disease and ESKD. In addition, millions of people are affected annually by nonfatal kidney diseases, most notably infections of the kidney or lower urinary tract, kidney stones, and urinary obstruction. The availability of dialysis and the success of renal transplantation have improved the outlook for patients.

The study of kidney diseases is facilitated by categorizing them according to their primary effects on one of the four basic anatomic components of the kidney: glomeruli, tubules, interstitium, and blood vessels. This approach is useful because the early manifestations of disease primarily affecting one of these components tend to be distinct. Further, some components seem to be more vulnerable to specific forms of renal injury; for example, **most glomerular diseases are immunologically mediated, whereas tubular and interstitial disorders are frequently caused by toxic or infectious agents.** However, some disorders affect more than one structure, and the anatomic and functional interdependence of the components of the kidney means that damage to one usually secondarily affects the others. Primary disorders of the blood vessels, for example, inevitably affect all structures supplied by these vessels. Severe glomerular damage impairs the flow through the peritubular vascular system; conversely, tubular destruction, by increasing intraglomerular pressure, induces glomerular injury. Thus, whatever the origin, all forms of chronic kidney disease ultimately damage all four components of the kidney, culminating in what are called *end-stage kidneys*. The functional reserve of the kidney is large, and much damage may occur before there is evident functional impairment. For these reasons, recognizing the early signs and symptoms is of high clinical importance.

CLINICAL MANIFESTATIONS OF RENAL DISEASES

The clinical manifestations of renal disease can be grouped into reasonably well-defined syndromes. Some are unique to glomerular

diseases, and others are present in diseases that affect any one of the components.

- **Azotemia** is a biochemical abnormality that refers to an elevation of blood urea nitrogen (BUN) and creatinine levels, related largely to a decreased glomerular filtration rate (GFR). Azotemia is a consequence of many renal diseases, but it may also arise from extra-renal disorders. It is a typical feature of both acute and chronic kidney injury. *Prerenal azotemia* is encountered when there is hypoperfusion of the kidneys that impairs renal function in the absence of parenchymal damage. It may be caused by hypotension, excessive fluid losses from any cause, or if the effective intravascular volume is decreased due to shock or congestive heart failure. *Postrenal azotemia* is seen whenever urine flow is obstructed distal to the kidney. Relief of the obstruction is followed by correction of the azotemia.
- When azotemia leads to clinical signs and symptoms associated with biochemical abnormalities, it is termed *uremia*. Uremia is characterized not only by failure of renal excretory function but also by a host of metabolic and endocrine alterations resulting from renal damage. Uremic patients frequently manifest secondary involvement of the gastrointestinal system (e.g., uremic gastroenteritis), peripheral nerves (e.g., peripheral neuropathy), and heart (e.g., uremic fibrinous pericarditis).
- **Nephritic syndrome** is a clinical entity caused by inflammatory glomerular disease and is dominated by the acute onset of either grossly visible or microscopic hematuria, diminished GFR, mild to moderate proteinuria, and hypertension. It is the classic presentation of acute poststreptococcal glomerulonephritis. *Rapidly progressive glomerulonephritis* (RPGN) is a form of nephritic syndrome in which there is rapid decline in GFR (within hours to days).
- **Nephrotic syndrome**, typically caused by severe dysfunction of the glomerular filter, is characterized by heavy proteinuria (more than 3.5 g/day), resulting in hypoalbuminemia and severe edema, and hyperlipidemia and lipiduria (lipids in the urine).
- **Asymptomatic hematuria or proteinuria**, or a combination of these two, is usually a manifestation of subtle or mild glomerular abnormalities.
- **Acute kidney injury** (previously called acute renal failure) is characterized by rapid decline in GFR (within hours to days) with concurrent dysregulation of fluid and electrolyte balance, and retention of metabolic waste products normally excreted by the kidney including urea and creatinine. In its most severe forms, it is manifested by *oliguria* or *anuria* (reduced or no urine flow). It can result from glomerular, interstitial, vascular, or acute tubular injury (ATI).
- **Chronic kidney disease** (previously called chronic renal failure) is defined by a diminished GFR that is persistently less than 60 mL/min/1.73 m² for at least 3 months and/or persistent albuminuria. It may present with a clinically silent decline in renal excretory function in milder forms or with prolonged symptoms and signs of uremia in more severe cases. Chronic kidney

disease is estimated to affect 11% of adults in the United States and is particularly common in those greater than 60 years of age. It is the end result of a variety of renal diseases, most commonly diabetic and vascular kidney disease, and is the major cause of death from renal disease. Chronic kidney disease causes significant systemic abnormalities (Table 20.1).

- *In end-stage kidney disease*, the GFR is less than 5% of normal; this is the terminal stage of uremia.
- *Renal tubular defects* are dominated by polyuria (excessive urine formation), nocturia, and electrolyte disorders (e.g., metabolic acidosis). They are the result of diseases that either directly affect tubular structures (e.g., nephronophthisis) or cause defects in specific tubular functions. The latter can be inherited (e.g., familial nephrogenic diabetes, cystinuria, renal tubular acidosis) or acquired (e.g., lead nephropathy).
- *Urinary tract obstruction* and *renal tumors* have varied clinical manifestations based on the anatomic location and nature of the lesion. *Urinary tract infection* is characterized by bacteriuria and pyuria (bacteria and leukocytes in the urine). The infection may be symptomatic or asymptomatic, and it may affect the kidney (*pyelonephritis*) or the bladder (*cystitis*).
- *Nephrolithiasis (renal stones)* is manifested by spasms of severe pain (renal colic) and hematuria, often with recurrent stone formation.

TABLE 20.1 Principal Systemic Manifestations of Chronic Kidney Disease and Uremia

Fluid and Electrolytes
Dehydration
Edema
Hyperkalemia
Metabolic acidosis
Calcium Phosphate and Bone
Hyperphosphatemia
Hypocalcemia
Secondary hyperparathyroidism
Renal osteodystrophy
Hematologic
Anemia
Abnormal bleeding
Cardiopulmonary
Hypertension
Congestive heart failure
Cardiomyopathy
Pulmonary edema
Uremic pericarditis
Gastrointestinal
Nausea and vomiting
Bleeding
Esophagitis, gastritis, colitis
Neuromuscular
Myopathy
Peripheral neuropathy
Encephalopathy
Dermatologic
Sallow color
Pruritus
Dermatitis

KEY CONCEPTS

Clinical Manifestations of Renal Diseases

- Azotemia is the biochemical manifestation of acute or chronic kidney injury and is characterized by elevated BUN and serum creatinine. It reflects a reduction in GFR.
- Kidney injury that results in azotemia can be either acute or chronic. Acute kidney injury can be reversible or progress to chronic kidney disease, which is generally irreversible.
- One major manifestation of kidney injury is the nephrotic syndrome, in which injury to the glomerulus results in abnormal filtration, leading to heavy proteinuria, edema, and metabolic disturbances.
- Nephritic syndromes are those in which hematuria, azotemia, hypertension, and subnephrotic proteinuria are the major manifestations.
- Diseases involving the tubules and interstitium may have clinical manifestations of the nephritic syndrome, specific defects in tubular function, or acute or chronic kidney disease without specific defining features.

GLOMERULAR DISEASES

Glomerular diseases constitute some of the major problems in nephrology. Glomeruli may be injured by a variety of factors and in the course of several systemic diseases. Disorders in which the kidney is the only or predominant organ involved, such as minimal change disease or membranous nephropathy, constitute the various types of *primary glomerulonephritis* or, because some do not have a cellular inflammatory component, *primary glomerulopathy*. Systemic immunologic diseases such as systemic lupus erythematosus (SLE), vascular disorders such as thrombotic angiopathies, metabolic diseases such as diabetes mellitus, and some hereditary conditions such as Fabry disease often affect the glomerulus as one manifestation. These are termed *secondary glomerular diseases*. However, both the clinical manifestations and glomerular histologic changes in primary and secondary forms can be similar.

In the following sections, we discuss the normal glomerulus and mechanisms of glomerular injury, and then the various primary glomerular diseases. We also briefly review the secondary forms covered in other parts of this book. Table 20.2 lists the most common forms of glomerulonephritis that have reasonably well-defined pathologic and clinical characteristics. The clinical manifestations of glomerular diseases are clustered into the five major groups summarized in Table 20.3. Both primary glomerular diseases and systemic diseases affecting the glomerulus can result in these syndromes. Because glomerular diseases are often associated with systemic disorders, including diabetes, SLE, vasculitis, and monoclonal paraproteinemias, in any patient with manifestations of glomerular disease, it is essential to consider these systemic conditions.

Structure of the Glomerulus

Many clinical manifestations of glomerular diseases result from perturbations of specific components of the glomerulus, so before discussing these diseases we describe the key anatomic structures of glomeruli. The glomerulus consists of an anastomosing network of capillaries lined by fenestrated endothelium invested by two layers of epithelial cells (Fig. 20.1). The visceral epithelial cells, or *podocytes*, are incorporated into and become an intrinsic part of the capillary wall, separated from endothelial cells by a common basement membrane. The parietal epithelium, located on the inner aspect of Bowman capsule, lines the urinary space, the cavity in which plasma filtrate first collects.

TABLE 20.2 Glomerular Diseases

Primary Glomerulopathies
Acute proliferative glomerulonephritis
Postinfectious
Other
Rapidly progressive (crescentic) glomerulonephritis
Membranous nephropathy
Minimal change disease
Focal segmental glomerulosclerosis
Membranoproliferative glomerulonephritis
Dense deposit disease
IgA nephropathy
Systemic Diseases With Glomerular Involvement
Systemic lupus erythematosus
Diabetes mellitus
Amyloidosis
Goodpasture syndrome
Microscopic polyarteritis/polyangiitis
Granulomatosis with polyangiitis
Henoch-Schönlein purpura
Hereditary Disorders
Alport syndrome
Thin basement membrane nephropathy
Fabry disease

TABLE 20.3 Clinical Manifestations of Glomerular Diseases

Disease Group	Manifestations
Nephritic syndrome	Hematuria, azotemia, variable proteinuria, oliguria, edema, and hypertension
Rapidly progressive glomerulonephritis	Acute nephritis, proteinuria, and acute renal failure
Nephrotic syndrome	>3.5 g/day proteinuria, hypoalbuminemia, hyperlipidemia, lipiduria
Chronic kidney disease	Azotemia → uremia progressing for months to years
Isolated urinary abnormalities	Glomerular hematuria and/or subnephrotic proteinuria

The glomerular capillary wall is the filtration barrier, and consists of the following (Fig. 20.2):

- A thin layer of fenestrated *endothelial cells*, with each fenestra being about 70 to 100 nm in diameter.
- The *glomerular basement membrane* (GBM) with a thick electron-dense central layer, the *lamina densa*, and thinner electron-lucent peripheral layers, the *lamina rara interna* and *lamina rara externa*. The GBM consists of collagen (mostly type IV), laminin, polyanionic proteoglycans (mostly heparan sulfate), fibronectin, entactin, and several other glycoproteins. Type IV collagen forms a network to which other glycoproteins attach. The building block (monomer) of this network is a triple-helical molecule composed of one or more of six types of α chains (COL4A1 to COL4A6). Each molecule consists of an N-terminal domain, a triple-helical domain in the middle, and a globular noncollagenous (NC1) domain at the C terminus. The biochemical properties of these structural components are critical to understanding glomerular diseases. For example, antigens in the NC1 domain are the targets of antibodies in anti-GBM nephritis; genetic defects in the α -chains underlie some forms of Alport syndrome; and the proteoglycan content of the GBM may contribute to its permeability characteristics (Fig. 20.3).

- The *visceral epithelial cells* (podocytes) form interdigitating processes and are adherent to the lamina rara externa of the basement membrane (see Fig. 20.1). Adjacent *foot processes* are separated by 20- to 30-nm-wide *filtration slits*, which are bridged by a unique cell-cell junction, the slit diaphragm (see Fig. 20.2).
- The entire glomerular tuft is supported by the mesangium, the tissue that lies between the capillaries and is composed of basement membrane–like *mesangial matrix* that forms a meshwork in which the *mesangial cells* are embedded (see Fig. 20.1). These cells, of mesenchymal origin, are contractile, phagocytic, and capable of proliferation, of both matrix and collagen production, and of secreting several biologically active mediators. Biologically, they are most akin to vascular smooth muscle cells and pericytes. Because no basement membrane separates mesangial cells from the capillary loops, glomerulonephritides often affect both the mesangial and the intracapillary compartments.

The normal glomerulus is highly permeable to water and small solutes because of the fenestrated nature of the endothelium, and it is impermeable to proteins of the size of albumin (~3.6-nm radius; 70 kilodaltons [kD] molecular weight) or larger. The permeability characteristics of the *glomerular filtration barrier* allow discrimination among various protein molecules, depending on their size (the larger, the less permeable) and charge (the more cationic, the more permeable). This size- and charge-dependent barrier function is accounted for by the structure of the capillary wall. The charge-dependent restriction is important in the virtually complete exclusion of albumin from the filtrate, because albumin is an anionic molecule.

The podocytes are important for the maintenance of glomerular barrier function and are largely responsible, under normal circumstances, for synthesis of GBM components. Their specialized junctional complex, the slit diaphragm, represents a size-selective distal barrier to the filtration of proteins, enabling effective charge- and size-dependent retention of plasma proteins. Proteins forming the slit diaphragm itself and molecules that are part of the intracellular aspect of the slit diaphragm complex are illustrated in Fig. 20.4. Nephlin is a transmembrane protein with a large extracellular portion composed of immunoglobulin (Ig)-like domains. Nephlin molecules extend toward each other from neighboring foot processes and dimerize across the filtration slit. Within the cytoplasm of the foot processes, nephlin forms molecular connections with intracellular slit diaphragm components such as podocin and CD2-associated protein (CD2AP), and ultimately links with the actin cytoskeleton, which anchors on focal adhesion hubs. The importance of the slit diaphragm proteins in maintaining glomerular barrier function is demonstrated by the observation that mutations in the genes encoding them give rise to defects in permeability and familial nephrotic syndromes (discussed later).

Pathologic Responses of the Glomerulus to Injury

Various types of glomerulopathies are characterized by one or more of four basic tissue reactions.

Hypercellularity

Some *inflammatory diseases* of the glomerulus are characterized by an increase in the number of cells in the glomerular tufts. This hypercellularity results from one or more of the following:

- *Proliferation* of mesangial or endothelial cells.
- *Infiltration of leukocytes*, including neutrophils, monocytes, and, in some diseases, lymphocytes.
- *Formation of crescents*. These are extracapillary accumulations of cells composed of proliferating glomerular epithelial cells. The epithelial cell

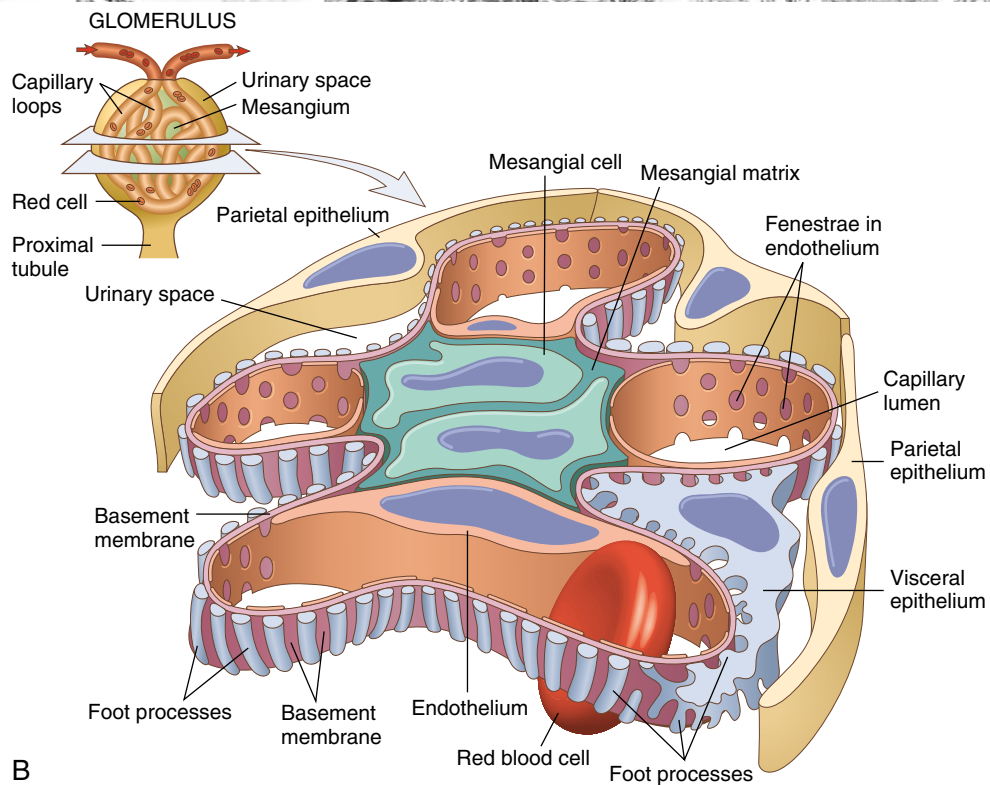
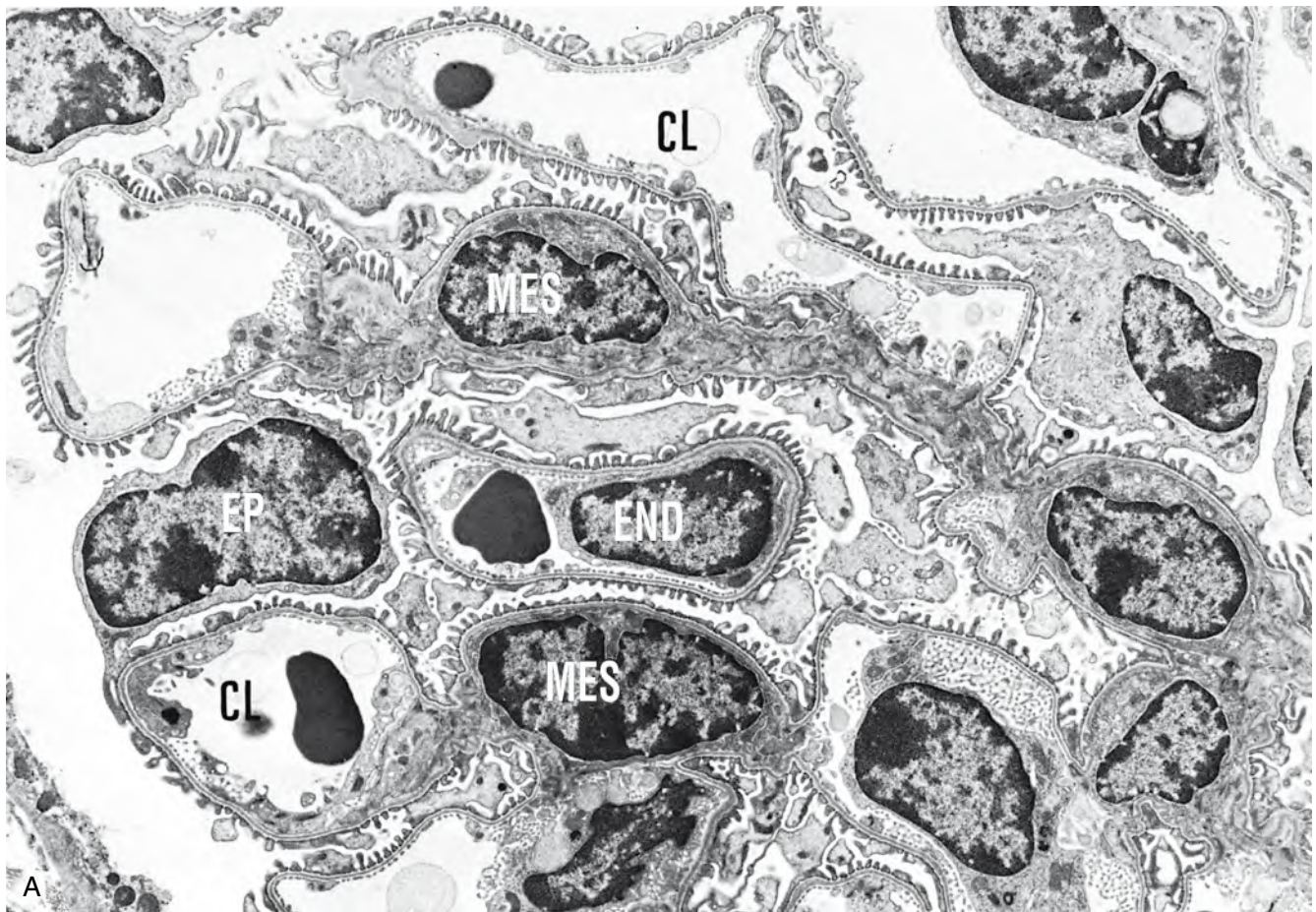


Fig. 20.1 (A) Low-power electron micrograph of renal glomerulus. *CL*, Capillary lumen; *EP*, visceral epithelial cells with foot processes; *END*, endothelium; *MES*, mesangium. (B) Schematic representation of a glomerulus. (Boston.)



Fig. 20.2 Glomerular filter consisting (from bottom to top) of fenestrated endothelium, basement membrane, and foot processes of epithelial cells. Note the filtration slits and diaphragm (arrows) between the foot processes. Note also that the basement membrane consists of a central lamina densa, sandwiched between two looser layers, the lamina rara interna and lamina rara externa. (Courtesy Dr. Helmut Rennke, Brigham and Women's Hospital, Boston.)

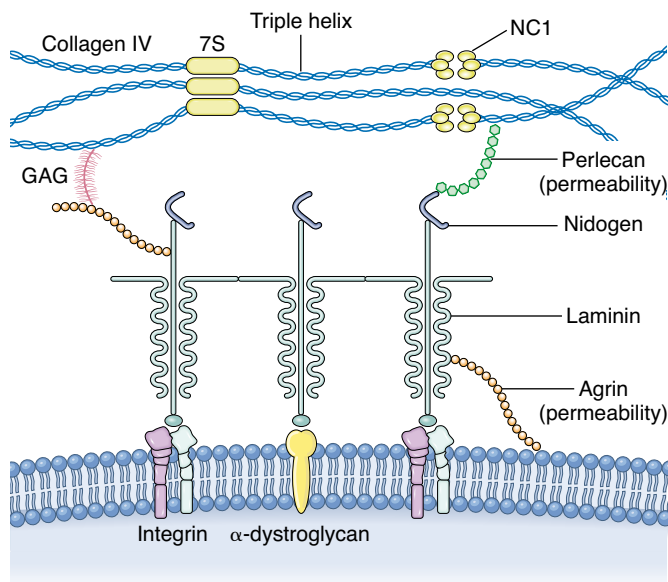


Fig. 20.3 A simplified diagram of the components of the glomerular basement membrane. GAG, Glycosaminoglycan; NC1, noncollagenous portion of the GBM

proliferation that characterizes crescent formation occurs following an immune/inflammatory injury that ruptures the glomerular basement membrane. Plasma proteins leak into the urinary space, which results in fibrin clot formation. Activation of coagulation factors such as thrombin is suspected of being a trigger for crescent formation, but the actual mechanisms are poorly understood.

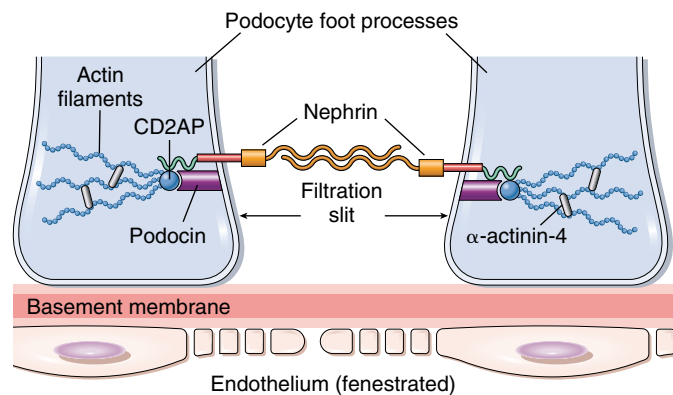


Fig. 20.4 A simplified diagram of some of the best-studied proteins of the glomerular slit diaphragm. CD2AP, CD2-associated protein.

Basement Membrane Thickening

By light microscopy, this change appears as thickening of the capillary walls, best seen in sections stained with periodic acid–Schiff (PAS). By electron microscopy, such thickening takes one of several forms:

- *Deposition of amorphous electron-dense material*, most often immune complexes, on the endothelial or epithelial side of the basement membrane or within the GBM itself. Fibrin, amyloid, cryoglobulins, and abnormal fibrillary proteins may also deposit in the GBM.
- *Increased synthesis of the protein components of the basement membrane*, as occurs in diabetic glomerulosclerosis.
- *Formation of additional layers of basement membrane matrices* (“double contours”), which most often occupy subendothelial locations and may range from loosely organized matrix to fully duplicated lamina densa, as occurs in the membranoproliferative pattern of glomerular injury, either following episodic endothelial injury in the context of subendothelial deposits or in chronic thrombotic microangiopathies.
- Mutations in genes encoding components of the GBM sometimes result in thickening, such as in Alport syndrome, in which mutations affecting type IV collagen impart a thickened basket-weave appearance to the GBM.

Hyalinosis and Sclerosis

Hyalinosis, as applied to the glomerulus, denotes the accumulation of material that is homogeneous and eosinophilic by light microscopy. *Hyaline* is an extracellular, amorphous material composed of high molecular weight plasma proteins that have leaked from the circulation into glomerular structures. When extensive, these deposits may obliterate the capillary lumens of the glomerular tuft. Hyalinosis can be a consequence of endothelial or capillary wall injury, and also occurs following extensive podocyte loss in diabetic nephropathy and collapsing glomerulosclerosis. It often is the end result of various forms of glomerular filter damage.

Sclerosis is characterized by deposition of extracellular collagenous matrix. It may be confined to mesangial areas, as is often the case in diabetic glomerulosclerosis, involve the capillary loops, or both. The sclerosing process may also result in obliteration of some (segmental sclerosis) or all (global sclerosis) of the capillary lumens in affected glomeruli.

Many primary glomerulopathies are classified by their histology. The histologic changes can be further subdivided by their distribution into the following categories: *diffuse*, involving all of the glomeruli in the

kidney; *global*, involving the entirety of individual glomeruli; *focal*, involving only a fraction of the glomeruli in the kidney; *segmental*, affecting a part of each glomerulus; and *capillary loop* or *mesangial*, affecting predominantly capillary or mesangial regions.

KEY CONCEPTS

Glomerular Injury

- The GBM is composed of type IV collagen molecules and other matrix proteins. These proteins can be the target of antibodies in some types of glomerulonephritis; genetic abnormalities in their composition are the basis for some forms of Alport syndrome.
- Visceral epithelial cells (podocytes) are a critical component of the glomerular filtration barrier, and their injury leads to leakage of plasma proteins, mainly albumin, into the urinary space (proteinuria).
- The acute glomerular response to injury includes hypercellularity with proliferation of mesangial and/or endothelial cells, influx of leukocytes, and, when severe, formation of crescents.
- Chronic glomerular responses to injury include basement membrane thickening, hyalinosis, and sclerosis.

Pathogenesis of Glomerular Injury

Immune mechanisms underlie most forms of primary glomerulopathies and many secondary glomerular disorders (Table 20.4). Glomerulonephritis can be induced experimentally by antigen-antibody reactions. Furthermore, glomerular deposits of immunoglobulins, often with components of complement, are found in most individuals with glomerulonephritis. Cell-mediated immune reactions may also play a role, usually in concert with antibody-mediated mechanisms. We begin this discussion with a review of antibody-instigated injury.

Two forms of antibody-mediated injury have been established: (1) injury by *antibodies reacting in situ within the glomerulus*, either binding to insoluble fixed (intrinsic) glomerular antigens or extrinsic molecules planted within the glomerulus, leading to deposition and/or

functional interference, and (2) injury resulting from *deposition of circulating antigen-antibody complexes in the glomerulus*.

Diseases Caused by In Situ Formation of Immune Complexes

In this form of injury, immune complexes are formed locally by antibodies that react with intrinsic tissue antigens or with extrinsic antigens “planted” in the glomerulus from the circulation (Fig. 20.5A). Membranous nephropathy is the classic example of glomerular injury resulting from local formation of immune complexes by antibodies reactive with endogenous antigens. The target antigens are spatially separated in the GBM and not continuous.

The pattern of deposition of antibodies and complement proteins by immunofluorescence microscopy is granular, reflecting localized accumulation of antigen-antibody complexes (Fig. 20.5D). On electron microscopy, the glomerulopathy is characterized by the presence of numerous discrete subepithelial electron-dense immune complex deposits. These subepithelial deposits, with resultant host responses, eventually result in a thickened basement membrane appearance by light microscopy; hence the term *membranous nephropathy*.

Some antibodies that bind to intrinsic antigens directly interfere with the function of glomerular proteins. An example are autoantibodies described in minimal change disease, which bind to the slit diaphragm component nephrin and disrupt the junctional complexes between podocytes. These antibody-antigen complexes appear to be short lived; as a result, they can be difficult to detect by immunostaining.

Antibodies can also react in situ with antigens that are not normally present in the glomerulus but are “planted” there. Such antigens may localize to the kidney by interacting with various intrinsic components of the glomerulus. Planted antigens include cationic molecules that bind to anionic components of the glomerulus; DNA, nucleosomes, and other nuclear proteins, which may have an affinity for GBM components; bacterial products; large aggregated proteins (e.g., aggregated immunoglobulins), which deposit in the mesangium due to their size and biochemical properties; and immune complexes themselves, because they continue to have reactive sites for further interactions with free antibody, free antigen, or complement. There is no dearth of other possible planted antigens, including viral, bacterial, and parasitic products and drugs.

TABLE 20.4 Immune Mechanisms of Glomerular Injury

In Situ Immune Complex Deposition
Fixed intrinsic tissue antigens
NC1 domain of type IV collagen antigen (anti-GBM nephritis)
PLA ₂ R, NELL1, THSD7A, EXT1/2 antigens (membranous glomerulopathy)
Mesangial antigens
Others
Planted antigens
Exogenous (infectious agents, drugs)
Endogenous (DNA, nuclear proteins, immunoglobulins, immune complexes, IgA)
Antibody-Mediated Injury
Circulating Immune Complex Deposition
Endogenous antigens (e.g., DNA, tumor antigens)
Exogenous antigens (e.g., infectious products)
Other Causes of Immune Injury
Activation of alternative complement pathway
T cell-mediated immune responses

EXT 1/2, Exostosin 1 and 2; GBM, glomerular basement membrane; NELL1, neural epidermal growth factor like 1; PLA₂R, phospholipase A2 receptor; THSD7A, thrombospondin type 1 domain-containing 7A.

Disease Caused by Antibodies Directed Against Normal Components of the Glomerular Basement Membrane

In anti-GBM antibody-induced glomerulonephritis, antibodies bind to intrinsic antigens distributed along the entire length of the GBM, resulting in a diffuse linear pattern of staining for the antibodies by immunofluorescence techniques (Fig. 20.5B,E). This contrasts with the granular pattern of immunofluorescence staining corresponding to the discrete immune complexes seen in membranous nephropathy, or other glomerular diseases in which large complexes of antigens and antibodies form in situ. Although anti-GBM antibody-induced glomerulonephritis accounts for less than 5% of cases of human glomerulonephritis, it is one of the more common causes of the clinical syndrome of RPGN.

Glomerulonephritis Resulting From Deposition of Circulating Immune Complexes

This type of glomerular injury is caused by the entrapment of circulating antigen-antibody complexes within glomeruli. The antibodies have no immunologic specificity for glomerular constituents, and the complexes localize to the glomeruli because of their physicochemical

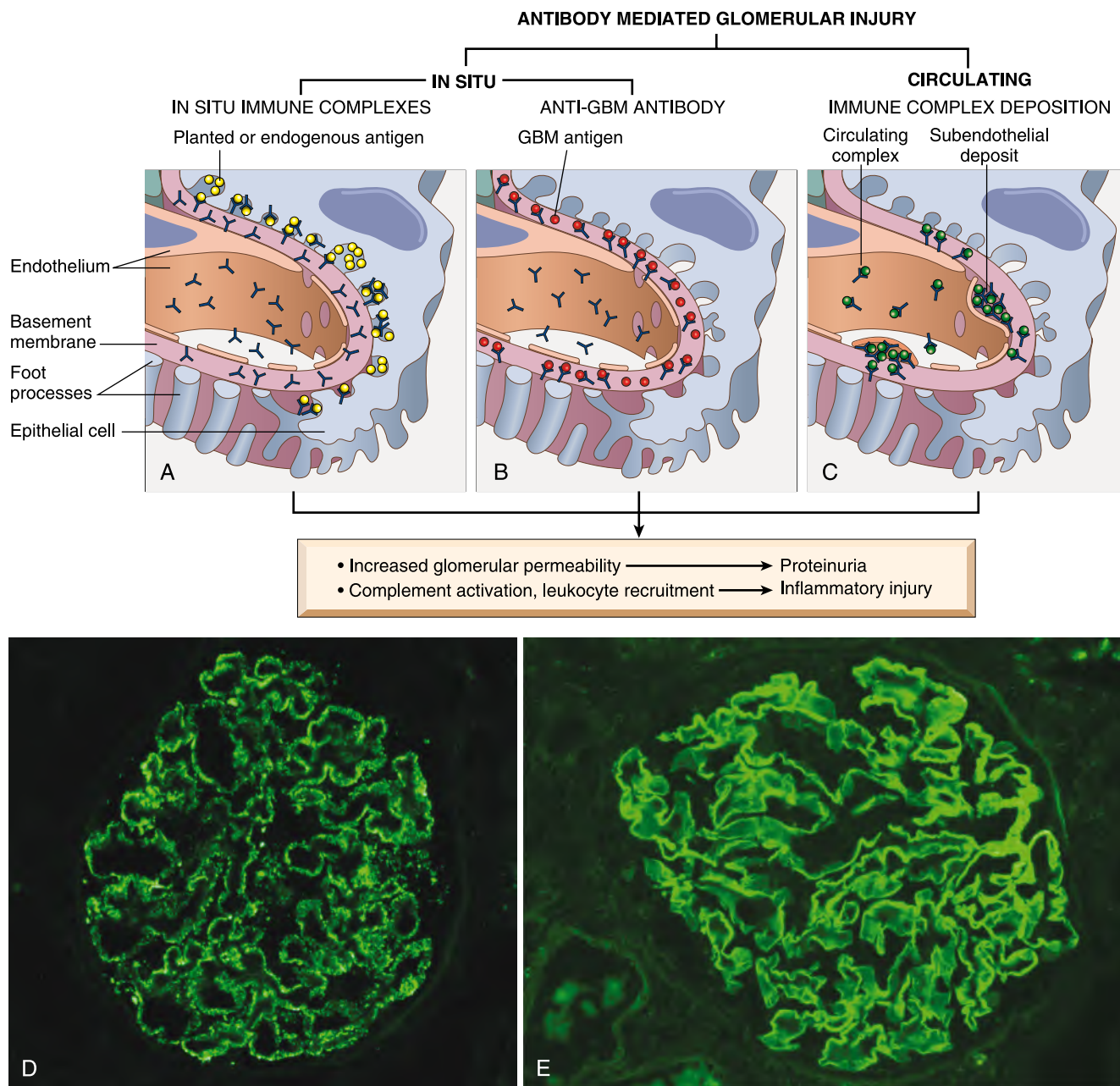


Fig. 20.5 Antibody-mediated glomerular injury can result from in situ formation of immune complexes (A), deposition of anti-glomerular basement membrane antibody (B), or deposition of circulating immune complexes (C). The patterns of deposition of antibody (IgG, shown here) or complement proteins, as seen by immunofluorescence microscopy reflect the underlying mechanisms: granular, characteristic of immune complex deposition (D), and linear, characteristic of anti-GBM disease (E). (D, Courtesy Dr. J. Kowalewska, Department of Pathology, University of Washington, Seattle. E, From Kumar V, Abbas AK, Aster JC. *Robbins Basic Pathology*. 10th ed. Elsevier; 2018.)

properties and hemodynamic factors specific to the glomerulus (Fig. 20.5C).

The antigens that trigger the formation of circulating immune complexes may be of endogenous origin, as in the glomerulonephritis associated with SLE or in IgA nephropathy, or they may be exogenous, as may occur in infection-associated glomerulonephritis. Microbial antigens that are implicated include bacterial products (streptococcal proteins), the surface antigen of hepatitis B virus, hepatitis C virus antigens, and antigens of *Treponema pallidum* and *Plasmodium falciparum*. Some tumor antigens

are also thought to cause immune complex-mediated nephritis. In many cases, the inciting antigen is unknown.

Mechanisms of Glomerular Injury Following Immune Complex Formation

The pathogenesis of immune complex diseases is discussed in Chapter 6. Here we briefly review the salient features that relate to glomerular injury. **Whatever the antigen may be, antigen-antibody complexes formed or deposited in the glomeruli may elicit a local inflammatory**

reaction that produces injury. The antibodies may activate complement and engage Fc receptors on leukocytes and perhaps glomerular mesangial or other cells, leading to inflammation. Generally, subendothelial immune complexes provoke leukocyte infiltration and glomerular crescent formation. Similarly, mesangial immune complex deposition can elicit an inflammatory response as the mesangium is not separated from the capillary lumen by a GBM; the glomerular lesions may exhibit leukocytic infiltration and proliferation of mesangial and endothelial cells. In contrast, subepithelial immune complexes are shielded from the circulation by the GBM and do not induce inflammatory responses

Electron microscopy reveals electron-dense deposits containing immune complexes that may lie in the mesangium, between the endothelial cells and the GBM (subendothelial deposits), or between the outer surface of the GBM and the podocytes (subepithelial deposits). Deposits may be located at more than one site in a given case. By immunofluorescence microscopy, the immune complexes are seen as granular deposits along the basement membrane (see Fig. 20.5D), in the mesangium, or in both locations. Once deposited in the kidney, immune complexes may eventually be degraded, mostly by infiltrating neutrophils and monocytes/macrophages, mesangial cells, and endogenous proteases, and the inflammatory reaction may then subside, allowing for complete recovery of the glomerulus. Such a course occurs when the exposure to the inciting antigen is short-lived and/or limited, as in most cases of poststreptococcal glomerulonephritis. However, if immune complexes are deposited repeatedly for prolonged periods, as may be seen with SLE or viral hepatitis, episodic cellular injury and tissue remodeling results in membranous (subepithelial deposition) or membranoproliferative (subendothelial deposition) patterns of glomerular injury.

Several factors affect the localization of antigen, antibody, or immune complexes in the glomerulus, such as the charge and size of the immune reactants, glomerular hemodynamics, mesangial function, and the integrity of the charge-selective capillary wall barrier, leading to variable patterns of immune-reactant deposition in various forms of glomerulonephritis (Fig. 20.6). In turn, the distinct patterns of localization of immune complexes are a key determinant of the injury response and the histologic features that subsequently develop.

In summary, **most cases of immune complex-mediated glomerulonephritis are a consequence of deposition of discrete immune complexes, which give rise to granular immunofluorescence staining along the peripheral capillary wall or in the mesangium.** However, it is difficult to determine whether the deposition has occurred in situ, by circulating complexes, or by both mechanisms because entrapment of circulating immune complexes can initiate further in situ complex formation. Single etiologic agents, such as hepatitis B and C viruses, can cause either a membranous pattern of glomerulonephritis, with subepithelial in situ deposition of antigens, or a membranoproliferative pattern indicative of subendothelial deposition of antigens or deposition of circulating complexes.

Cell-Mediated Immunity in Glomerulonephritis

Although antibody-mediated mechanisms initiate most forms of glomerulonephritis, there is evidence that sensitized T cells may also cause glomerular injury and contribute to the progression of some glomerulonephritides. Clues to involvement of cellular immunity include the presence of activated macrophages and T cells and their products in the glomerulus in some forms of human and experimental glomerulonephritis; in vitro and in vivo evidence of lymphocyte activation on exposure to antigen in human and experimental glomerulonephritis; abrogation of glomerular injury by lymphocyte depletion; and experiments in which glomerular injury may be induced by transfer of T cells

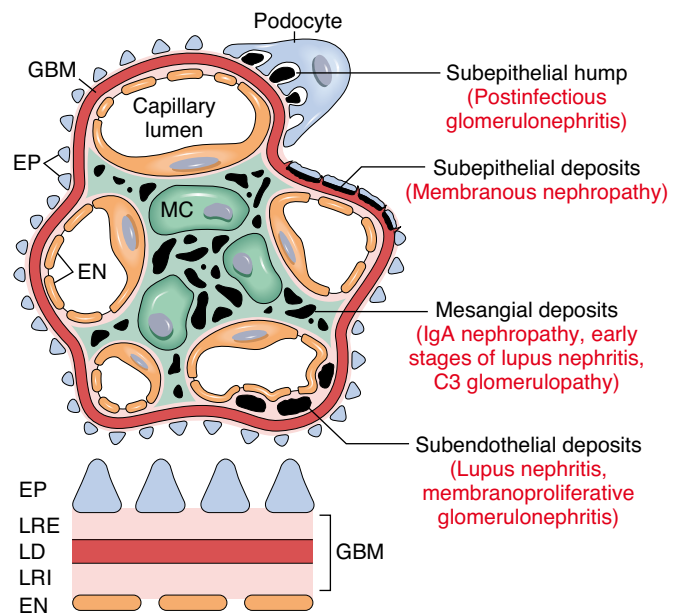


Fig. 20.6 Localization of immune complexes in the glomerulus: isolated subepithelial humps, as in acute glomerulonephritis; frequent regularly spaced subepithelial deposits, as in membranous nephropathy; subendothelial deposits, as in lupus nephritis and membranoproliferative glomerulonephritis; mesangial deposits, as in IgA nephropathy. EN, Endothelium; EP, epithelium; GBM, glomerular basement membrane; LD, lamina densa; LRE, lamina rara externa; LRI, lamina rara interna; MC, mesangial cell; MM, mesangial matrix. (Modified from Couser WG. Mediation of immune glomerular injury. *J Am Soc Nephrol.* 1990;1[1]:13-29.)

from nephritic animals to normal recipients. The evidence is most compelling for certain types of experimental crescentic glomerulonephritis in which antibodies to GBM initiate glomerular injury and activated T lymphocytes propagate the inflammation. Despite this body of suggestive evidence, there is no proof that glomerulonephritis in humans results primarily from T-cell activation.

Activation of Alternative Complement Pathway

Alternative complement pathway activation occurs in the clinicopathologic entity called *C3 glomerulopathy*, which encompasses the entities of dense deposit disease and C3 glomerulonephritis. This is discussed later.

Mediators of Glomerular Injury

Once immune reactants or sensitized T cells have localized in the glomerulus, how does the glomerular damage ensue? The mediators—both cells and molecules—are the usual suspects involved in acute and chronic inflammation, described in Chapter 3, and only a few are highlighted here.

Cells. Numerous recruited and intrinsic cells produce mediators that are responsible for glomerular injury.

- **Neutrophils** and **monocytes** infiltrate glomeruli in certain types of glomerulonephritis, largely because of activation of complement, resulting in generation of chemotactic molecules (mainly C5a), but also by Fc receptor-mediated activation. Neutrophils release proteases, which cause GBM degradation; oxygen-derived free radicals, which cause cell damage; and arachidonic acid metabolites, which cause vascular changes that contribute to reductions in GFR.
- **Macrophages** and **T lymphocytes**, which infiltrate glomeruli in antibody- and cell-mediated reactions, release a vast number of biologically active molecules when activated.

- *Platelets* may aggregate in glomeruli during immune-mediated injury. Their release of eicosanoids, growth factors, and other mediators may contribute to vascular injury and proliferation of glomerular cells. Antiplatelet agents have beneficial effects in both human and experimental glomerulonephritis.
- *Resident glomerular cells*, particularly mesangial cells, can be stimulated to produce inflammatory mediators, including reactive oxygen species (ROS), cytokines, chemokines, growth factors, eicosanoids, nitric oxide, and endothelin. They may initiate inflammatory responses in glomeruli, even in the absence of leukocytic infiltration.

Soluble Mediators. Virtually all known inflammatory chemical mediators (Chapter 3) have been implicated in glomerular injury.

- *Complement activation* leads to the generation of chemotactic products that induce leukocyte influx (complement-neutrophil-dependent injury) and the formation of C5b–C9, the membrane attack complex. C5b–C9 causes cell lysis and may also stimulate mesangial cells to produce oxidants, proteases, and other mediators. Thus, even in the absence of neutrophils, C5b–C9 can cause proteinuria, as has been demonstrated in experimental membranous glomerulopathy. In some diseases, collectively called C3 glomerulopathies, there is evidence of complement activation resulting from defective regulation of the alternative complement pathway, which can be triggered or exacerbated by the presence of immune complexes or monoclonal immunoglobulins.
- *Eicosanoids, nitric oxide, angiotensin, and endothelin* are involved in hemodynamic changes.
- *Cytokines*, particularly IL-1 and tumor necrosis factor (TNF), which may be produced by infiltrating leukocytes and resident glomerular cells, induce leukocyte adhesion and a variety of other effects.
- *Chemokines* such as monocyte chemoattractant protein 1 (also known as CCL2) promote monocyte and lymphocyte influx.
- *Growth factors* such as platelet-derived growth factor (PDGF) are involved in mesangial cell proliferation. TGF- β , connective tissue growth factor, and fibroblast growth factor seem to be critical in the extracellular matrix (ECM) deposition and hyalinization leading to glomerulosclerosis in chronic injury. Vascular endothelial growth factor (VEGF) maintains endothelial integrity and may help regulate capillary permeability.
- The *coagulation system* is also a mediator of glomerular damage. Fibrin is frequently present in the glomeruli and Bowman space in glomerulonephritis, indicative of coagulation cascade activation, and activated coagulation factors, particularly thrombin, may be a stimulus for crescent formation.

Epithelial Cell Injury

Podocyte injury is common to many forms of both primary and secondary glomerular diseases, of both immune and nonimmune etiologies. The term *podocytopathy* has been applied to diseases with different etiologies whose principal manifestation is injury to podocytes. Podocyte injury can be induced by antibodies to podocyte antigens in membranous nephropathy and minimal change disease; by toxins, as in an experimental model of proteinuria induced by puromycin aminonucleoside, which inhibits ribosome function; possibly by certain cytokines; in the context of viral infections such as human immunodeficiency virus (HIV), parvovirus and SARS-CoV-2; or by unknown circulating factors postulated to underlie some forms of minimal change disease and primary/idiopathic focal segmental glomerulosclerosis (FSGS). Podocyte injury is reflected by morphologic changes, which include effacement of foot processes, disassembly of the junctional slit diaphragm complex, vacuolization, and

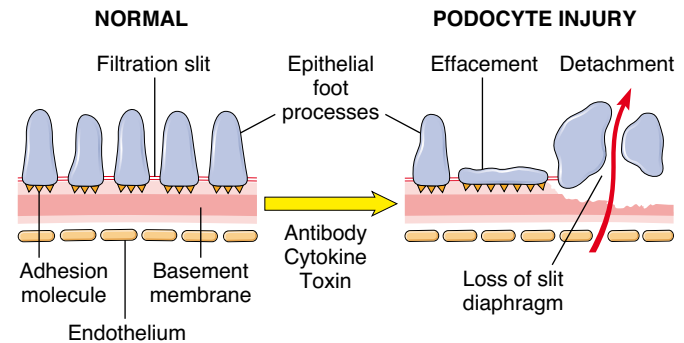


Fig. 20.7 Epithelial cell injury. The postulated sequence is a consequence of antibodies specific to epithelial cell antigens, toxins, cytokines, or other factors causing injury; this results in foot process effacement and sometimes detachment of epithelial cells and protein leakage through defective glomerular basement membrane and filtration slits.

detachment of cells from the GBM via loss of adhesion or cell death, and leads to proteinuria (Fig. 20.7).

Podocytes have a limited capacity for replication and repair, and their loss may be a feature of multiple types of glomerular injury, including FSGS, collapsing glomerulopathy, and diabetic nephropathy. As remaining podocytes undergo compensatory hypertrophy, mild podocyte injury often goes unrecognized in pathologic specimens unless morphometric techniques are applied. In most forms of glomerular injury, loss of slit diaphragms is a key event in the development of proteinuria (see Fig. 20.7). Functional abnormalities of the slit diaphragm may also result from mutations in genes encoding its components, such as nephrin and podocin. Such mutations are implicated in hereditary forms of nephrotic syndrome.

Mechanisms of Progression in Glomerular Diseases

Thus far, the immunologic mechanisms and mediators that *initiate* glomerular injury have been discussed. The outcome of such injury depends on several factors, including the severity of renal damage, the nature and persistence of the antigens, and the immune status, age, and genetic predisposition of the host.

It has long been known that **once any renal disease, glomerular or otherwise, destroys sufficient nephrons to reduce the GFR to about 30% to 50% of normal, progression to end-stage kidney disease proceeds at a steady rate, independent of the original insult or activity of the underlying disease.** The secondary factors that lead to progression are of great clinical interest because they are potential targets of therapy that could delay or prevent the inexorable journey to dialysis or transplantation.

The two major histologic characteristics of such progressive renal damage are *glomerulosclerosis* and *tubulointerstitial fibrosis*.

Glomerulosclerosis. Sclerosis involving portions of some glomeruli (also referred to as FSGS) develops after many types of renal injury and can lead to proteinuria and increasing functional impairment. Glomerulosclerosis may be seen even in cases in which the primary disease was nonglomerular. The mechanisms of glomerulosclerosis in this setting are described in the discussion of FSGS later in this chapter.

Tubular Injury and Interstitial Fibrosis. Tubulointerstitial injury, manifested by tubular damage and interstitial inflammation, is a component of many acute and chronic glomerulonephritides. Tubulointerstitial fibrosis contributes to progression in both immune and nonimmune glomerular diseases, for example, diabetic nephropathy.

Indeed, **the extent of tubulointerstitial damage often shows better correlation with decline in renal function than the severity of glomerular injury.** Many factors may lead to such tubulointerstitial injury, including ischemia of tubular segments downstream from sclerotic glomeruli, acute and chronic inflammation in the adjacent interstitium, and reduced peritubular capillary blood supply. It appears that proteinuria can also cause direct injury to and activation of tubular cells. Activated tubular cells in turn express adhesion molecules and elaborate proinflammatory cytokines, chemokines, and growth factors that contribute to interstitial fibrosis. Filtered proteins that may produce these tubular effects include cytokines, complement products, the iron in hemoglobin, immunoglobulins, lipid moieties, and oxidized plasma proteins.

KEY CONCEPTS

Pathogenesis of Glomerular Injury and Progression of Glomerular Disease

- Antibody-mediated injury is an important mechanism of glomerular damage, mainly via complement- and leukocyte-mediated pathways. Antibodies may also be directly cytotoxic to glomerular cells.
- The most common forms of antibody-mediated glomerulonephritis are caused by the formation and accumulation of immune complexes, which may involve either endogenous antigens (e.g., PLA₂R in membranous nephropathy) or exogenous antigens (e.g., microbial). Immune complexes show a granular pattern of deposition by immunofluorescence.
- Binding of antibodies to intrinsic glomerular proteins can also directly lead to glomerular dysfunction (e.g., nephrin in minimal change disease). In some cases, immune complexes are small and short-lived and may not be readily detectable by immunofluorescence.
- Autoantibodies against components of the GBM are the cause of anti-GBM antibody-mediated disease, often associated with severe injury. The immunofluorescence pattern of antibody deposition is linear.
- Activation of the alternative complement pathway is an important mechanism of injury in C3 glomerulopathy that includes dense deposit disease and C3 glomerulonephritis.
- Soluble inflammatory mediators, such as cytokines, chemokines, growth factors, eicosanoids, nitric oxide, and activated coagulation factors, also contribute to the glomerular injury.
- Epithelial cell (podocyte) injury induced by antibodies, toxins, cytokines, infections, and poorly characterized circulating factors is a common manifestation of various forms of glomerular diseases.
- Progressive glomerular injury, resulting from either primary or secondary causes, ensues from a cycle of glomerular and nephron loss with compensatory changes, which lead to further glomerular injury, glomerulosclerosis, tubulointerstitial fibrosis, and eventually end-stage kidney disease.

Having discussed the factors involved in the initiation and progression of glomerular injury, we now turn to individual glomerular diseases. [Table 20.5](#) summarizes the main clinical and pathologic features of the major forms of primary glomerulopathies.

Nephritic Syndrome

Glomerular diseases presenting with a nephritic syndrome are characterized by inflammation in the glomeruli. The main clinical features of nephritic syndrome include:

- *Hematuria* (red blood cells and red cell casts in urine)
- *Proteinuria* (usually subnephrotic range) with or without edema

- *Azotemia*
- *Hypertension*

Nephritic syndrome is the typical clinical presentation of most proliferative types of glomerulonephritis (GN) such as postinfectious GN, rapidly progressive (crescentic) GN, and proliferative lupus GN (described later, since it also causes nephrotic syndrome). The lesions that cause the nephritic syndrome have in common proliferation of the cells within the glomeruli, often accompanied by infiltration of inflammatory leukocytes. This inflammatory reaction severely injures the capillary walls, permitting blood to pass into the urine and inducing hemodynamic changes that lead to a reduction in GFR. The reduced GFR is manifested clinically by oliguria, fluid retention, and azotemia. Hypertension probably is a result of both the fluid retention and renin release from the ischemic kidneys.

Acute Proliferative (Postinfectious and Infection-Associated) Glomerulonephritis

Postinfectious GN is caused by deposition or formation of immune complexes containing microbial antigens, which induce infiltration of leukocytes and proliferation of glomerular cells. The most common underlying infections are streptococcal, but the disorder may also be associated with other infections. Poststreptococcal GN is decreasing in frequency in the United States due to early antibiotic treatment but continues to be a fairly common disorder worldwide. It usually appears 1 to 4 weeks after a streptococcal infection of the pharynx or skin (impetigo). Skin infections are commonly associated with overcrowding and poor hygiene. Poststreptococcal glomerulonephritis occurs most frequently in children 6 to 10 years of age, but children and adults of any age can be affected.

Pathogenesis

Poststreptococcal GN is caused by immune complexes containing streptococcal antigens and specific antibodies. Only certain strains of group A β -hemolytic streptococci are nephritogenic, more than 90% of cases being traced to types 1, 4, and 12, which can be identified by typing of the M protein of the bacterial cell walls.

Many lines of evidence support an immunologic basis for poststreptococcal glomerulonephritis. The latent period between infection and onset of nephritis is compatible with the time required for the production of antibodies and the formation of immune complexes. Elevated titers of antibodies against one or more streptococcal antigens are present in a great majority of patients. Serum complement levels are low, compatible with activation of the complement system and consumption of complement components. There are granular immune deposits in the glomeruli, indicative of an immune complex-mediated mechanism.

The preponderance of evidence suggests streptococcal pyogenic exotoxin B (SpeB) is the principal antigen in most but not all cases of poststreptococcal glomerulonephritis. This protein is commonly secreted by nephritogenic strains of streptococci, and has been localized to the “humplike” deposits characteristic of this disease (described later). At the outset, the inciting antigens are exogenously planted from the circulation in subendothelial locations in glomerular capillary walls, leading to in situ formation of immune complexes, where they elicit an inflammatory response. Subsequently, through mechanisms that are not well understood, the antigen-antibody complexes dissociate, migrate across the GBM, and reform on the subepithelial side of the GBM. Deposition of

TABLE 20.5 Summary of Major Primary Glomerulonephritides

Disease	Most Frequent Clinical Presentation	Pathogenesis	Glomerular Pathology		
			Light Microscopy	Fluorescence Microscopy	Electron Microscopy
Postinfectious glomerulonephritis	Nephritic syndrome	Immune complex–mediated; circulating or planted antigen	Diffuse endocapillary proliferation; leukocytic infiltration	Granular IgG and C3 in GBM and mesangium; Granular IgA in some cases	Primarily subepithelial humps; subendothelial deposits in early disease stages
Crescentic (rapidly progressive) glomerulonephritis	Nephritic syndrome; rapid progression	Anti-GBM antibody–mediated; immune complex–mediated; ANCA mediated and unknown	Extracapillary proliferation with crescents; necrosis	Linear IgG and C3 in anti-GBM antibody–mediated GN; granular IgG, other Igs, and/or complement in immune complex–mediated GN; or no deposits in ANCA mediated GN	No deposits in anti-GBM and ANCA mediated GN; immune complexes at various locations in immune complex–mediated GN
Membranous nephropathy	Nephrotic syndrome	In situ immune complex PLA ₂ R antigen in most cases of primary disease	Diffuse capillary wall thickening	Granular IgG and C3; diffuse	Subepithelial deposits
Minimal change disease	Nephrotic syndrome	Anti-nephrin antibodies in most cases; loss of glomerular polyanion; podocyte injury	Normal; lipid in tubules	Diffuse fine granular podocyte IgG associated with nephrin clustering in most cases	Effacement of foot processes; no deposits
Focal segmental glomerulosclerosis	Nephrotic syndrome; nonnephrotic proteinuria	Anti-nephrin antibodies in some cases; Genetic predisposition; Ablation nephropathy podocyte injury	Focal and segmental sclerosis and hyalinosis	Diffuse fine granular podocyte IgG associated with nephrin clustering in some cases; IgM + C3 in many cases	Effacement of foot processes; epithelial denudation
Membranoproliferative glomerulonephritis (MPGN) type I	Nephritic/ nephrotic syndrome	Immune complex	Mesangial proliferative or membranoproliferative patterns of proliferation; GBM thickening; splitting	IgG ++ C3; C1q ++ C4	Subendothelial deposits
Dense deposit disease (MPGN type II)	Hematuria Chronic renal failure	Acquired or genetic dysregulation of the alternative complement pathway	Mesangial proliferative or membranoproliferative patterns of injury; GBM thickening; splitting	C3; no C1q or C4	Dense deposits
IgA nephropathy	Recurrent hematuria or proteinuria	Unknown	Focal mesangial proliferative glomerulonephritis; mesangial widening	IgA ± IgG, IgM, and C3 in mesangium	Mesangial and paramesangial dense deposits

ANCA, Antineutrophil cytoplasmic antibodies; GBM, glomerular basement membrane.

circulating immune complexes also may contribute to the lesions. A similar form of glomerulonephritis occurs sporadically in association with other infections, including those of bacterial (e.g., staphylococcal endocarditis, pneumococcal pneumonia, and meningococcemia), viral (e.g., hepatitis B, hepatitis C, mumps, HIV infection, varicella, and infectious mononucleosis), and parasitic (malaria, toxoplasmosis) origin. In these settings, granular immunofluorescent deposits and subepithelial humps characteristic of immune complex nephritis are also present.

MORPHOLOGY

The classic histologic picture is one of **enlarged, hypercellular glomeruli** (Fig. 20.8B). The hypercellularity is caused by (1) infiltration by leukocytes, both neutrophils and monocytes; (2) proliferation of endothelial and mesangial cells; and (3) in severe cases by crescent formation. The proliferation and leukocyte infiltration are typically global and diffuse, that is, involving all lobules of all glomeruli. There is also swelling of endothelial cells, and the combination of proliferation, swelling, and leukocyte infiltration obliterates the capillary lumens. There may be interstitial edema and inflammation, and the tubules often contain red cell casts.

By immunofluorescence microscopy, there are **granular deposits of IgG and C3** (Fig. 20.8C), and sometimes IgM in the mesangium and along the GBM. Although immune complex deposits are almost universally present, they are often focal and sparse. The characteristic electron microscopic findings are discrete, amorphous, **electron-dense deposits** on the epithelial side of the membrane, often having the appearance of “humps” (Fig. 20.8D), presumably representing the antigen-antibody complexes at the subepithelial cell surface. Subendothelial deposits are also commonly seen, typically early in the disease course, and mesangial and intramembranous deposits may be present.

Clinical Features

In the typical case, a young child abruptly develops malaise, fever, nausea, oliguria, and hematuria (smoky or cola-colored urine) 1 to 2 weeks after recovery from a sore throat. The patients have dysmorphic red cells or red cell casts in the urine, mild proteinuria (usually less than 1 g/day), periorbital edema, and mild to moderate hypertension. In adults the onset is more likely to be atypical, such as the sudden appearance of hypertension or edema, frequently with elevation of BUN. The glomerulonephritis is subclinical in some infected individuals, and is discovered only on screening for microscopic hematuria carried out during epidemic outbreaks. Important labora-

tory findings include elevations of antistreptococcal antibody titers and a decline in the serum concentration of C3 and other components of the complement cascade.

More than 95% of affected children eventually recover renal function with conservative therapy aimed at maintaining sodium and water balance. A small minority of children (perhaps less than 1%) do not improve, become severely oliguric, and develop a rapidly progressive form of glomerulonephritis (described later). Some of the remaining patients undergo slow progression to chronic glomerulonephritis with or without recurrence of an active nephritic picture. Prolonged and persistent heavy proteinuria and abnormal GFR mark patients with an unfavorable prognosis.

In adults, the disease is less benign. Although the overall prognosis in epidemics is good, in only about 60% of sporadic cases do the patients recover promptly. In the remainder, the glomerular lesions fail to resolve quickly, as manifested by persistent proteinuria, hematuria, and hypertension. In some of these patients, the lesions eventually clear, but others develop chronic glomerulonephritis or even RPGN.

Rapidly Progressive (Crescentic) Glomerulonephritis

Rapidly progressive glomerulonephritis (RPGN) is a clinical syndrome associated with severe glomerular injury. It is characterized

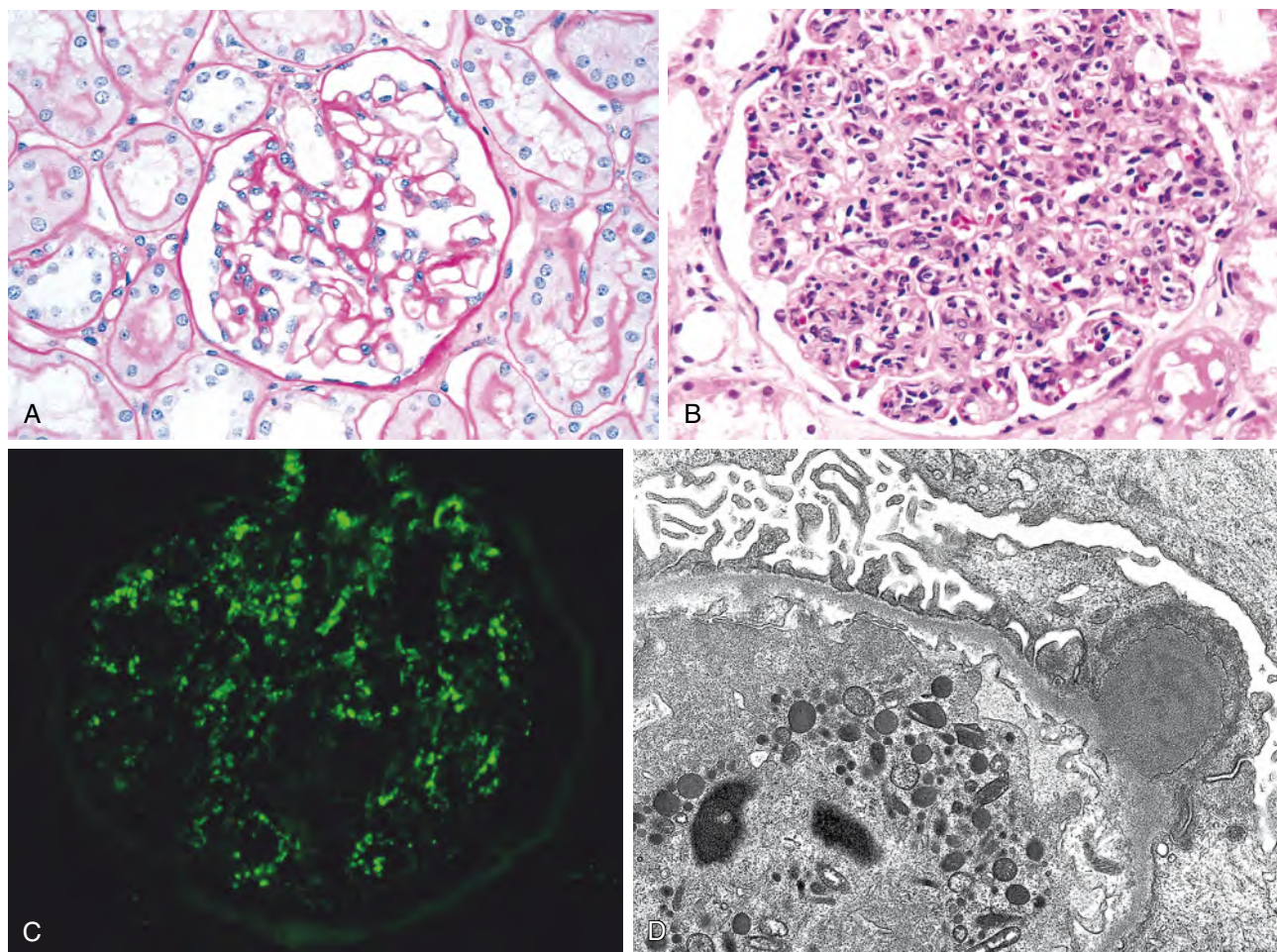


Fig. 20.8 Acute proliferative glomerulonephritis. (A) Normal glomerulus. (B) Glomerular hypercellularity is due to infiltration of leukocytes and proliferation of intrinsic glomerular cells. (C) Immunofluorescent stain demonstrates discrete, coarsely granular deposits of complement protein C3 (stain for IgG was similar), corresponding to “humps” illustrated in part D. (D) Typical electron-dense subepithelial “hump” and a neutrophil in the lumen. (A–C, Courtesy Dr. H Rennke, Brigham and Women’s Hospital, Boston. D, Courtesy DJ Kowalewska, Cedars-Sinai Medical Center, Los Angeles.)

by relatively rapid and progressive loss of renal function associated with severe oliguria and signs of nephritic syndrome. The most common histologic picture is the presence of crescents in most of the glomeruli, hence the name *crescentic glomerulonephritis*. As discussed earlier, crescents are produced predominantly by the proliferation of the epithelial cells lining the Bowman capsule. The presence of fibrous (healed) crescents indicates that the injury has been present for some time or is episodic; this presentation is often associated with slower disease progression.

Pathogenesis

RPGN, typically associated with crescent formation, may be a manifestation of several different diseases, some restricted to the kidney and others systemic. Although no single mechanism can explain all cases, there is little doubt that **in most cases the glomerular injury is immunologically mediated**. A practical classification divides crescentic GN into several groups based on immunologic findings (Table 20.6), which have been described as:

- *Anti-GBM antibody-mediated disease, characterized by linear deposits of IgG and, in many cases, C3 in the GBM.* In some of these patients, the anti-GBM antibodies cross-react with pulmonary alveolar basement membranes to produce the clinical picture of pulmonary hemorrhage associated with renal failure (*Goodpasture syndrome*). Plasmapheresis to remove the pathogenic circulating antibodies is the mainstay of treatment, which also includes therapy to suppress the underlying immune response.
The antigen common to the alveoli and GBM is a peptide within the noncollagenous portion of the α_3 chain of collagen type IV. What triggers the formation of these autoantibodies is unclear. Exposure to viruses or hydrocarbon solvents (found in paints and dyes) has been implicated in some patients, as have various drugs and cancers. There is a high prevalence of certain human leukocyte antigen (HLA) alleles (e.g., HLA-DRB1) in affected patients, a finding consistent with a genetic predisposition to autoimmunity.
- *Diseases caused by immune complex deposition, with granular deposits of antibodies and complement by immunofluorescence.* Crescentic GN can be a complication of any of the immune complex nephritides, including postinfectious glomerulonephritis, lupus nephritis, IgA nephropathy, fibrillary GN and IgA vasculitis/Henoch-Schönlein purpura. These patients usually cannot be helped by plasmapheresis, and they require treatment for the underlying disease.

TABLE 20.6 Crescentic (Rapidly Progressive) Glomerulonephritides

Anti-GBM Antibody (Type I)
Renal limited Goodpasture syndrome
Immune Complex (Type II)
Idiopathic Postinfectious glomerulonephritis Lupus nephritis Henoch-Schönlein purpura IgA nephropathy Fibrillary glomerulonephritis C3 glomerulonephritis
Pauci-Immune (Type III)
ANCA-associated Idiopathic Granulomatosis with polyangiitis Microscopic polyangiitis

ANCA, Antineutrophil cytoplasmic antibodies; GBM, glomerular basement membrane.

- *C3-only or C3-dominant crescentic GN related to abnormalities of the alternative complement-regulatory pathway.* While C3-related glomerular disease is mostly associated with endocapillary proliferative or membranoproliferative patterns of injury, it can rarely also present with crescentic GN. Autoantibodies to complement-regulatory factors or mutations in *CFHR5*, which encodes a putative complement regulatory protein, may be underlying causes. Ultrastructurally, glomeruli often show ill-defined mildly electron-dense deposits with strong immunoreactivity for C3 by immunofluorescence.
- *Pauci-immune crescentic GN, defined by the lack of detectable anti-GBM antibodies or a paucity of immune complexes by immunofluorescence and electron microscopy.* Most patients with this type of crescentic GN have circulating antineutrophil cytoplasmic antibodies (ANCA) that produce a cytoplasmic (PR3-ANCA) or perinuclear (MPO-ANCA) staining pattern and are known to play a role in some vasculitides (Chapter 11). This type of crescentic GN may be a component of a systemic vasculitis such as granulomatosis with polyangiitis (formerly called Wegener granulomatosis) or microscopic polyangiitis. In many cases, however, pauci-immune crescentic glomerulonephritis is limited to the kidneys. More than 90% of cases have PR3-ANCA (previously c-ANCA, so called because the antibodies are specific for the neutrophil granule protein proteinase-3) or MPO-ANCA (previously p-ANCA, indicating reactivity with neutrophil myeloperoxidase) in the serum. ANCA have proved to be invaluable as a highly sensitive diagnostic marker for pauci-immune crescentic GN, but proof of their role as a direct cause has been elusive. Recent evidence of their pathogenic potential has been obtained by studies in mice showing that transferring antibodies against myeloperoxidase (MPO) induces a form of crescentic GN. B-cell targeting therapies therefore have a critical role in treatment of ANCA-associated vasculitides.

To summarize, about one-fifth of patients with crescentic GN have anti-GBM antibody-mediated glomerulonephritis without lung involvement; another one-quarter have immune complex-mediated crescentic glomerulonephritis; and the remainder (which constitute the majority) are of the pauci-immune type.

MORPHOLOGY

The kidneys are enlarged and pale, often with petechial hemorrhages on the cortical surfaces. Depending on the underlying cause, the glomeruli may show focal and segmental necrosis, diffuse or focal endothelial proliferation, and mesangial proliferation. Segmental glomerular fibrinoid necrosis with destruction of tuft segments and distinctive extracapillary **crescents** (Fig. 20.9A) adjacent to unaffected glomeruli are the features most typical of pauci-immune RPGN. Crescents consist of proliferating glomerular epithelial cells and monocytes and macrophages that have migrated into the urinary space. Neutrophils and lymphocytes may be present. The crescents may obliterate the urinary space and compress the glomerular tuft. **Fibrin strands** are frequently prominent between the cellular layers in the crescents. By immunofluorescence microscopy, immune complex-mediated cases show granular immune deposits; anti-GBM-mediated cases show linear GBM fluorescence for Ig and complement; C3GN may have glomerular reactivity for C3 only; and pauci-immune cases have little or no deposition of immune reactants. Electron microscopy discloses deposits in those cases due to immune complex deposition or complement dysregulation. Regardless of type, electron microscopy may show **ruptures in the GBM**, which allows leukocytes, plasma proteins such as coagulation factors and complement, and inflammatory mediators to reach the urinary space, where they trigger crescent formation (Fig. 20.9B). In time, most crescents undergo organization and foci of segmental necrosis resolve as segmental scars (a type of segmental sclerosis), but restoration of normal glomerular architecture may be achieved with early aggressive therapy.

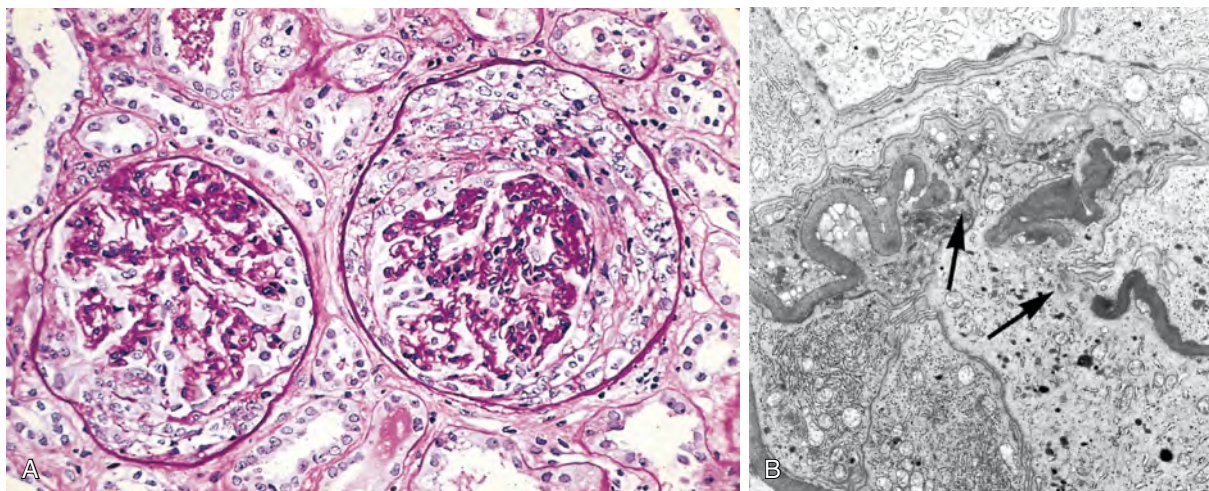


Fig. 20.9 Crescentic glomerulonephritis. (A) Note the compressed glomerular tufts and the crescent-shaped mass of proliferating epithelial cells and leukocytes within the Bowman capsule (periodic acid-Schiff stain). (B) Electron micrograph showing characteristic wrinkling of glomerular basement membrane with focal disruptions (arrows). (A, Courtesy Dr. MA Venkatachalam, University of Texas Health Sciences Center, San Antonio.)

Clinical Features

The renal manifestations of all forms of crescentic glomerulonephritis include hematuria with red blood cell casts in the urine, moderate proteinuria occasionally reaching the nephrotic range, and variable hypertension and edema. In Goodpasture syndrome, the course may be dominated by recurrent hemoptysis or even life-threatening pulmonary hemorrhage. Measurement of serum anti-GBM antibodies, antinuclear antibodies and ANCA, and analysis of complement-regulatory factors are helpful in the diagnosis of specific subtypes. Although milder forms of glomerular injury may subside, the renal involvement is usually progressive over a matter of weeks and culminates in severe oliguria. Recovery of renal function may follow early intensive plasmapheresis (plasma exchange) combined with steroids and cytotoxic agents in Goodpasture syndrome. This therapy can reverse both pulmonary hemorrhage and renal failure. Other forms of crescentic GN also respond well to steroids and cytotoxic agents. However, despite therapy, many patients eventually require chronic dialysis or transplantation, particularly if the disease is discovered at a late stage.

Nephrotic Syndrome

Certain glomerular diseases virtually always produce the nephrotic syndrome. In addition, many other forms of primary and secondary glomerulopathies discussed in this chapter may underlie the syndrome. Before the major diseases associated with nephrotic syndrome are presented, the causes and pathophysiology of this clinical complex are briefly discussed.

Pathophysiology

Nephrotic syndrome is caused by a derangement in glomerular capillary walls resulting in increased permeability to plasma proteins.

The manifestations of the syndrome include:

- **Massive proteinuria**, with the daily loss of 3.5 g or more of protein (less in children)
- **Hypoalbuminemia**, with plasma albumin levels less than 3 g/dL
- **Generalized edema**
- **Hyperlipidemia and lipiduria**

The various components of nephrotic syndrome bear a logical relationship to one another. The glomerular capillary wall, composed of the endothelium, GBM and podocytes, acts as a size and charge barrier through which the plasma filtrate passes. Increased permeability resulting from either structural or physicochemical alterations in this barrier allows proteins to escape from the plasma into the urinary space, resulting in proteinuria.

Heavy proteinuria depletes serum albumin levels at a rate beyond the compensatory synthetic capacity of the liver, resulting in hypoalbuminemia. Increased renal catabolism of filtered albumin also contributes to hypoalbuminemia. The generalized edema is a direct consequence of *decreased intravascular colloid osmotic pressure* because of low plasma albumin. There is also *sodium and water retention*, which aggravates the edema (Chapter 4). This seems to be due to several factors, including compensatory secretion of aldosterone, mediated by the hypovolemia-enhanced renin secretion; stimulation of the sympathetic nervous system; and a reduction in the secretion of natriuretic factors such as atrial natriuretic peptides. Edema is characteristically soft and pitting, and is most marked in the periorbital regions and dependent portions of the body. If severe, hypoalbuminemia may also lead to pleural effusions and ascites.

KEY CONCEPTS

Nephritic Syndrome

- Nephritic syndrome is characterized by hematuria, oliguria with azotemia, proteinuria, and hypertension.
- The most common causes are immunologically mediated glomerular injury; lesions are characterized by proliferative changes and leukocyte infiltration.
- Acute postinfectious glomerulonephritis typically occurs after streptococcal infection in children and young adults but may occur following infection with many other organisms; it is caused by deposition of immune complexes, mainly in the subepithelial spaces, with abundant neutrophils and proliferation of glomerular cells. Most affected children recover; the prognosis is worse in adults.
- RPGN is a clinical entity with features of nephritic syndrome and rapid loss of renal function.
- RPGN (commonly called crescentic GN) is associated with severe glomerular injury with fibrinoid necrosis, GBM breaks, and subsequent proliferation of parietal epithelial cells forming crescents.
- Crescentic GN may be mediated by autoantibodies to the GBM or result from immune complex deposition. It can also occur in the absence of significant antibody deposition; most patients with this type of crescentic GN have ANCA-associated vasculitis with circulating ANCA.

The largest proportion of protein lost in the urine is albumin, but globulins are also excreted in some diseases. The ratio of low to high molecular weight proteins in the urine in various cases of nephrotic syndrome is a manifestation of the selectivity of proteinuria. A highly selective proteinuria consists mostly of low-molecular-weight proteins (albumin, transferrin), whereas a poorly selective proteinuria consists of higher-molecular-weight globulins in addition to albumin.

The genesis of hyperlipidemia is complex. Most patients with nephrotic syndrome have increased blood levels of cholesterol, triglyceride, very-low-density lipoprotein, low-density lipoprotein, lipoprotein(a), and apoprotein, and there is a decrease in high-density lipoprotein concentration in some patients. These alterations seem to be due to a combination of increased synthesis of lipoproteins in the liver, abnormal transport of circulating lipid particles, and decreased lipid catabolism. Lipiduria follows hyperlipidemia, because lipoproteins also leak across the glomerular capillary wall. The lipid appears in the urine either as free fat or as *oval fat bodies*, representing lipoprotein resorbed by tubular epithelial cells and then shed along with injured tubular cells that have detached from the basement membrane.

Nephrotic patients are particularly vulnerable to infection, especially staphylococcal and pneumococcal infections, probably due to loss of immunoglobulins in the urine. Thrombotic and thromboembolic complications are also common in nephrotic syndrome, due in part to loss of endogenous anticoagulants (e.g., antithrombin III) in the urine. *Renal vein thrombosis*, once thought to be a cause of nephrotic syndrome, is most often a consequence of this hypercoagulable state, particularly in patients with membranous nephropathy (see later).

Pathogenesis

The incidences of the several causes of nephrotic syndrome vary according to age and geography. In children younger than 17 years in North America, for example, nephrotic syndrome is almost always caused by a lesion primary to the kidney; among adults, in contrast, it is often associated with a systemic disease. Table 20.7 represents a composite derived from several studies of the causes of nephrotic syndrome

and is therefore only approximate. The most frequent *systemic causes* of nephrotic syndrome are diabetes, amyloidosis, and SLE. The most important of the *primary glomerular lesions* are minimal change disease, membranous nephropathy, and primary/idiopathic FSGS. The first is most common in children in North America, the second is most common in older adults, and primary FSGS occurs at all ages. These three lesions are discussed individually in the following sections. Other less common causes of nephrotic syndrome include the various proliferative glomerulonephritides such as membranoproliferative glomerulonephritis (MPGN) and IgA nephropathy.

Membranous Nephropathy

Membranous nephropathy is characterized by diffuse thickening of the glomerular capillary wall due to the accumulation of antibody-containing deposits along the subepithelial side of the basement membrane. Approximately 75% of cases of membranous nephropathy are considered primary. The remaining cases usually occur in association with other systemic diseases, and hence are referred to as secondary membranous nephropathy. The most notable of these associations are:

- *Drugs* (penicillamine, captopril, gold, nonsteroidal antiinflammatory drugs [NSAIDs]). From 1% to 7% of patients with rheumatoid arthritis treated with penicillamine or gold (drugs now used infrequently for this disease) develop membranous nephropathy.
- *Underlying malignant tumors*, particularly carcinomas of the lung and colon, and melanoma. According to some studies, these are present in as many as 5% to 10% of adults with membranous nephropathy.
- *SLE*. About 10% to 15% of glomerulonephritis in SLE is of the membranous type.
- *Infections* (chronic hepatitis B, hepatitis C, syphilis, schistosomiasis, malaria).
- *Other autoimmune disorders* such as thyroiditis can be associated with secondary membranous nephropathy.

Pathogenesis

Membranous nephropathy (MN) is a form of chronic immune complex-mediated disease, in which the inciting antigens can sometimes be identified in the immune complexes. The antigens may be endogenous or exogenous. The endogenous antigens may be renal or nonrenal (described later). Exogenous antigens include those derived from hepatitis B virus and *Treponema pallidum* in patients infected with these microbes.

MN is *caused in most cases by antibodies to a renal autoantigen*. A major advance came from the identification of the M-type phospholipase A₂ receptor (PLA₂R) as the antigen that underlies ~60% of human MN. Autoantibody binding to PLA₂R, a membrane protein present on the basal surface of glomerular epithelial cells, is followed by complement activation and then shedding of the immune aggregates from the cell surface to form characteristic deposits of immune complexes along the *subepithelial aspect* of the basement membrane (see Fig. 20.5C). Less often, the target antigens are neural epidermal growth factor like 1 (NELL1), thrombospondin type-1 domain-containing 7A (THSD7A), exostocin 1/2 (EXT1/2), neural cell adhesion molecule 1 (NCAM1) or neutral endopeptidase (CD10), a membrane protein recognized by placentally transferred maternal antibodies in cases of neonatal membranous nephropathy. The identification of the most common target antigens is of diagnostic and prognostic value, as associations with systemic autoimmune disease (exostocin) and malignancy (e.g., NELL1) have been described.

How does the glomerular capillary wall become leaky in membranous nephropathy? There is a paucity of neutrophils, monocytes, or platelets in glomeruli. The C5b-9 membrane attack complex of complement may be detected in lesions, but its role in pathogenesis is unclear.

TABLE 20.7 Causes of Nephrotic Syndrome

Causes	Approximate Prevalence (%) ^a	
	Children	Adults
Primary Glomerular Diseases		
Membranous nephropathy	3	30
Minimal change disease	75	8
Focal segmental glomerulosclerosis	10	35
Membranoproliferative glomerulonephritis and dense deposit disease ^b	10	10
Other proliferative glomerulonephritides (focal, "pure mesangial," IgA nephropathy) ^b	2	17
Systemic Diseases		
Diabetes mellitus		
Amyloidosis		
Systemic lupus erythematosus		
Drugs (nonsteroidal antiinflammatory, penicillamine, heroin)		
Infections (malaria, syphilis, hepatitis B and C, HIV)		
Malignant disease (carcinoma, lymphoma)		
Miscellaneous (bee-sting allergy, hereditary nephritis)		

^aApproximate prevalence of primary disease = 95% of nephrotic syndrome in children, 60% in adults. Approximate prevalence of systemic disease = 5% in children, 40% in adults.

^bMembranoproliferative and other proliferative glomerulonephritides may result in mixed nephrotic/nephritic syndromes.

A subclass of IgG, IgG4, which differs from other IgG subclasses in being a poor activator of the classical complement pathway, is the principal immunoglobulin deposited in many cases of MN, PLA₂R-associated MN in particular. How IgG4 may be associated with complement activation is not clear.

MORPHOLOGY

By light microscopy, the glomeruli either appear normal in the early stages of the disease or exhibit **uniform, diffuse thickening of the glomerular capillary wall** (Fig. 20.10A). By electron microscopy, the thickening is seen to be caused by irregular electron-dense deposits containing immune complexes between the basement membrane and the overlying epithelial cells, with effacement of podocyte foot processes (Fig. 20.10B and D). Basement membrane material is laid down between these deposits by the podocytes, appearing as irregular spikes protruding from the GBM. These spikes are best seen by silver stains, which color the basement membrane, but not the deposits, black. In time, these spikes thicken to produce domelike protrusions after the immune complex formation subsides and eventually close over the immune deposits, burying them within a markedly thickened, irregular membrane. Immunofluorescence microscopy demonstrates that the granular deposits contain both immunoglobulins and complement (Fig. 20.10C). Immunostains reveal presence of the relevant target antigens in the deposits in most patients. As the disease advances, segmental sclerosis may occur and glomeruli may become totally sclerosed. The epithelial cells of the proximal tubules contain protein reabsorption droplets, and there may be considerable interstitial mononuclear cell inflammation.

Clinical Features

This disorder usually presents with the insidious onset of nephrotic syndrome or, in 15% of patients, with sub-nephrotic proteinuria. Hematuria and mild hypertension are present in 15% to 35% of cases. It is necessary in any patient to first rule out secondary causes of nephrotic syndrome, because treatment of the underlying condition (malignant neoplasm, infection, or SLE) or discontinuance of the offending drug can reverse or ameliorate the injury.

The course of the disease is variable but generally indolent. In contrast to minimal change disease, described later, the proteinuria is nonselective and usually does not respond well to corticosteroid therapy. Complete or partial remissions may occur in up to 40% of patients, even in some patients without therapy. Progression is associated with increasing sclerosis of glomeruli, rising serum creatinine reflecting renal insufficiency, and development of hypertension. Although proteinuria persists in more than 60% of patients, only about 10% die or progress to renal failure within 10 years, and no more than 40% eventually develop severe chronic kidney disease or ESKD. The disease recurs in up to 40% of patients who undergo transplantation for ESKD. Spontaneous remissions and a relatively benign outcome occur more commonly in females and in those with proteinuria in the subnephrotic range.

Because of the variable course of the disease, it has been difficult to evaluate the overall effectiveness of corticosteroids or other immunosuppressive therapy in controlling the proteinuria or progression. B-cell targeted therapeutics appear to be the most effective form of therapy to date. It is thought that serologic typing of circulating antibodies to the various antigens as biomarkers of disease activity may aid in the diagnosis, prognosis, and management of membranous nephropathy.

Minimal Change Disease (Diffuse Podocytopathy With Minimal Changes)

This disorder is characterized by **diffuse effacement of foot processes of podocytes, detectable by electron microscopy, in glomeruli that appear normal by light microscopy**. It is the most frequent cause of

nephrotic syndrome in children, but it is less common in adults (see Table 20.7). The peak incidence is between 2 and 6 years of age. The disease sometimes follows a respiratory infection or routine prophylactic immunization.

Pathogenesis

Historically, this disease was characterized by absence of immune deposits in the glomerulus, which led to dismissal of antibody-mediated causes. However, recent evidence points to an autoantibody-mediated pathogenesis. The current hypothesis is that a subset of minimal change disease is caused by circulating antibodies directed against nephrin, a transmembrane podocyte protein and essential slit diaphragm component, which cause podocyte injury through interference with nephrin crosslinking. Proteinuria ensues due to the rapid breakdown of the podocyte slit diaphragm accompanied by diffuse podocyte foot process effacement, dramatically altering the permeability and selectivity of the glomerular filter. The ultrastructural changes support a primary *podocyte injury* (podocytopathy), and studies in animal models also suggest the loss of glomerular polyanions. Thus, defects in the charge barrier may contribute to the proteinuria.

MORPHOLOGY

The glomeruli are normal by light microscopy (Fig. 20.11A). By electron microscopy the GBM appears normal, and no electron-dense deposits are seen. **The principal lesion is in the podocytes, which show a uniform, diffuse effacement of foot processes**, along with loss of intervening slit diaphragms (Fig. 20.11B). This change, often incorrectly termed “fusion” of foot processes, actually represents simplification of the epithelial cell architecture with flattening, retraction, and swelling of foot processes, which is the result of a collapse of their central actin cytoskeleton. Foot process effacement is also present in other proteinuric states (e.g., membranous glomerulopathy, diabetic nephropathy); however, it usually occurs at later stages of the disease and in the context of significant architectural distortion of the glomerulus. It is only when effacement is associated with normal glomeruli by light microscopy that the diagnosis of minimal change disease can be made. In most patients, these podocyte changes are completely reversible after corticosteroid therapy, concomitant with remission of the proteinuria and disappearance of circulating anti-nephrin antibodies. The cells of the proximal tubules are often laden with lipid and protein, reflecting tubular reabsorption of lipoproteins passing through diseased glomeruli (hence the historical name **lipoid nephrosis** for this disease). Immunofluorescence studies can be negative, but often show subtle diffuse, global fine granular staining for IgG over the podocytes, in the absence of complement reactivity (Fig. 20.11C). IgG staining typically colocalizes with clustered nephrin and represents binding of anti-nephrin antibodies in the tissue.

Clinical Features

Despite massive proteinuria, renal function most often remains stable, and there is typically no hypertension or hematuria. The proteinuria usually is highly selective, most of the protein being albumin. **A characteristic feature is a dramatic response to corticosteroid therapy**. Most children (>90%) with minimal change disease respond rapidly to this treatment. However, proteinuria recurs in a significant proportion of patients, and the disease sometimes becomes steroid-dependent or -resistant, and/or progresses to FSGS (discussed later). Nevertheless, the long-term prognosis for patients is excellent, and even steroid-dependent disease may resolve when children reach puberty. Although adults are slower to respond, their long-term prognosis is also excellent. Steroid-dependent and frequently relapsing disease has been reported to respond well to B-cell depletion therapy.

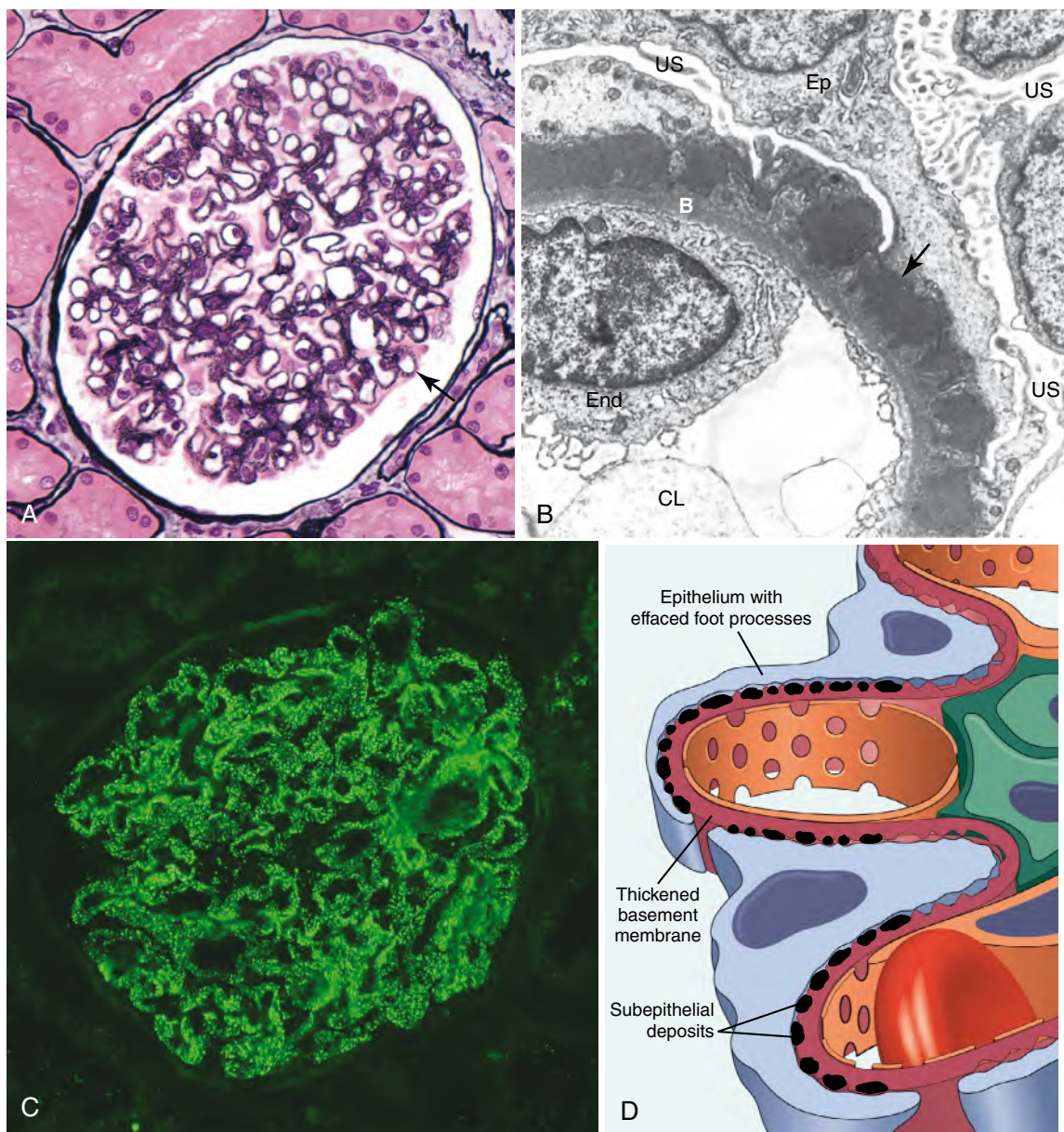


Fig. 20.10 Membranous nephropathy. (A) Silver methenamine stain. Note the marked diffuse thickening of the capillary walls without an increase in the number of cells. There are prominent “spikes” of silver-staining matrix (*arrow*) projecting from the basement membrane lamina densa toward the urinary space, which separate and surround deposited immune complexes that lack affinity for the silver stain. (B) Electron micrograph showing electron-dense deposits (*arrow*) along the epithelial side of the basement membrane (*B*). Note the effacement of foot processes overlying deposits. (C) Characteristic granular immunofluorescent deposits of IgG along the glomerular basement membrane. (D) Diagrammatic representation of membranous nephropathy. *CL*, Capillary lumen; *End*, endothelium; *Ep*, epithelium; *US*, urinary space. (A, Courtesy Dr. Charles Lassman, UCLA School of Medicine, Los Angeles.)

Minimal change disease in adults can be associated with Hodgkin lymphoma and, less frequently, other lymphomas and leukemias. In addition, minimal change disease may follow vaccination and infection, and has also been described associated with NSAID therapy.

Focal Segmental Glomerulosclerosis

FSGS (both primary and secondary forms) is the most common cause of nephrotic syndrome in adults in the United States, particularly in individuals of Hispanic and African descent. The disease is

characterized by a primary and diffuse podocytopathy and therefore can be considered in the same spectrum as minimal change disease. In contrast to minimal change disease, primary FSGS, as the name implies, is characterized by sclerosis of only a portion of the capillary tuft of only a subset of glomeruli. Primary FSGS typically manifests clinically by the acute or subacute onset of nephrotic syndrome, but in later stages may be associated with sub-nephrotic proteinuria. Hypertension, microscopic hematuria, and some degree of azotemia are commonly present when the disease is first recognized clinically.

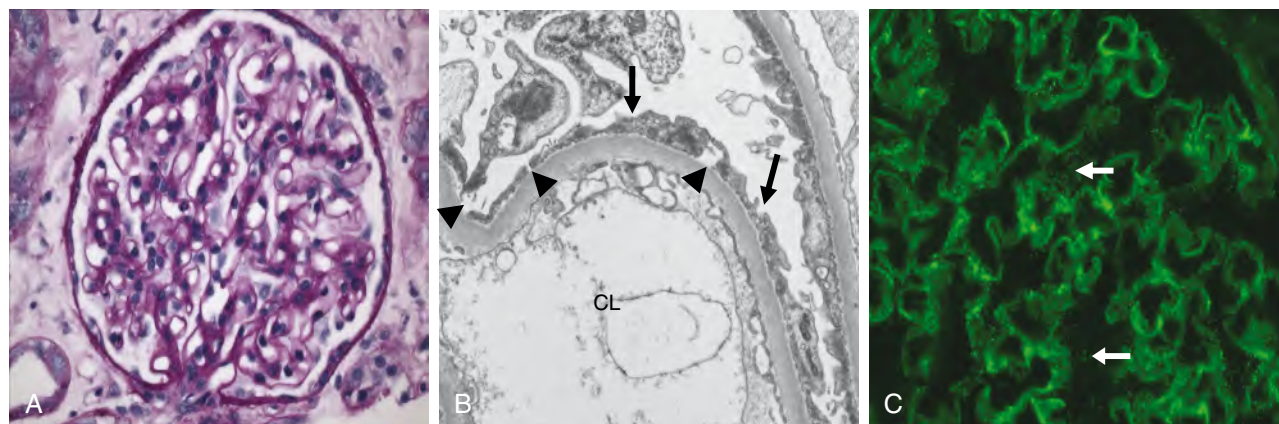


Fig. 20.11 Minimal change disease (MCD). (A) Glomerulus stained with periodic acid–Schiff (PAS) stain. Note normal basement membranes and absence of proliferation. (B) Ultrastructural characteristics of MCD, showing diffuse podocyte foot process effacement (arrows), absence of electron-dense deposits, and slit diaphragm loss (arrowheads). CL, Capillary lumen. (C) Glomerulus, IgG immunofluorescence. There is subtle fine punctate staining of podocytes (arrows).

Classification and Types. FSGS is a result of podocyte loss in the following settings, each of which is discussed in more detail later:

- **Primary or idiopathic:** A primary diffuse podocytopathy
- **Viral infections and other toxic injuries:** Podocyte loss in the setting of viral infection (such as HIV, SARS-CoV-2), heroin addiction (heroin nephropathy), sickle cell disease, or other forms of direct toxic podocyte injury
- **Adaptive changes:** Podocyte damage due to the adaptive response to high body mass index, loss of renal tissue (after nephrectomy, unilateral renal agenesis, renal dysplasia, or reflux nephropathy), or in advanced stages of any other renal disorder with significant nephron loss (most commonly vascular disease)
- **Inherited forms** of FSGS, which are uncommon, are associated with mutations in slit diaphragm- and actin cytoskeleton-associated proteins
- **High-risk APOL1 alleles:** Following podocyte loss in patients carrying certain APOL1 alleles
- **Secondary:** As a secondary event, reflecting scarring and nephron dropout in cases of focal glomerulonephritis (e.g., IgA nephropathy, ANCA-associated vasculitis)

So-called *idiopathic FSGS*, a form of primary podocytopathy, accounts for 10% and 35% of cases of nephrotic syndrome in children and adults, respectively. The clinical signs differ from those of minimal change disease in the following respects: (1) there is a higher incidence of hematuria, reduced GFR, and hypertension; (2) proteinuria is more often nonselective; (3) response to corticosteroid therapy is poor; and (4) there is progression to chronic kidney disease, with at least 50% of patients developing ESKD within 10 years.

Pathogenesis of Primary FSGS

Primary FSGS is marked by the same characteristic diffuse podocyte injury that is seen in minimal change disease (diffuse podocytopathy). However, in primary FSGS, the podocytopathy leads to a degree of podocyte loss that exceeds the ability of the surviving podocytes to compensate. The initiating mechanism likely involves undefined circulating factors and genetically determined defects affecting components of the slit diaphragm. Once areas of the tuft are devoid of covering podocytes, hyalinosis and sclerosis ensue, stemming from entrapment of high molecular weight plasma proteins in hyperpermeable foci, capillary collapse, and increased ECM deposition. The recurrence of

massive proteinuria after transplantation, sometimes within 24 hours, with subsequent progression to overt lesions of FSGS, recapitulating the disease in the native kidneys, points to persistence of the causative circulating factor in some patients.

The discovery of a genetic basis for some cases of FSGS and other causes of the nephrotic syndrome has improved the understanding of the pathogenesis of proteinuria in the nephrotic syndrome and has provided new methods for diagnosis and prognosis. Inherited forms of FSGS may be associated with:

- The first relevant gene identified, *NPHS1*, encodes the protein nephrin, a key component of the slit diaphragm (see Fig. 20.4). Several mutations of the *NPHS* gene have been identified that give rise to *congenital nephrotic syndrome of the Finnish type*, producing a minimal change disease–like glomerulopathy with extensive foot process effacement.
- A distinctive pattern of autosomal recessive FSGS results from mutations in the *NPHS2* gene, which encodes the intracellular slit diaphragm associated protein podocin. Mutations in *NPHS2* are one cause of *steroid-resistant nephrotic syndrome of childhood onset*. Some implicated podocin mutations are relatively common variants and show incomplete penetrance; it is suspected that these variants may increase susceptibility to forms of FSGS that are triggered by acquired factors.
- Mutations in the gene encoding the podocyte actin-binding proteins α -actinin 4 (*ACTN4*) and inverted formin-2 (*INF2*) underlie some cases of *autosomal dominant FSGS*, which can be insidious in onset but have a high rate of progression to renal insufficiency.
- Mutations in the gene encoding TRPC6 have been found in some families with adult-onset FSGS. This protein is widely expressed, including in podocytes, and the pathogenic mutations may perturb podocyte function by increasing calcium flux in these cells.

What the affected proteins have in common is their localization to the slit diaphragm and to adjacent podocyte cytoskeletal structures. Their specific functions and interactions are incompletely understood, but they all contribute to the integrity of the glomerular filter.

FSGS Associated With High-Risk Alleles of APOL1. In 2010, it was discovered that two sequence variants (G1/G2 risk alleles) in the apolipoprotein L1 gene (*APOL1*) are strongly associated with an increased risk of chronic kidney disease and renal failure in individuals of African

descent, although the mechanisms underlying this association are not yet completely understood. These sequence variants are remarkable because they appear to convey resistance to trypanosome infection, creating a selective pressure that explains their high prevalence in individuals of African descent. One recent line of evidence suggests a pathogenic role of increased APOL1 expression in podocyte cell death.

FSGS in the Setting of Adaptive Changes. Adaptive forms of FSGS occur as a complication of glomerular and nonglomerular diseases leading to structural or functional reduction in renal tissue. Particularly striking examples in which this occurs are reflux nephropathy and unilateral renal agenesis. These may lead to progressive glomerulosclerosis and renal failure. The glomerulosclerosis seems to be initiated by the adaptive change that occurs in the relatively unaffected glomeruli of diseased kidneys. Such a mechanism is suggested by experiments in rats subjected to subtotal nephrectomy. *Compensatory hypertrophy* of the remaining glomeruli initially maintains renal function in these animals, but proteinuria and segmental glomerulosclerosis soon develop, eventually leading to diffuse global glomerulosclerosis and uremia. The glomerular hypertrophy is associated with *hemodynamic changes*, including increases in glomerular blood flow, filtration, and transcapillary pressure (glomerular hypertension), and often with systemic hypertension. A similar mechanism underlies the development of hypertrophy and FSGS in obese individuals with relative insufficiency of nephron mass compared to body mass (obesity-related FSGS).

The sequence of events (Fig. 20.12) that is thought to lead to sclerosis in this setting entails endothelial and podocyte injury; podocyte loss leading to segments of GBM denuded of overlying foot processes and consequent increased glomerular permeability to proteins similar to the process seen in primary FSGS; and accumulation of proteins in the mesangial matrix. This is followed by proliferation of mesangial cells,

infiltration by macrophages, increased accumulation of ECM, and segmental and eventually global sclerosis of glomeruli. With increasing reduction in nephron mass and ongoing compensatory changes, a vicious cycle of continuing glomerulosclerosis sets in. Other than the elimination of identifiable causes of podocyte injury, the most successful interventions to interrupt these mechanisms of progressive glomerulosclerosis involve treatment with inhibitors of the renin-angiotensin system, which not only reduce intraglomerular hypertension, but also have direct effects on each of the mechanisms identified earlier. Importantly, these agents have been shown to ameliorate progression of sclerosis in both animal and human studies.

FSGS in HIV-Associated and COVID-19-Associated Nephropathy.

HIV infection can directly or indirectly cause several renal complications, including acute renal failure or acute interstitial nephritis induced by drugs or infection, thrombotic microangiopathies, postinfectious glomerulonephritis, and, most commonly, a severe form of the collapsing variant of FSGS, termed *HIV-associated nephropathy*. A similar form of collapsing glomerulopathy has also been reported in patients with COVID-19. Both diseases are seen more frequently in people of African ancestry than in individuals of European descent, likely due to the higher prevalence of high-risk APOL1 alleles in those of African descent. With the advent of antiretroviral therapy for HIV infection and widespread vaccination for COVID-19, the incidence of these disorders has fallen sharply.

MORPHOLOGY

By light microscopy, the **focal and segmental lesions may involve only a minority of the glomeruli** (Fig. 20.13A) and may be missed if the biopsy specimen contains a small number of glomeruli. In the sclerotic segments, there is collapse of capillary loops, increase in matrix deposition, and segmental entrapment of plasma proteins along the capillary wall (hyalinosis), which may become so pronounced as to occlude capillary lumens. Lipid droplets and foamy macrophages are often present (Fig. 20.13B). Glomeruli that do not show segmental lesions usually appear normal by light microscopy but may show increased mesangial matrix. By electron microscopy, both sclerotic and nonsclerotic areas show **effacement of foot processes, which is diffuse in primary forms of FSGS during acute nephrotic syndrome**, and there may also be focal detachment of the epithelial cells and denudation of the underlying GBM. By immunofluorescence microscopy, IgM and C3 may be present in the sclerotic areas and/or in the mesangium. In addition to the focal sclerosis, there may be pronounced hyalinosis and thickening of afferent arterioles, particularly in those forms associated with primary vascular injury. With progression of the disease, increased numbers of glomeruli become involved, and sclerosis spreads within each glomerulus as more podocytes are lost. In time, this leads to total (i.e., global) sclerosis of glomeruli, with pronounced tubular atrophy and interstitial fibrosis.

A morphologic variant of FSGS, called **collapsing glomerulopathy**, is characterized by retraction and/or collapse of the entire glomerular tuft, with or without additional FSGS lesions of the type described earlier (Fig. 20.14). This variant of FSGS is the result of accelerated and abundant podocyte loss, followed by capillary collapse and activation and proliferation of parietal epithelial cells, which will eventually migrate onto the denuded segment(s) of the tuft. Collapsing glomerulopathy can be seen in patients with high-risk APOL1 alleles, in those with vaso-occlusive disease and renal infarction, as a feature of drug-induced podocyte injury (e.g., pamidronate), and as the most characteristic lesion of HIV-associated nephropathy (HIVAN) and COVID-19-associated nephropathy (COVAN). Collapsing glomerulopathy is typically associated with prominent tubular injury with formation of microcysts containing stagnant

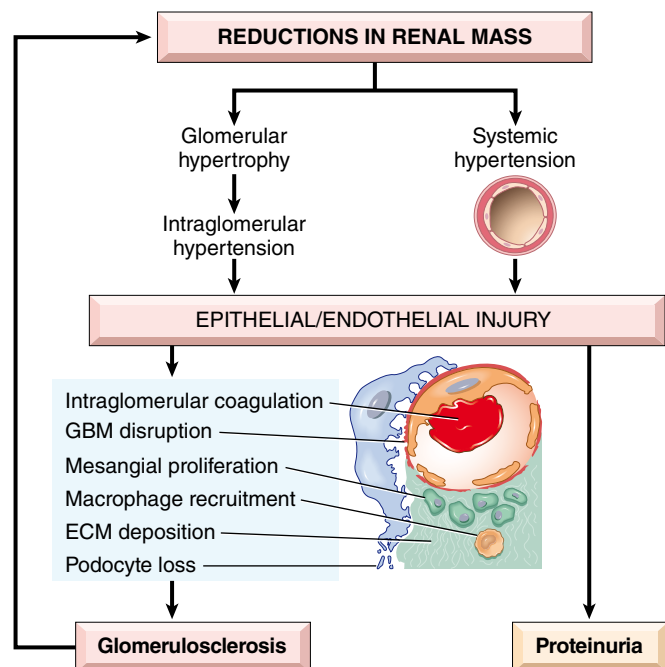


Fig. 20.12 Focal segmental glomerulosclerosis associated with loss of renal mass. The adaptive changes in glomeruli (hypertrophy and glomerular capillary hypertension), as well as systemic hypertension, cause epithelial and endothelial injury that culminate in progressive glomerulosclerosis (see text). *ECM*, Extracellular matrix; *GBM*, glomerular basement membrane.

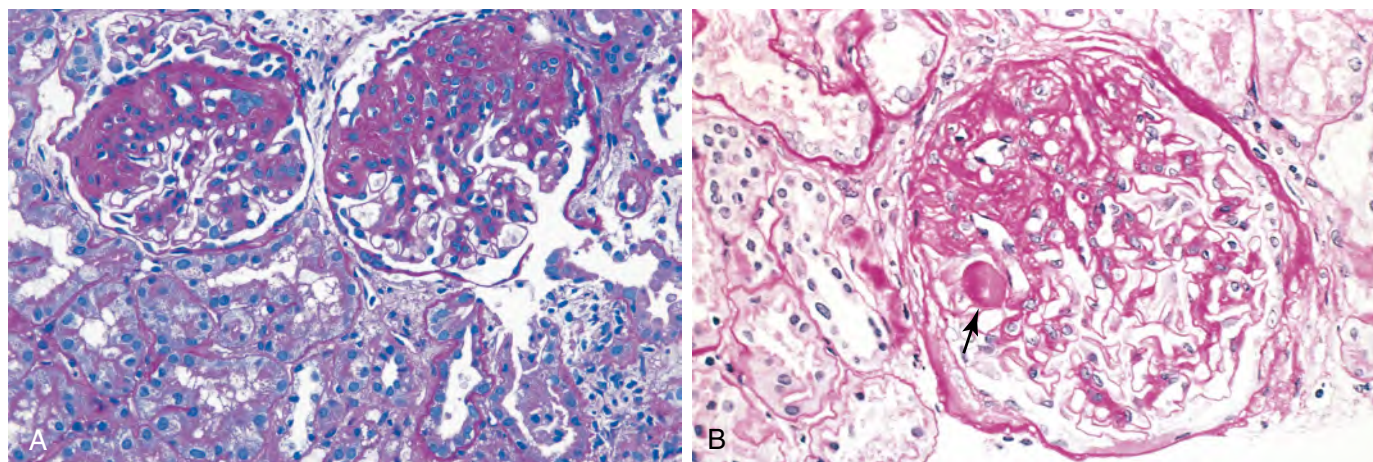


Fig. 20.13 Focal segmental glomerulosclerosis, periodic acid-Schiff stain. (A) Segmental sclerosis involves the upper half of both glomeruli. (B) High-power view showing hyaline insudation (arrow) and lipid (small vacuoles) in sclerotic area.

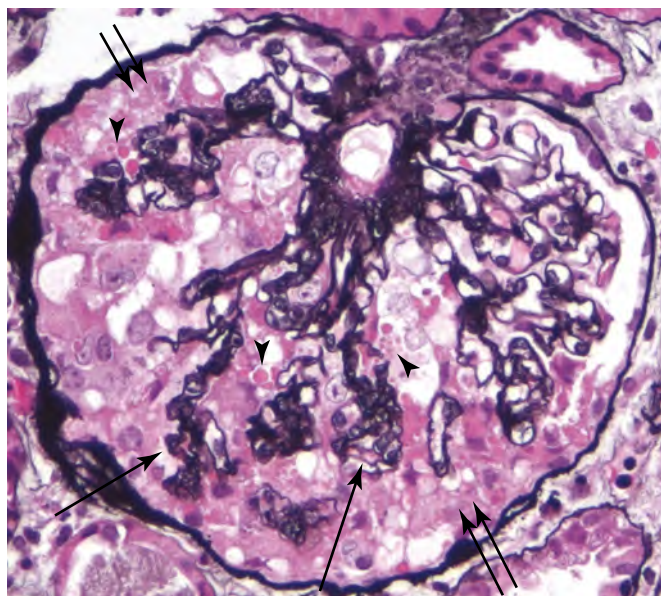


Fig. 20.14 Collapsing glomerulopathy. Visible are retraction of the glomerular tuft (arrows), narrowing of capillary lumens, proliferation and swelling of visceral epithelial cells (double arrows), and prominent accumulation of intracellular protein absorption droplets in the visceral epithelial cells (arrowheads). Silver methenamine stain. (Courtesy Dr. Jolanta Kowalewska, Cedars-Sinai Medical Center, Los Angeles.)

protein casts and prominent interstitial inflammation. It has a particularly poor prognosis.

Other features of HIVAN and COVAN include striking focal cystic dilation of tubule segments, which are filled with proteinaceous material, often accompanied by tubulointerstitial inflammation and fibrosis, and tubuloreticular inclusions within endothelial cells detected by electron microscopy. Such inclusions have been shown to be modifications of endoplasmic reticulum induced by high levels of type I interferons, which are seen in these disorders. They are not usually present in idiopathic FSGS and therefore may have diagnostic value.

Clinical Features

Spontaneous remission is rare in idiopathic FSGS, and responses to corticosteroid therapy are variable. In general, children have a better prognosis than adults. Progression to renal failure occurs at variable rates. About 20% of patients follow an unusually rapid course, with intractable massive proteinuria ending in renal failure within 2 years. Factors associated with rapid progression include the degree of proteinuria, the degree of renal insufficiency at diagnosis, and histologic subtype (the collapsing variant has an unfavorable course). Recurrences are seen in 25% to 50% of patients receiving kidney transplants.

Membranoproliferative Glomerulonephritis

MPGN is best considered a pattern of glomerular injury rather than a specific disease. It is caused by deposition of immune complexes containing IgG and complement (previously known as type I MPGN), or overactivation of complement without immune complex deposition (now including C3-dominant glomerulopathies, previously called type II MPGN).

The MPGN pattern is characterized histologically by remodeling and duplication of the GBM and proliferation of glomerular cells (with or without leukocyte infiltration), in the presence of deposits along glomerular capillary walls and in the mesangium. Mesangial proliferation and accumulation of matrix is another feature frequently present; therefore, an older term for this pattern of injury is *mesangiocapillary* (or *mesangioproliferative*) glomerulonephritis.

The MPGN pattern presents as nephrotic syndrome in 10% of cases in both children and young adults. Some patients present only with hematuria or proteinuria in the subnephrotic range, but many others have a combined nephrotic-nephritic picture. MPGN is increasingly recognized to be associated with systemic disorders and various known etiologic agents; therefore, careful and thorough clinical work-up of these patients is required.

Pathogenesis

Immune complex deposition in the glomerulus and activation of both classical and alternative complement pathways are frequent findings. The antigens involved in so-called “idiopathic” MPGN are unknown. In secondary forms of MPGN, described later, they are

believed to be proteins derived from infectious agents such as hepatitis C and B viruses.

MORPHOLOGY

The **glomeruli are large and hypercellular** due to proliferation of cells in the mesangium and the presence of infiltrating leukocytes. The glomeruli have an accentuated “lobular” appearance due to the **proliferating mesangial cells, increased mesangial matrix and inflammatory leukocytes in capillary lumens** (Fig. 20.15). The GBM is thickened, and often shows a “**double-contour**” or “**tram-track**” appearance, especially evident on silver or PAS stains. This is caused by “**duplication**” of the basement membrane (also commonly referred to as splitting), usually as the result of new basement membrane synthesis in response to subendothelial deposits of immune complexes. Between the duplicated basement membranes, there is inclusion or interposition of cellular elements, which can be of mesangial, endothelial, or leukocytic origin. Such interposition also gives rise to the appearance of “split” basement membranes (Fig. 20.16A). Crescents are present in many cases.

Immune complex-mediated (formerly type I) **MPGN** is characterized by the presence of discrete **subendothelial electron-dense deposits**. Mesangial and occasional subepithelial deposits may also be present (Fig. 20.16A). By immunofluorescence, IgG and C3 are most commonly deposited in a granular, or less commonly, in a confluent pattern, and early complement components (C1q and C4) are often also present, indicative of an immune complex-mediated pathogenesis.

Clinical Features

Most patients with immune complex-mediated MPGN present in adolescence or as young adults with nephrotic syndrome and a nephritic component manifested by hematuria or, more insidiously, as mild proteinuria. Few remissions occur spontaneously in either type, and the disease follows a slowly progressive but unremitting course. Some patients develop numerous crescents and a clinical picture of RPGN. About 50% develop chronic renal failure within 10 years. Treatments with steroids, immunosuppressive agents, and antiplatelet drugs have not proven to be of any benefit and current therapy generally focuses on the underlying disorder.

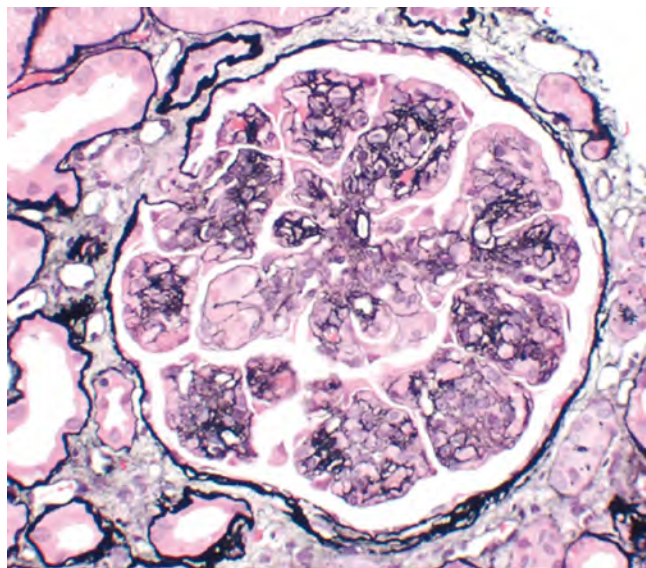


Fig. 20.15 Membranoproliferative glomerulonephritis, showing mesangial cell proliferation, increased mesangial matrix (staining black with silver stain), basement membrane thickening with segmental splitting, accentuation of lobular architecture, swelling of cells lining peripheral capillaries, and influx of leukocytes (endocapillary hypercellularity).

Secondary Membranoproliferative Glomerulonephritis. Secondary MPGN (invariably immune complex-mediated) is more common in adults and arises in the following settings:

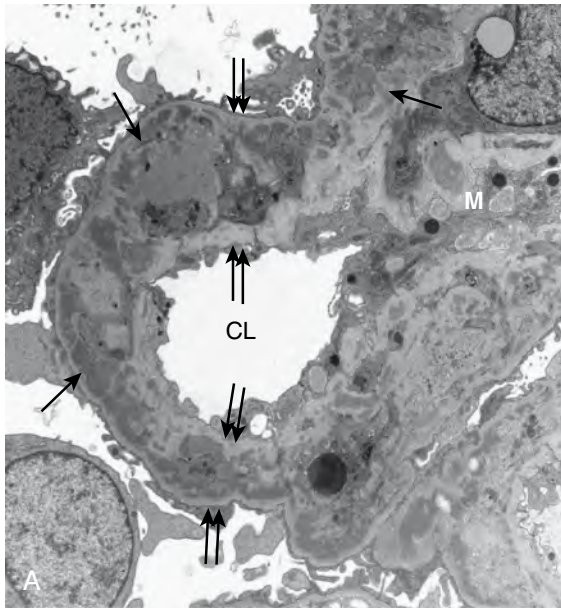
- Chronic immune complex disorders, such as SLE; hepatitis B infection; hepatitis C infection, usually with cryoglobulinemia; endocarditis; infected ventriculoatrial shunts; chronic visceral abscesses; HIV infection; and schistosomiasis. In viral infections, the glomerular deposits are believed to contain viral proteins, which presumably behave either as “planted” antigens after first binding to or becoming trapped within glomerular structures, or are contained in preformed immune complexes deposited from the circulation.
- Malignant diseases complicated by development of autoantibodies or by monoclonal immunoglobulins, typically in the setting of plasma cell neoplasms or related disorders (Chapter 13)

C3-Dominant Glomerulopathy

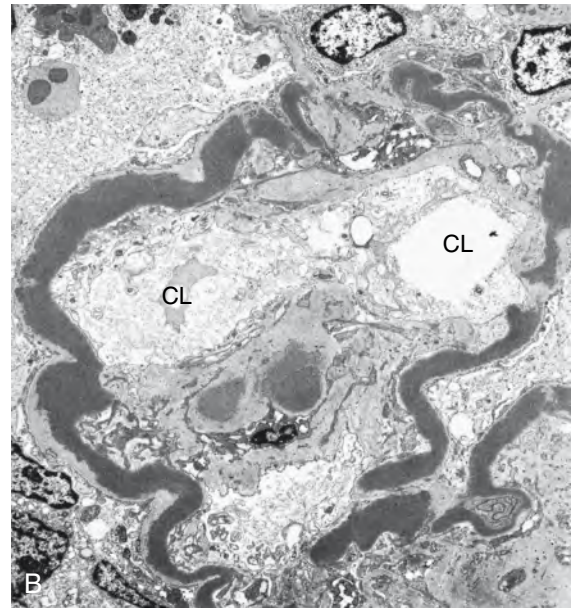
This entity includes dense deposit disease (formerly called type II MPGN) and C3 glomerulonephritis (C3GN), and is the result of genetic or acquired abnormalities of the alternative complement pathway, leading to its excessive activation and deposition. The disease is consistently associated with decreased serum C3 accompanied by diminished levels of Factor B and properdin, both components of the alternative complement pathway. By contrast, serum levels of the early classical complement pathway components C1 and C4 are normal. The glomeruli contain deposits of C3 and properdin but not IgG. Recall that in the alternative complement pathway, C3 is directly cleaved to C3b (Fig. 20.17; see also Fig. 3.11). This reaction depends on the activation of C3 by such substances as bacterial polysaccharides, endotoxin, and aggregates of IgA via a pathway involving Factors B and D, and leads to the generation of C3bBb, the alternative pathway C3 convertase. Normally, C3bBb is labile, but most patients with acquired C3GN have a circulating autoantibody termed *C3 nephritic factor* (C3NeF) that binds C3bBb and protects it from inactivation (see Fig. 20.17), resulting in persistent C3 activation and hypocomplementemia. There is also decreased C3 synthesis by the liver, further contributing to the profound hypocomplementemia. A recent study has shown that apolipoprotein E is enriched in dense deposits and serves as a biomarker of dense deposit disease. Mutations in components of the alternate pathway such as Factor H have also been associated with dense deposit disease and C3GN.

MORPHOLOGY

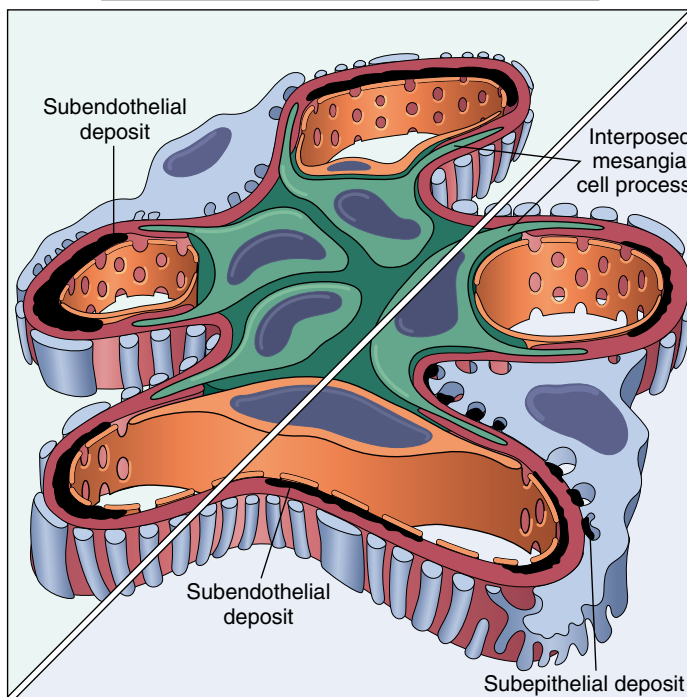
While some cases of C3-dominant glomerulopathies share histologic features with MPGN, they can show a wider spectrum of histologic alterations. Many cases have a predominately mesangial proliferative pattern of injury, while others have an inflammatory and focally crescentic appearance. In some cases, dense deposits of a cellular material can be seen permeating the GBMs in histologic sections. The defining feature of **dense deposit disease** is revealed by electron microscopy, which demonstrates permeation of the lamina densa of the GBM by a ribbonlike, homogeneous, extremely electron-dense material of unknown composition (see Fig. 20.16B). In contrast, **C3GN** is characterized ultrastructurally by ill-defined, cloudy mesangial, intramembranous and subendothelial deposits. Both can show subepithelial humps. By immunofluorescence, C3 is present in glomeruli in all forms of C3GN, typically two to three times the intensity of any other routine reactants, and in irregular granular or linear foci in the basement membranes on either side but not within the dense deposits in dense deposit disease. IgG is usually absent, as are components of the classical pathway of complement activation (such as C1q and C4). As aberrant C3 activation can occur in the setting of circulating paraproteins, it is prudent to rule out masked monoclonal immunoglobulin deposits.



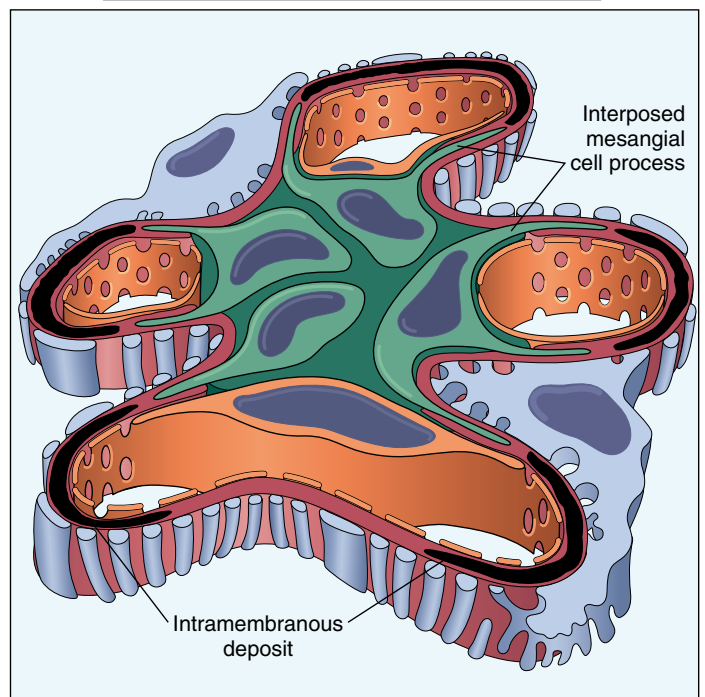
IMMUNE COMPLEX-MEDIATED MPGN



DENSE DEPOSIT DISEASE



C



D

Fig. 20.16 Membranoproliferative glomerulonephritis. (A) Immune complex-mediated membranoproliferative glomerulonephritis (formerly type I MPGN). Note discrete electron-dense deposits (*arrows*) incorporated into the glomerular capillary wall between duplicated (split) basement membranes (*double arrows*), and in mesangial regions (*M*); (B) Complement-mediated MPGN in dense deposit disease (formerly type II MPGN). There are dense homogeneous deposits within the basement membrane. In both, mesangial interposition gives the appearance of split basement membranes when viewed in the light microscope. (C–D) Schematic representation of patterns in the two types of membranoproliferative glomerulonephritis. (C) In the immune complex-mediated form, subendothelial deposits with or without rare subepithelial deposits are present. (D) C3 glomerulopathies, including dense deposit disease, are often characterized by intramembranous dense or ill-defined deposits. *CL*, Capillary lumen. (A, Courtesy Dr. Jolanta Kowalewska, Cedars-Sinai Medical Center, Los Angeles.)

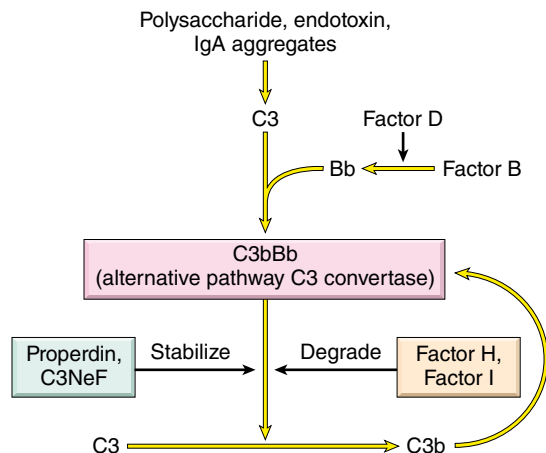


Fig. 20.17 Regulation of the alternative complement pathway. C3 nephritic factor (C3NeF), an antibody present in individuals with membranoproliferative glomerulonephritis, mimics the activity of properdin by stabilizing the alternative pathway C3 convertase, thus enhancing C3 activation.

Clinical Features

Dense deposit disease primarily affects children and young adults. The clinical presentation of nephritic syndrome with hematuria and/or nephrotic syndrome with proteinuria overlaps with that of MPGN. C3GN has broad age range, with forms mediated by genetic variants in complement-regulatory proteins presenting in children and younger adults, while acquired forms typically reflect the age distribution of the associated diseases (e.g., lymphoproliferative diseases more commonly affecting older adults). The prognosis is poor, with over one-half of patients progressing to ESRD. There is a high incidence of recurrence in transplant recipients with recurrence rates of up to 90% in dense deposit disease.

Fibrillary Glomerulonephritis

Fibrillary glomerulonephritis is a rare disease of unknown pathogenesis characterized by fibrillar deposits in the mesangium and glomerular capillary walls that differ from amyloid fibrils ultrastructurally in size and appearance, and typically do not stain with Congo red. The glomerular lesions usually show membranoproliferative and/or crescentic patterns by light microscopy. By immunofluorescence microscopy, there is deposition of polyclonal IgG, often of the IgG4 subclass, complement C3, and Igκ and Igλ light chains, but cases with monoclonal immunoglobulin deposits have been described. The deposits also contain DNAJB9, a member of the heat shock protein family; immunostaining for DNAJB9 is a sensitive and specific test for this entity. Clinically, patients can develop nephrotic syndrome, hematuria, and progressive renal insufficiency. The disease recurs in kidney transplants.

KEY CONCEPTS

Nephrotic Syndrome

- Membranous nephropathy is caused by an autoimmune response, most often directed against PLA₂R on podocytes; it is characterized by granular subepithelial deposits of antibodies with GBM thickening and loss of foot processes but little or no inflammation; the disease is often resistant to steroid therapy.
- The nephrotic syndrome is characterized by proteinuria, which results in hypoalbuminemia and edema.

- Podocyte injury is an underlying mechanism of proteinuria and may be the result of autoantibodies or other factors that disrupt the function of podocyte proteins (as in minimal change disease and FSGS) or immune complex deposition (as in membranous nephropathy).
- Minimal change disease is the most frequent cause of nephrotic syndrome in children; it is manifested by proteinuria and effacement of glomerular foot processes without antibody deposits; the pathogenesis can be mediated by autoantibodies directed against the slit diaphragm protein nephrin; the disease can be triggered by infections, immunization, drugs, and certain neoplastic lesions; the disease responds well to steroid therapy.
- FSGS may be primary (podocyte injury caused by mechanisms likely similar to minimal change disease, and/or due to genetic causes) or secondary (e.g., as a consequence of prior glomerulonephritis, hypertension, or infection such as HIV); glomeruli show focal and segmental obliteration of capillary lumens, and loss of foot processes; the disease is often resistant to therapy and may progress to ESKD.
- MPGN in most cases is the result of immune complex deposition in both mesangial regions and capillary walls. It may be associated with systemic infections, cryoglobulins, or circulating paraproteins.
- C3 glomerulopathies (including dense deposit diseases and C3GN), are defined by a unique permeation of the glomeruli by deposits of variable density. Dense deposit disease primarily affects children and young adults, and its deposits are enriched in apolipoprotein E. C3 glomerulopathies are associated with acquired or genetic dysregulation of the alternative complement pathway.
- Fibrillary glomerulonephritis is a rare disease characterized by Congo-red negative fibrillary glomerular deposits containing DNAJB9.

Other Glomerular Diseases

IgA Nephropathy

IgA nephropathy, characterized by the presence of prominent IgA deposits in the mesangial regions and recurrent hematuria, is the most common type of glomerulonephritis worldwide. The disease can be suspected by light microscopic examination, but diagnosis requires detection of glomerular IgA deposition. Mild proteinuria is usually present, and the nephrotic syndrome may occasionally develop in cases of widespread podocyte injury. Rarely, patients present with crescentic GN.

Whereas IgA nephropathy is typically an isolated renal disease, similar IgA deposits are present in a systemic disorder of children, *Henoch-Schönlein purpura*, to be discussed later, which has many overlapping features with IgA nephropathy. In addition, *secondary IgA nephropathy* may occur in patients with liver and intestinal diseases, as discussed later, as well as in patients with staphylococcal infections.

Pathogenesis

Current evidence favors a “multi-hit” etiology for this disorder involving several steps. IgA, the main Ig in mucosal secretions, is present in plasma at low concentrations, mostly in monomeric form, the polymeric forms being catabolized in the liver. In patients with IgA nephropathy, levels of plasma polymeric IgA are increased, but increased production is not sufficient to cause this disease. A clue comes from the observation that in IgA nephropathy, the glomerular deposits consist predominantly of polymeric IgA molecules with aberrant glycosylation. It is believed that a key facet of IgA nephropathy is a hereditary or acquired defect in the formation or attachment of galactose-containing sugar chains called O-linked glycans to the hinge region of the IgA molecule (particularly to those of the IgA1 subclass) prior to their secretion by B cells. This aberrantly glycosylated IgA is either

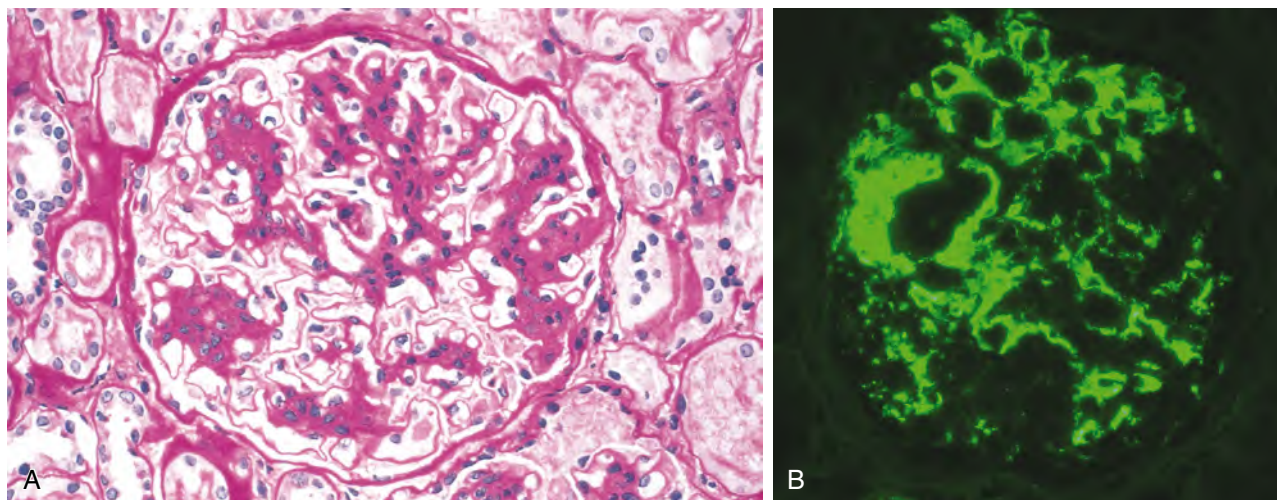


Fig. 20.18 IgA nephropathy. (A) Light microscopy showing mesangial proliferation and matrix increase. (B) Characteristic deposition of IgA, principally in mesangial regions, detected by immunofluorescence. (Courtesy Dr. Sujal Shah, Brigham and Women's Hospital, Boston.)

deposited by itself in glomeruli, or it elicits an immune response that leads to production of IgG autoantibodies directed against the abnormal IgA molecules. Binding of these antibodies to the abnormal IgA forms immune complexes that are deposited in the mesangium; alternatively, the abnormal IgA1 is deposited in the mesangium with subsequent formation of immune complexes in situ. The mesangial immune deposits activate mesangial cells, which proliferate, produce increased amounts of ECM, and secrete numerous cytokines and growth factors. These secreted mediators may not only participate in further mesangial cell activation but may also recruit inflammatory cells into the glomeruli. The recruited leukocytes contribute to glomerular injury and to a reparative response, which can include opsonization and removal of the immune complexes. The deposited IgA and IgA-containing immune complexes may activate the complement system via the alternative pathway, and hence the presence of C3 and the absence of C1q and C4 in glomeruli are typical of this disorder. A genetic influence is suggested by the occurrence of this condition in families and in HLA-identical siblings, the increased frequency of certain HLA and complement genotypes in affected individuals, and genome-wide association studies linking specific MHC class II loci to disease susceptibility.

Epidemiologic features of this disorder indicate that the increased synthesis of abnormal IgA may occur in response to respiratory or gastrointestinal exposure to environmental agents (e.g., viruses, bacteria, food proteins). The specific initiating antigens are unknown, and several infectious agents and food products have been implicated. IgA nephropathy occurs with increased frequency in individuals with *gluten enteropathy (celiac disease)*, in whom intestinal mucosal defects are seen, and in *liver disease*, in which there is defective hepatobiliary clearance of IgA complexes (*secondary IgA nephropathy*).

MORPHOLOGY

On histologic examination, the lesions vary considerably. The glomeruli may be normal or may show mesangial expansion and endocapillary proliferation (mesangioproliferative glomerulonephritis) (Fig. 20.18A), segmental proliferation confined to some glomeruli (focal proliferative glomerulonephritis), or rarely, overt crescentic glomerulonephritis. The presence of leukocytes within

glomerular capillaries is a variable feature. The mesangial widening may be the result of cell proliferation, accumulation of matrix, immune deposits, or some combination of these abnormalities. Healing of the focal proliferative lesion may lead to secondary focal segmental sclerosis. The characteristic immunofluorescence finding is **mesangial deposition of IgA** (Fig. 20.18B), often with C3 and properdin and lesser amounts of IgG or IgM. Early complement components are usually absent. Electron microscopy confirms the presence of electron-dense deposits predominantly in the mesangium; capillary wall deposits, if present, are usually sparse.

Clinical Features

The disease affects people of any age, most commonly older children and young adults. Many patients present with gross hematuria after an infection of the respiratory or, less commonly, gastrointestinal or urinary tract; 30% to 40% have only microscopic hematuria, with or without proteinuria; and 5% to 10% develop acute nephritic syndrome, including some with RPGN. The hematuria typically lasts for several days and then subsides, only to return every few months. The subsequent course is highly variable. Many patients maintain normal renal function for decades. Slow progression to chronic renal failure occurs in 15% to 40% of cases over a period of 20 years. Onset in old age, heavy proteinuria, hypertension, and the extent of glomerulosclerosis on biopsy are clues to an increased risk of progression. Recurrence of IgA deposits in transplanted kidneys is frequent, and in approximately 15% of those with recurrent IgA deposits, the disease runs the same slowly progressive course as that of primary IgA nephropathy.

Alport Spectrum Disorders

Alport spectrum disorders (formally called **hereditary nephritis**) refers to a group of heterogeneous familial kidney diseases associated with mutations in genes encoding basement membrane components that manifest primarily with glomerular injury. Classic severe *Alport syndrome* and its variants are described next

Alport Syndrome. Alport syndrome is caused by mutations affecting type IV collagen that result in hematuria with progression to chronic renal failure, accompanied by sensorineural deafness and

various eye disorders, including lens dislocation, posterior cataracts, and corneal dystrophy. The disease is inherited as an X-linked trait in approximately 85% of cases. In the X-linked form, males express the full syndrome, while female heterozygotes typically present with hematuria. The disease is 100% penetrant, with approximately 90% of affected males progressing to ESRD by 40 years of age. Autosomal recessive and autosomal dominant forms also exist, in which males and females are equally susceptible to the full syndrome.

Pathogenesis

The disease manifestations are due to mutations in one of several genes coding for subunits of the collagen IV molecule. More than 500 pathogenic mutations have been identified, all of which interfere with assembly of type IV collagen, which is crucial for function of the GBM, the lens of the eye, and the cochlea. Because the GBM consists of networks of trimeric collagen IV molecules composed of α_3 , α_4 , and α_5 chains, mutations affecting any one chain result in defective assembly of the collagen network. Because type IV collagen chains are encoded on autosomes (chromosomes 2 and 13) and the X-chromosome, the inheritance pattern can be autosomal or X-linked. Missense and splice site mutations, insertions, and deletions have all been identified. Genetic analysis has shown that in patients with X-linked disease, large deletions in the collagen IV α_5 chain (COL4A5) are associated with ESRD at an earlier age.

MORPHOLOGY

Fully developed Alport syndrome has characteristic electron microscopic findings. The GBM shows irregular foci of thickening alternating with attenuation (thinning), and pronounced splitting and lamination of the lamina densa, often producing a distinctive **basket-weave appearance** (Fig. 20.19). Similar alterations can be found in the tubular basement membranes. In the milder heterozygous X-linked or autosomal dominant Alport syndrome, basement membrane changes can be more subtle, with diffuse or segmental attenuation and focal absence of a lamina densa.

Immunohistochemistry can be helpful in classic Alport syndrome in early stages or in absence of electron microscopy examination, because antibodies to α_3 , α_4 , and α_5 collagen fail to stain both glomerular and tubular basement membranes in the classic X-linked form. There is also absence of α_5 staining in skin biopsy specimens from these patients. As the disease progresses, focal segmental and global glomerulosclerosis can be seen in all forms of Alport syndrome, accompanied by other changes of progressive renal injury, including vascular sclerosis, tubular atrophy, and interstitial fibrosis. Basement membrane abnormalities can also make podocytes more susceptible to injury and detachment.

Clinical Features

The most common presenting sign is gross or microscopic hematuria, frequently accompanied by red cell casts. Proteinuria may develop later, and rarely, the nephrotic syndrome develops. Symptoms appear at 5 to 20 years of age, and the onset of overt renal failure is between 20 and 50 years of age in males. The auditory defects may be subtle, requiring sensitive testing.

Variants of Alport Syndrome. A milder form of Alport syndrome is manifested clinically by familial hematuria usually uncovered on routine urinalysis, and morphologically by diffuse or segmental attenuation of the GBM to widths between 150 and 225 nm (compared with 300 to 400 nm in healthy adults). Mild or moderate proteinuria may also be present, and renal function may be normal

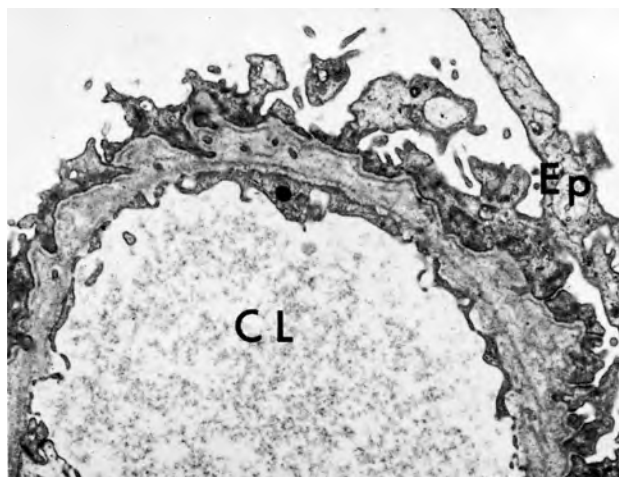


Fig. 20.19 Alport syndrome. Electron micrograph of a glomerular capillary with irregular thickening of the basement membrane, lamellation of the lamina densa, and foci of rarefaction. Such changes may be present in other diseases but are most pronounced and widespread in hereditary nephritis. CL, Capillary lumen; Ep, epithelium.

or there may be progressive renal failure, often with development of FSGS. In some cases, relatives of these patients manifest classic severe Alport syndrome. The disorder is distinguished from IgA nephropathy, another common cause of hematuria, based on biopsy findings and the presence of hearing loss and ocular abnormalities in Alport syndrome.

Thin basement membrane nephropathy is a related disorder associated with mutations in genes encoding α_3 or α_4 chains of type IV collagen. The disease most often has an autosomal inheritance, and most patients are heterozygous for the defective gene. As these diseases represent a continuum of changes resulting from mutations in collagen type IV genes, they should be considered a spectrum disorder requiring careful genetic counseling and clinical follow up.

KEY CONCEPTS

Glomerular Diseases with Isolated Hematuria

- IgA nephropathy, characterized by mesangial deposits of IgA-containing immune complexes, is the most common cause of glomerulonephritis worldwide. It is a common cause of both isolated and frequently recurrent hematuria and of various degrees of proteinuria and sometimes of nephritic syndrome; it commonly affects children and young adults and has a variable course.
- Alport syndrome is caused by mutations in genes encoding type IV collagen. It manifests as hematuria and slowly progressing proteinuria and declining renal function; glomeruli appear normal by light microscopy until late in the disease course.
- A milder clinical course can be seen in patients heterozygous for autosomal and X-linked Alport syndrome-related variants; these patients also carry an elevated risk of disease progression. Hence, those may be considered as part of the Alport syndrome spectrum.

Glomerular Lesions Associated With Systemic Diseases

Many immunologically mediated, metabolic, or hereditary systemic disorders are associated with glomerular injury; in some (e.g., SLE and

diabetes), the glomerular involvement is a major clinical manifestation. Most of these diseases are discussed elsewhere in this book. Here we briefly recall some of the lesions and discuss only those not considered in other sections.

Lupus Nephritis

As discussed in Chapter 6, SLE gives rise to a wide variety of renal lesions and clinical presentations. **Up to 50% of patients with SLE develop clinically significant renal involvement mainly in the form of autoimmune-mediated glomerulopathies and tubulointerstitial nephritis.** Clinical manifestations can include recurrent microscopic or gross hematuria, nephritic syndrome, RPGN, nephrotic syndrome, acute and chronic renal failure, and hypertension. The glomerular lesions are most often the result of deposition of immune complexes on the glomerular basement membrane, in the mesangium, and sometimes throughout the glomerulus. Immune complexes are usually reactive for IgG, often accompanied by IgA, IgM, C3, and C1q.

MORPHOLOGY

According to the currently accepted classification, six patterns of glomerular disease are seen in SLE. It should be noted that there is some overlap between these classes, and over time lesions may evolve from one class to another. Thus, the exact percentage of patients with each of the six classes of lesions is difficult to determine. Suffice it to say that class I is the least common and class IV is the most common pattern.

- *Minimal mesangial lupus nephritis* (class I) is very uncommon and is characterized by immune complex deposition in the mesangium, identified by immunofluorescence and by electron microscopy, but without structural changes by light microscopy. It can have other features, including lupus podocytopathy (similar to minimal change disease), tubulointerstitial disease, or thrombotic microangiopathy (discussed later).
- *Mesangial proliferative lupus nephritis* (class II) is characterized by mesangial cell proliferation, often accompanied by accumulation of mesangial matrix, and granular mesangial deposits of immunoglobulin and complement without involvement of glomerular capillaries or active endocapillary inflammation. Similar to class I lupus nephritis, this class can also have other features, as mentioned earlier.
- *Focal lupus nephritis* (class III) is defined by involvement of fewer than 50% of glomeruli. The lesions may be segmental (affecting only a portion of the glomerulus) or global (involving the entire glomerulus). Affected glomeruli may exhibit swelling and proliferation of endothelial and mesangial cells, leukocyte accumulation, capillary necrosis, and hyaline thrombi. There is also often extracapillary proliferation associated with focal necrosis and crescent formation (Fig. 20.20A). The clinical presentation ranges from mild hematuria and proteinuria to acute renal insufficiency. Red cell casts in the urine are common when the disease is active and correlate with a nephritic sediment. Some patients progress to diffuse glomerulonephritis. The active (or proliferative) inflammatory lesions can heal completely or lead to chronic global or segmental glomerular scarring.
- *Diffuse lupus nephritis* (class IV) is the most common and severe form of lupus nephritis. The lesions are similar to those in class III, but differ in extent; typically, in class IV nephritis greater than 50% of the glomeruli are affected. Involved glomeruli show proliferation of endothelial, mesangial, and epithelial cells (Fig. 20.20B), with the latter producing cellular crescents that fill the Bowman space. Subendothelial immune complex deposits may cause circumferential thickening of the capillary wall, forming “wire loop” structures on light microscopy (Fig. 20.20C). Immune complexes can be readily detected by electron microscopy (Fig. 20.20D) and immunofluorescence (Fig. 20.20E). Lesions may progress to scarring of glomeruli. Patients with diffuse glomerulonephritis are usually symptomatic, showing hematuria as well as proteinuria. Hypertension and mild to severe renal insufficiency are also common.

- *Membranous lupus nephritis* (class V) is characterized by diffuse thickening of the capillary walls due to deposition of subepithelial immune complexes, similar to idiopathic membranous nephropathy. The immune complexes are usually accompanied by increased production of basement membrane–like material. This lesion typically causes severe proteinuria or nephrotic syndrome and may occur concurrently with focal or diffuse lupus nephritis. Mesangial immune complex deposits are frequently seen.
- *Advanced sclerosing lupus nephritis* (class VI) is characterized by global sclerosis of 90% or more of the glomeruli.

Changes in the interstitium and tubules are frequently present in lupus nephritis. Rarely, tubulointerstitial lesions may be the dominant abnormality. Immune complexes similar to those in glomeruli are present in the tubular or peritubular capillary basement membranes in many lupus nephritis patients. Occasionally, there are well-organized B-cell follicles in the interstitium, with plasma cells that may be sources of autoantibodies.

Vascular involvement can also be present in isolation or together with other renal manifestations. While deposition of immune complexes in the intima and media of arteries is frequently seen, active vasculitis, rarely with necrosis of the vessel wall or presenting in conjunction with a thrombotic microangiopathy, is less common.

Henoch-Schönlein Purpura

This childhood syndrome consists of purpuric skin lesions, abdominal pain and intestinal bleeding, and arthralgias along with renal abnormalities. Skin lesions characteristically involve the extensor surfaces of arms and legs as well as buttocks; abdominal manifestations include pain, vomiting, and intestinal bleeding. Renal manifestations occur in one-third of patients and include gross or microscopic hematuria, nephritic syndrome, nephrotic syndrome, or some combination of these. A small number of patients, mostly adults, develop a rapidly progressive form of glomerulonephritis with many crescents. The disease is most common in children 3 to 8 years of age, but it also occurs in adults, in whom the renal manifestations are usually more severe. There is a background of atopy in about one-third of patients, and onset often follows an upper respiratory infection. IgA is deposited in the glomerular mesangium in a distribution similar to that of IgA nephropathy. This has led to the concept that *IgA nephropathy and Henoch-Schönlein purpura are manifestations of the same disease*. The finding of Ig and C3 deposits in glomeruli suggests that immune complexes are involved in the disease.

MORPHOLOGY

On histologic examination, the renal lesions vary from mild focal mesangial proliferation to diffuse mesangial proliferation and/or endocapillary proliferation to crescentic glomerulonephritis. Whatever the histologic lesions, the pathognomonic feature by immunofluorescence microscopy is the **deposition of IgA, sometimes with IgG and C3, in the mesangial region**, sometimes with deposits extending to the capillary loops. The skin lesions consist of subepidermal hemorrhages and a necrotizing vasculitis involving the small vessels of the dermis. Deposits of IgA, along with IgG and C3, are also present in such vessels. Vasculitis also occurs in other organs, such as the gastrointestinal tract, but is rare in the kidney.

The course of the disease is variable, but recurrences of hematuria may persist for many years after onset. Most children have an excellent prognosis. Patients with more diffuse lesions, crescents, or the nephrotic syndrome have a somewhat poorer prognosis.

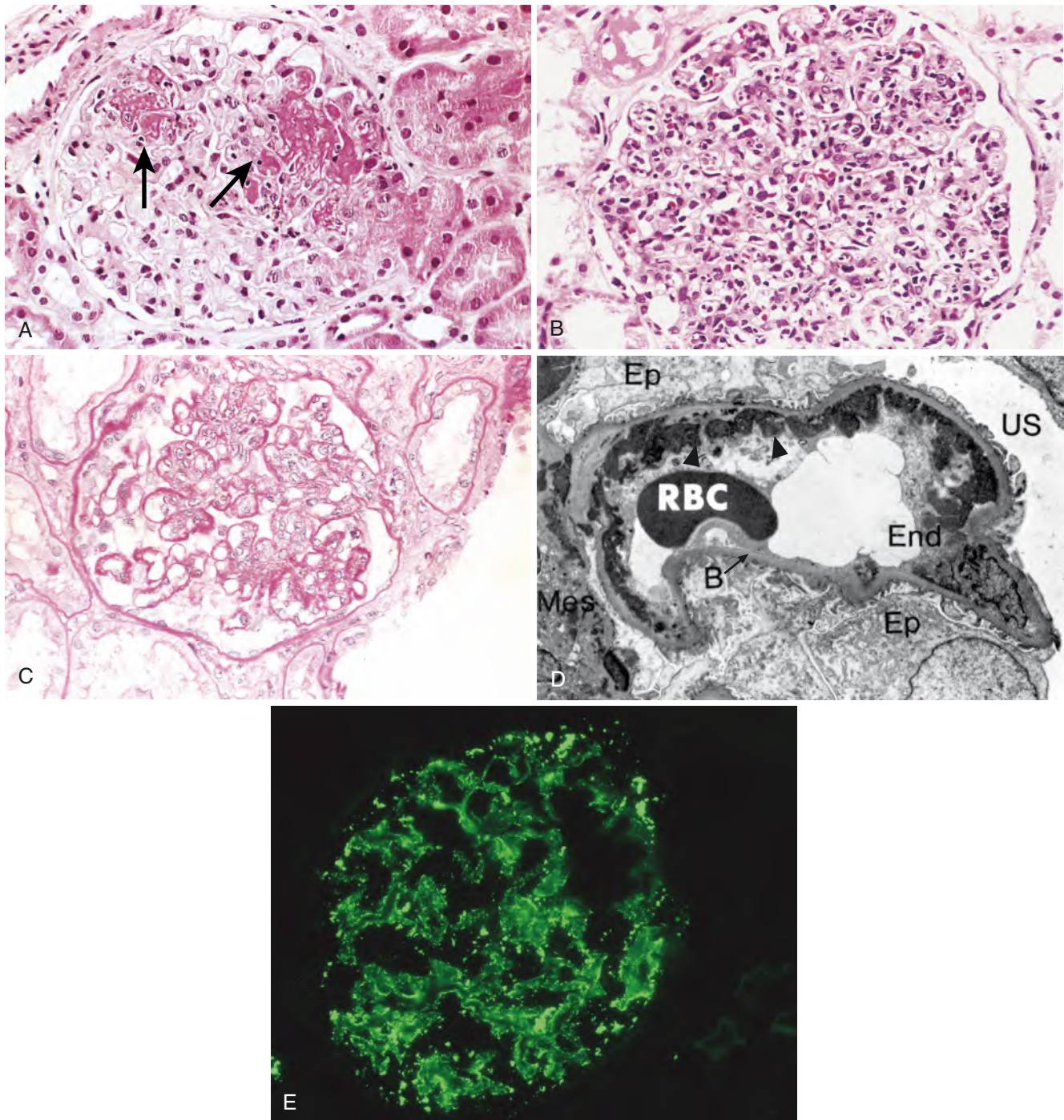


Fig. 20.20 Lupus nephritis. (A) Focal proliferative glomerulonephritis, with two necrotizing lesions (*arrows*) (H&E stain). Cellular proliferation is not prominent in this case. (B) Diffuse proliferative glomerulonephritis. Note the marked increase in cellularity throughout the glomerulus (H&E stain). (C) Membranous lupus nephritis showing a glomerulus with several “wire loop” lesions representing extensive subendothelial deposits of immune complexes (periodic acid–Schiff stain). (D) Electron micrograph of a renal glomerular capillary loop from a patient with lupus nephritis. Subendothelial dense deposits (*arrowheads*) correspond to “wire loops” seen by light microscopy. (E) Deposition of IgG antibody in a granular pattern, detected by immunofluorescence. *B*, Basement membrane; *End*, endothelium; *Ep*, epithelial cell with foot processes; *Mes*, mesangium; *RBC*, red blood cell in capillary lumen; *US*, urinary space. (A–C, Courtesy Dr. Helmut Rennke, Department of Pathology, Brigham and Women’s Hospital, Boston. D, Courtesy Dr. Edwin Eigenbrodt, Department of Pathology, University of Texas, Southwestern Medical School, Dallas. E, Courtesy Dr. Jean Olson, Department of Pathology, University of California, San Francisco.)

Diabetic Nephropathy

Diabetes is a major cause of renal morbidity and mortality, and diabetic nephropathy is the leading cause of chronic kidney failure in the United States. Advanced or end-stage kidney disease occurs in as many as 40% of both type 1 diabetics and type 2 diabetics. The pathology and pathogenesis of this disorder are discussed in Chapter 24.

Other Systemic Disorders

Goodpasture syndrome (Chapter 15), *microscopic polyangiitis*, and *granulomatosis with polyangiitis* (Chapter 11) are commonly associated with glomerular lesions, as described in the discussion of these diseases. Suffice it to say here that the glomerular lesions in these three conditions can be histologically similar and are principally characterized by foci of glomerular necrosis and crescent formation. In the early or mild forms of renal involvement, there is focal and segmental, sometimes necrotizing, glomerulonephritis, and most patients have hematuria with a mild decline in GFR. In the more severe cases, which may be associated with RPGN, there is more extensive necrosis, fibrin deposition, and extensive formation of epithelial (cellular) crescents, which can become organized to form fibrocellular and fibrous crescents if the glomerular injury evolves into segmental or global scarring (sclerosis).

Essential mixed cryoglobulinemia is another systemic condition in which deposits of cryoglobulins composed principally of IgG-IgM complexes induce cutaneous vasculitis, synovitis, and a proliferative glomerulonephritis, typically MPGN. Most cases of essential mixed cryoglobulinemia have been associated with infection with hepatitis C virus, and this condition in particular is associated with glomerulonephritis, usually MPGN.

Immunoglobulins secreted by plasma cell neoplasms may also produce glomerular lesions, including amyloidosis (see Fig. 6.46C).

TUBULAR AND INTERSTITIAL DISEASES

Most forms of tubular injury involve the interstitium as well; therefore, diseases affecting these two components are discussed together. Under this heading, we consider three major processes: (1) infection of the kidney and urinary tract (*pyelonephritis*), (2) inflammation of the tubules and interstitium (*tubulointerstitial nephritis*), and (3) *ischemic or toxic tubular injury*, leading to ATI and the clinical syndrome of acute tubular necrosis.

Pyelonephritis and Urinary Tract Infection

Pyelonephritis is one of the most common diseases of the kidney and is defined as inflammation caused by infection affecting the tubules, interstitium, and renal pelvis. *Acute pyelonephritis* is generally caused by bacterial infection and is associated with urinary tract infection. *Chronic pyelonephritis* can ensue from episodes of acute pyelonephritis that recur due to the presence of predisposing factors (such as vesicoureteral reflux and urinary tract obstruction).

Pyelonephritis is a serious complication of urinary tract infections that affect the bladder (cystitis), the kidneys and their collecting systems (pyelonephritis), or both. Bacterial infections of the lower urinary tract may be asymptomatic (asymptomatic bacteriuria) and remain localized to the bladder, but may also spread to the kidney.

Pathogenesis

More than 85% of urinary tract infections are caused by enteric gram-negative bacilli that are normal inhabitants of the intestinal tract. By far the most common is *Escherichia coli*, followed by *Proteus*, *Klebsiella*, and *Enterobacter spp.* *Streptococcus faecalis*, also of

enteric origin, and staphylococci; however, virtually any bacterial and fungal agent can cause lower urinary tract and renal infection. Mycobacterial and fungal organisms induce caseating and noncaseating granulomatous inflammation, respectively. In immunocompromised persons, particularly those with transplanted organs, viruses such as polyomavirus, cytomegalovirus, and adenovirus can also cause renal infection.

There are two routes by which bacteria can reach the kidneys (Fig. 20.21): (1) the lower urinary tract (ascending infection); and (2) through the bloodstream (hematogenous infection). **Ascending infection is the most common cause of pyelonephritis.** The hematogenous route is less common and results from seeding of the kidneys by bacteria from distant foci such as infective endocarditis. Hematogenous infection is more likely to occur in the presence of ureteral obstruction, and in debilitated patients. Typically, in patients receiving immunosuppressive therapy, nonenteric organisms, such as staphylococci and certain fungi and viruses, are involved.

Normal human bladder and bladder urine are sterile; therefore, a number of alterations must happen for renal infection to occur by the ascending route:

- The first step is *colonization of the distal urethra and introitus* (in the female) by coliform bacteria. This colonization is influenced by the degree of bacterial adherence to urethral mucosal epithelia, as discussed in Chapter 8, which involves adhesive molecules (adhesins) on the P-fimbriae (pili) of bacteria that interact with receptors on the surface of urothelial cells.
- *From the urethra to the bladder*, organisms gain entrance during urethral catheterization or other instrumentation. Long-term catheterization in particular carries a risk of infection. In the absence of instrumentation, *urinary infections are much more common in females*. This has been ascribed to a combination of factors, including the shorter urethra in females, the absence of the antibacterial properties of prostatic fluid, hormonal changes affecting bacterial adherence to the mucosa, and urethral trauma during sexual intercourse. Several conditions predispose to the movement of microbes from the bladder to the kidneys:
 - *Urinary tract obstruction and stasis of urine.* Ordinarily, organisms introduced into the bladder are cleared by voiding and by antibacterial mechanisms. However, outflow obstruction or bladder dysfunction results in incomplete emptying and residual urine. In the presence of stasis, bacteria introduced into the bladder can multiply unhindered. Accordingly, urinary tract infection is frequent among patients with lower urinary tract obstruction, such as may occur with benign prostatic hypertrophy, tumors, calculi, or neurogenic bladder dysfunction caused by diabetes or spinal cord injury.
 - *Vesicoureteral reflux.* Although obstruction is an important predisposing factor in ascending infection, *incompetence of the vesicoureteral valve* allows bacteria to ascend the ureter into the renal pelvis. The normal ureteral insertion into the bladder is a one-way valve that prevents retrograde flow of urine when the intravesical pressure rises, as in micturition. An incompetent vesicoureteral orifice allows the reflux of bladder urine into the ureters (Fig. 20.22). Reflux is most often due to a congenital absence or shortening of the intravesical portion of the ureter, such that the ureter is not compressed during micturition. In addition, it may be acquired due to bladder infection. It is postulated that bacteria or the associated inflammation promote reflux by affecting ureteral contractility, particularly in children. Vesicoureteral reflux is estimated to affect 1% to 2% of otherwise normal children. *Acquired vesicoureteral reflux* in adults can result from persistent bladder

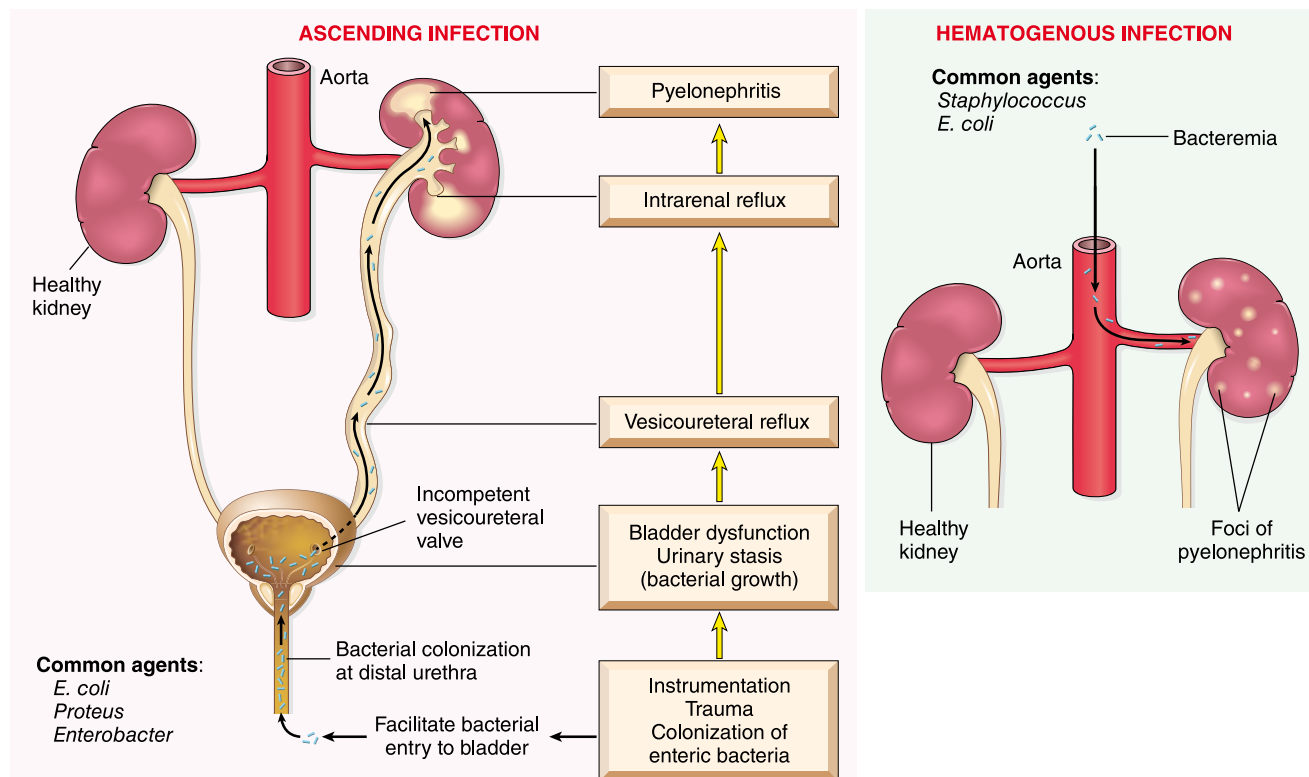


Fig. 20.21 Schematic representation of pathways of renal infection. Hematogenous infection results from bacteremic spread. More common is ascending infection, which results from a combination of urinary bladder infection, vesicoureteral reflux, and intrarenal reflux.



Fig. 20.22 Vesicoureteral reflux demonstrated by a voiding cystourethrogram. Dye injected into the bladder refluxes into both dilated ureters, filling the pelvis and calyces.

atony caused by spinal cord injury. Similar to that of an obstruction, the residual urine in the urinary tract after voiding favors bacterial growth.

- **Intrarenal reflux.** Vesicoureteral reflux also affords a mechanism to propel infected bladder urine up to the renal pelvis and deep into the renal parenchyma through open ducts at the tips of the papillae (intrarenal reflux). Intrarenal reflux is most common in the upper and lower poles of the kidney, where papillae tend to have flattened or concave tips rather than the convex pointed type present in the midzones of the kidney. Reflux can be demonstrated by voiding cystourethrography, in which the bladder is filled with a radiopaque dye and images are taken during micturition. Vesicoureteral reflux can be demonstrated in about 30% of infants and children with urinary tract infection (see Fig. 20.22).

In the absence of vesicoureteral reflux, infection usually remains localized in the bladder. Thus, most individuals with repeated or persistent bacterial colonization of the urinary tract experience cystitis and urethritis (lower urinary tract infection) rather than pyelonephritis.

Acute Pyelonephritis

Acute pyelonephritis is a suppurative inflammation of the kidney caused by bacterial and sometimes fungal or mycobacterial infection, which can reach the kidney by hematogenous spread or, more commonly, through the ureters in association with vesicoureteral reflux.

MORPHOLOGY

The hallmarks of acute pyelonephritis are **patchy interstitial suppurative inflammation, intratubular aggregates of neutrophils, neutrophilic tubulitis, and tubular injury**. The suppuration may occur as discrete focal abscesses or large wedge-like areas and can involve one or both kidneys (Fig. 20.23A).

In the early stages, the neutrophilic infiltration is limited to the tubules. The tubular lumens are a conduit for the extension of the infection, which may extend to the interstitium and produce abscesses that destroy the involved tubules (Fig. 20.23B). Characteristically, glomeruli are relatively resistant to the infection. Extensive disease, however, eventually also destroys the glomeruli. Fungal pyelonephritis (e.g., *Candida*) often results in granulomatous interstitial inflammation.

Three complications may be superimposed on acute pyelonephritis.

- **Papillary necrosis** is seen mainly in diabetics, sickle cell disease, and in those with urinary tract obstruction. Papillary necrosis is usually bilateral but may be unilateral. One or all of the pyramids of the affected kidney may be involved. On cut section, the tips or distal two-thirds of the pyramids have areas of gray-white to yellow necrosis (Fig. 20.24A). On microscopic examination, the necrotic tissue shows characteristic ischemic coagulative necrosis, with preservation of outlines of tubules (Fig. 20.24B). The leukocytic response is limited to the junctions between preserved and destroyed tissue.
- **Pyonephrosis** is seen when there is total or almost complete obstruction, particularly when it is high in the urinary tract. The suppurative exudate is unable to drain and fills the renal pelvis, calyces, and ureter with pus.
- **Perinephric abscess** is an extension of suppurative inflammation through the renal capsule into the perinephric tissue.

After the acute phase of pyelonephritis, healing occurs. The neutrophilic infiltrate is replaced by macrophages, plasma cells, and lymphocytes, and eventually by irregular scars that can be seen on the cortical surface as fibrous depressions. Such scars are characterized microscopically by tubular atrophy, interstitial fibrosis, and a lymphocytic infiltrate in a characteristic patchy, jigsaw pattern with intervening preserved parenchyma. **The pyelonephritic scar is almost always associated with inflammation, fibrosis, and deformation of the underlying calyx and pelvis**, reflecting the role of ascending infection and vesicoureteral reflux in the pathogenesis of the disease.

Clinical Features

Acute pyelonephritis is often associated with:

- **Urinary tract obstruction**, either congenital or acquired
- **Instrumentation** of the urinary tract, most commonly catheterization
- **Vesicoureteral reflux**
- **Pregnancy**. Between 4% and 6% of pregnant females develop bacteriuria sometime during pregnancy, and 20% to 40% of these eventually develop symptomatic urinary infection if not treated.
- **Sex and age**. After the first year of life (when congenital anomalies in males commonly become evident) and up to around 40 years of age, infections are much more frequent in females. With increasing age, the incidence in males rises as a result of prostatic hypertrophy and instrumentation.
- **Preexisting renal lesions**, causing intrarenal scarring and obstruction
- **Diabetes**, in which increased susceptibility to infection, neurogenic bladder dysfunction, and more frequent instrumentation are predisposing factors
- **Immunosuppression and immunodeficiency**

Acute pyelonephritis usually presents with a sudden onset of pain at the costovertebral angle and systemic evidence of infection, such as fever and malaise. There are often indications of bladder and urethral irritation, such as dysuria, frequency, and urgency. The urine contains many leukocytes (pyuria) derived from the inflammatory infiltrate, but pyuria does not differentiate upper from lower urinary tract infection. The finding of leukocyte casts, typically rich in neutrophils (pus casts), indicates renal involvement, because casts are formed only in tubules. The diagnosis of infection is established by urine culture.

Uncomplicated acute pyelonephritis follows a benign course, and symptoms disappear within a few days after the institution of appropriate antibiotic therapy. Bacteria, however, may persist in the urine, or there may be recurrence of infection with new serologic types of *E. coli* or other organisms. In the presence of unrelieved urinary obstruction, diabetes, or immunodeficiency, acute pyelonephritis may be more serious, leading to repeated septicemic episodes. The superimposition of papillary necrosis may lead to acute renal failure.

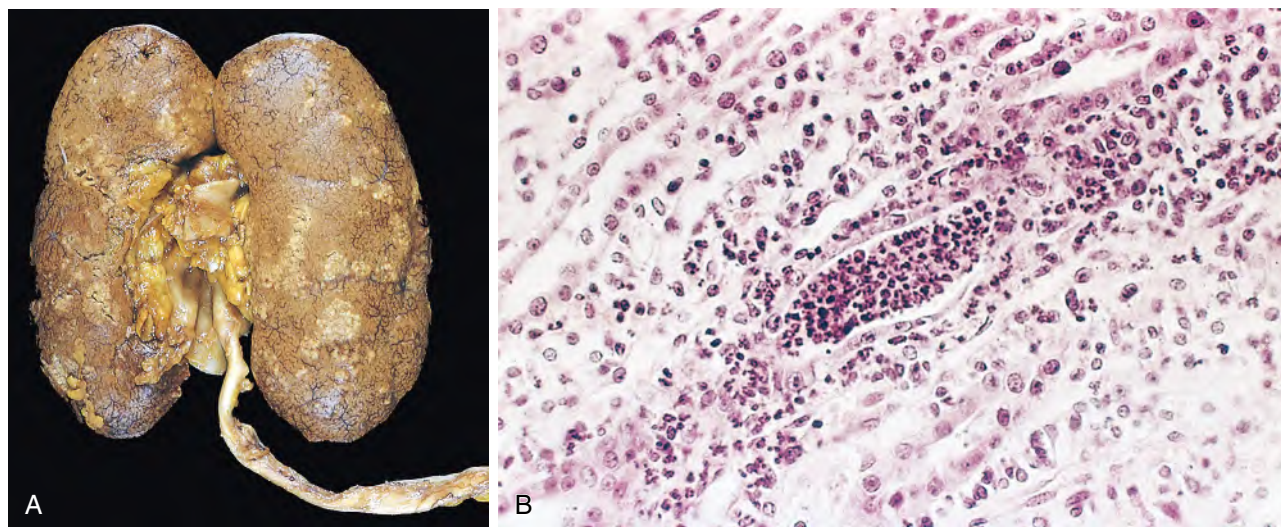


Fig. 20.23 Acute pyelonephritis. (A) Cortical surface shows grayish white areas of inflammation and abscess formation. (B) Neutrophilic exudate within tubules and interstitial inflammation.

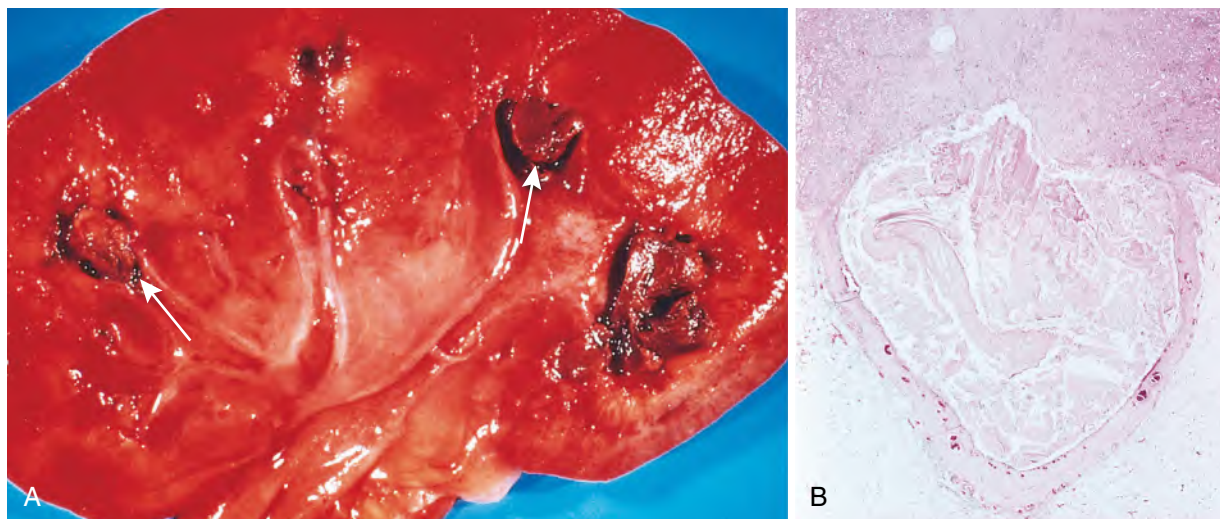


Fig. 20.24 Papillary necrosis. (A) Areas of hemorrhage and necrosis involve the papillae (arrows). (B) Low-power view showing a large area of renal papillary necrosis.

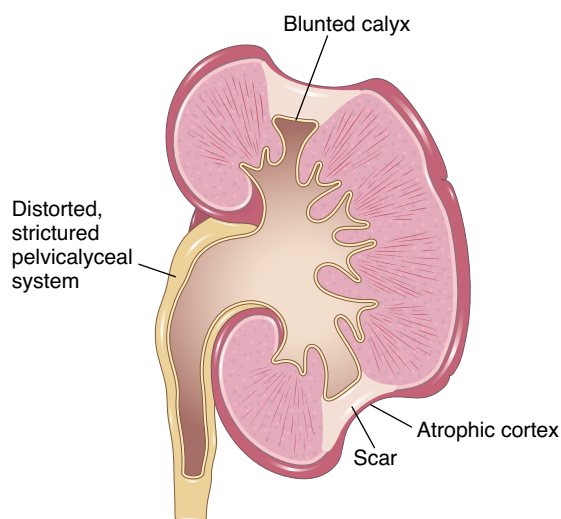


Fig. 20.25 Typical coarse scars of chronic pyelonephritis associated with vesicoureteral reflux. The scars are usually polar and are associated with underlying blunted calyces.

Chronic Pyelonephritis and Reflux Nephropathy

Chronic pyelonephritis is a disorder in which chronic tubulointerstitial inflammation and scarring involve the calyces and pelvis (Fig. 20.25). Although several diseases produce chronic tubulointerstitial alterations (see Table 20.8), only chronic pyelonephritis and analgesic nephropathy affect the calyces, making pelvocalyceal damage an important diagnostic clue. Chronic pyelonephritis at one time accounted for 10% to 20% of patients in renal transplant or dialysis units, until predisposing conditions such as reflux became better recognized and treated. This condition remains an important cause of kidney destruction in children with severe lower urinary tract abnormalities.

MORPHOLOGY

The characteristic changes of chronic pyelonephritis are seen on gross examination (Fig. 20.26A). The kidneys usually are irregularly scarred; if bilateral, the involvement is asymmetric. In contrast, both kidneys in chronic glomerulonephritis

are diffusely and symmetrically scarred. The hallmarks of chronic pyelonephritis are **coarse, discrete, corticomedullary scars overlying dilated, blunted, or deformed calyces, and flattening of the papillae**. The scars vary from one to several, and most are in the upper and lower poles, consistent with the frequency of reflux in these sites.

The microscopic changes predominantly involve tubules and interstitium (Fig. 20.26B). The tubules show atrophy in some areas and hypertrophy or dilation in others. Dilated tubules with flattened epithelium may be filled with proteinaceous casts resembling thyroid colloid (thyroidization). There are varying degrees of interstitial inflammation and fibrosis in the cortex and medulla. There is often fibrosis around the calyceal epithelium as well as a marked chronic inflammatory infiltrate. Glomeruli appear normal except for a variety of ischemic changes, including periglomerular fibrosis, fibrous obliteration, and secondary changes related to hypertension.

Xanthogranulomatous pyelonephritis is a rare form of chronic pyelonephritis characterized by accumulation of foamy macrophages intermingled with plasma cells, lymphocytes, polymorphonuclear leukocytes, and occasional giant cells. Often associated with *Proteus* infections and obstruction, the lesions sometimes produce large, yellowish orange nodules that may be grossly confused with renal cell carcinoma.

Clinical Features

Chronic pyelonephritis may have a silent onset or present with manifestations of acute recurrent pyelonephritis, such as back pain, fever, pyuria, and bacteriuria. These patients receive medical attention late in their disease course because of the gradual onset of renal insufficiency and hypertension. Reflux nephropathy is often discovered in children when the cause of hypertension is investigated. Loss of tubular function, in particular loss of concentrating ability, gives rise to polyuria and nocturia. Radiographic studies show asymmetrically contracted kidneys with characteristic coarse scars and blunting and deformity of the calyceal system. Significant bacteriuria may be present, but it is often absent in the late stages.

Although proteinuria is usually mild, some individuals with pyelonephritic scars develop secondary FSGS with significant proteinuria, even in the nephrotic range, usually several years after the scarring has occurred and often in the absence of continued infection or persistent vesicoureteral reflux. The onset of proteinuria is a poor prognostic sign since it indicates an increased likelihood of progression to ESRD. The glomerulosclerosis, as discussed, is attributable to the adaptive

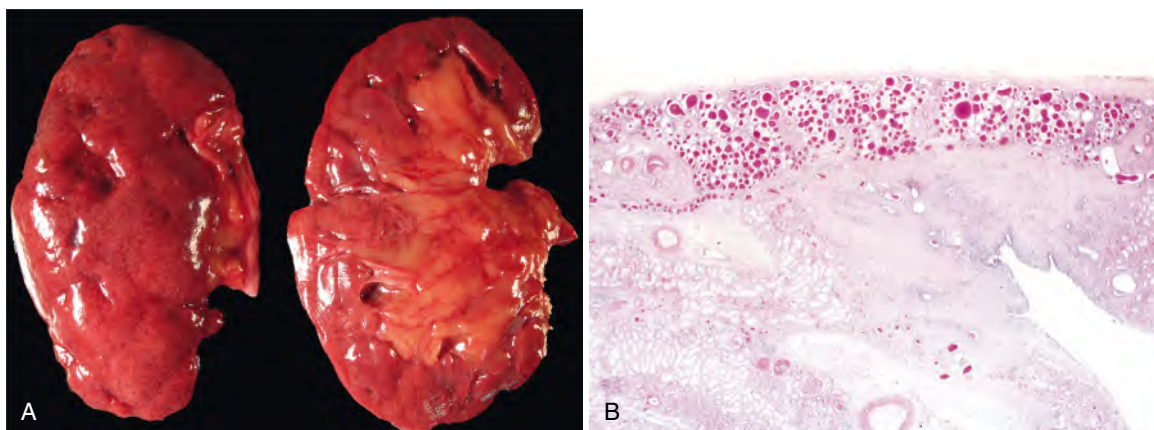


Fig. 20.26 Chronic pyelonephritis. (A) The surface (*left*) is irregularly scarred. The cut section (*right*) reveals blunting and loss of several papillae. (B) Low-power view shows a corticomedullary renal scar with an underlying dilated deformed calyx. Note the thyroidization of tubules in the cortex.

glomerular alterations secondary to loss of renal mass caused by pyelonephritic scarring.

KEY CONCEPTS

Pyelonephritis

- Acute and chronic pyelonephritis are caused by infection via the ascending (more common) or hematogenous route. Obstructive lesions of the urinary tract are important predisposing factors.
- Bacteria are the most common infectious agent in acute pyelonephritis and induce a prominent neutrophilic inflammatory response; granulomatous interstitial inflammation is characteristic of fungal or mycobacterial infections.
- Chronic pyelonephritis ensues when anatomic anomalies result in urine reflux or urine outflow obstruction; multiple episodes of this injury with or without acute pyelonephritis lead to irregular scarring of the kidney that is typically more prominent at the upper or lower poles where reflux is more common.

Tubulointerstitial Nephritis

This group of renal diseases involves inflammatory injuries of the tubules and interstitium that are often insidious in onset and are principally manifest by azotemia. Tubulointerstitial nephritis can be acute, but if the injury recurs or persists, it can result in chronic forms of interstitial nephritis. Glomerular and vascular alterations may also be present in advanced stages of these diseases. Conversely, chronic tubulointerstitial damage is an important consequence of progression in diseases that primarily affect the glomerulus.

Tubulointerstitial disorders are distinguished clinically from the glomerular diseases by the following hallmarks:

- Absence of nephritic or nephrotic syndrome
- Presence of defects in tubular function. The latter may be subtle and include impaired ability to concentrate urine, evidenced clinically by polyuria or nocturia; salt wasting; diminished ability to excrete acids (metabolic acidosis); and isolated defects in tubular reabsorption or secretion. Advanced forms, however, may be difficult to distinguish clinically from other causes of renal insufficiency.

Specific conditions listed in [Table 20.8](#) are discussed elsewhere in this book. This section deals principally with drug-induced tubulointerstitial nephritis.

TABLE 20.8 Causes of Tubulointerstitial Nephritis

Infections
Acute bacterial pyelonephritis
Chronic pyelonephritis (including reflux nephropathy)
Other infections (e.g., viruses, parasites)
Toxins
Drugs
Acute hypersensitivity interstitial nephritis
Analgesics
Heavy metals
Lead, cadmium
Metabolic Diseases
Urate nephropathy
Nephrocalcinosis (hypercalcemic nephropathy)
Oxalate nephropathy
Physical Factors
Chronic urinary tract obstruction
Neoplasms
Multiple myeloma (light-chain cast nephropathy)
Immunologic Reactions
Transplant rejection
Sjögren syndrome
Sarcoidosis
Miscellaneous
Nephronophthisis
“Idiopathic” interstitial nephritis

Tubulointerstitial Nephritis Induced by Drugs and Toxins

Drug- and toxin-induced tubulointerstitial nephritis is the second most common cause of acute kidney injury (after pyelonephritis). Toxins and drugs can injure kidneys in at least three ways: (1) by triggering an interstitial immunologic reaction, exemplified by the acute hypersensitivity nephritis induced by drugs such as methicillin; (2) by causing acute tubular injury (ATI), as described later; and (3) by causing subclinical but cumulative injury to tubules that takes years to result in chronic kidney disease. The last type of damage is especially problematic, because it may not be recognized until the renal damage is irreversible.

Acute Drug-Induced Interstitial Nephritis

First reported after the use of sulfonamides, acute interstitial nephritis most frequently occurs with synthetic penicillins (methicillin, ampicillin), other synthetic antibiotics (rifampin), diuretics (thiazides), NSAIDs, and miscellaneous other drugs (allopurinol, cimetidine, immune checkpoint inhibitors). The chronic interstitial nephritis caused by phenacetin-containing analgesics, termed *analgesic nephropathy*, is of historical interest as its incidence has substantially diminished due to the withdrawal or restriction of phenacetin in most countries.

Drug-induced acute interstitial nephritis begins 2 to 40 days after drug exposure and is characterized by *fever*, *eosinophilia* (which may be transient), *rash* (in about 25% of patients), and *renal abnormalities*. The latter takes the form of hematuria, mild proteinuria, and leukocyturia (often including eosinophils). A rising serum creatinine or acute kidney injury with oliguria develops in about 50% of cases.

Pathogenesis

Many features of the disease suggest an immune mechanism. Clinical evidence of hypersensitivity includes the latent period, eosinophilia and rash, the fact that the onset of nephropathy is not dose-related, and recurrence of clinical and pathologic manifestations after re-exposure to the same or a chemically related drug. In some patients, serum IgE levels are increased, and IgE-containing plasma cells and basophils are present in the lesions, suggesting that the late-phase reaction of an IgE-mediated (type I) hypersensitivity is involved in the pathogenesis (Chapter 6). In other cases, a mononuclear or granulomatous reaction, together with positive results of skin tests to drugs, suggest a T cell-mediated (type IV) delayed-type hypersensitivity reaction.

The most likely sequence of events is that, by binding to some plasma membrane or extracellular component of tubular cells, the drugs create modified self antigens that evoke an immune response. The resultant injury is due to IgE or cell-mediated immune reactions directed against the tubular cells or their basement membranes.

MORPHOLOGY

The interstitium shows variable but frequently pronounced **edema and infiltration by mononuclear cells**, principally lymphocytes and macrophages. Eosinophils and neutrophils may be present (Fig. 20.27), often in clusters and large numbers, and smaller numbers of plasma cells and mast cells are sometimes also present. Inflammation may be more prominent in the medulla where the inciting agent is concentrated. Tubulitis, the infiltration of tubules by lymphocytes, is common. Variable degrees of tubular injury and regeneration are present. With some drugs (e.g., methicillin, thiazides), interstitial nonnecrotizing granulomas may be seen when tubules rupture. The glomeruli are normal except in some cases caused by NSAIDs, when minimal change disease and the nephrotic syndrome develop concurrently (see later).

Clinical Features

It is important to recognize drug-induced acute interstitial nephritis because withdrawal of the offending drug is followed by recovery, although it may take several months, and irreversible damage can occur. While drugs are the leading cause of acute interstitial nephritis, in many affected patients (approximately 30% to 40%) an offending drug or mechanism cannot be identified.

On occasion, necrotic papillae are excreted and may cause gross hematuria or renal colic due to ureteric obstruction. Papillary necrosis is not specific for analgesic nephropathy and is also seen in diabetes, as well as in urinary tract obstruction, sickle cell disease or trait (described later), and focally in renal tuberculosis. In all cases, it is

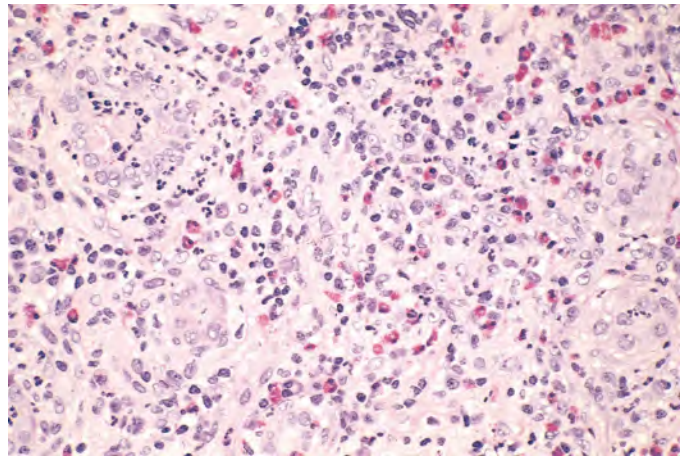


Fig. 20.27 Drug-induced interstitial nephritis, with prominent eosinophilic and mononuclear cell infiltrate. (Courtesy Dr. H Rennke, Brigham and Women's Hospital, Boston.)

caused by ischemia resulting from compression or obstruction of small blood vessels in the medulla. Compression may be caused by interstitial edema (as in inflammatory reactions and urinary tract obstruction) and vascular compromise by microvascular disease (as in diabetes).

Nephropathy Associated With NSAIDs. NSAIDs, one of the most commonly used classes of drugs, produce several forms of renal injury. Although these complications are uncommon, they should be kept in mind because NSAIDs are frequently administered to patients with other potential causes of renal disease. Many NSAIDs are nonselective cyclooxygenase inhibitors, and their adverse renal effects are related to their ability to inhibit cyclooxygenase-dependent prostaglandin synthesis. The selective COX-2 inhibitors, while sparing the gastrointestinal tract, do affect the kidneys because COX-2 is expressed in human kidneys. NSAID-associated renal syndromes include:

- *Acute kidney injury*, due to the decreased synthesis of vasodilatory prostaglandins and resultant ischemia. This is particularly likely to occur in the setting of other renal diseases or conditions causing volume depletion.
- *Acute hypersensitivity interstitial nephritis*, resulting in renal failure.
- *Minimal change disease*. This podocyte injury can occur either with or without acute interstitial nephritis, but is also consistent with a hypersensitivity reaction affecting the interstitium and possibly the glomeruli.
- *Membranous nephropathy*, with the nephrotic syndrome, is a rare complication of uncertain pathogenesis.

KEY CONCEPTS

Tubulointerstitial Nephritis Induced by Drugs and Toxins

- Drug-induced tubulointerstitial nephritis is the second most common cause of acute kidney injury after pyelonephritis.
- Prominent interstitial inflammation with associated tubular injury, which may or may not be accompanied by eosinophils or granulomatous inflammation, can be induced by almost any pharmacologic agent.
- NSAIDs can cause tubulointerstitial nephritis and/or glomerular injury, such as minimal change disease or membranous nephropathy.

Acute Tubular Injury/Necrosis

ATI is characterized by acute renal failure and often, but not invariably, morphologic evidence of tubular injury, in the form of necrosis of tubular epithelial cells. The term ATI is preferred by pathologists over the historical term *acute tubular necrosis* (ATN), because necrosis may be absent. ATI is the most common cause of acute kidney injury and accounts for some 50% of cases of acute kidney injury in hospitalized patients.

Depending on the magnitude and duration of injury, ATI is generally a reversible process that arises in a variety of clinical settings. Most of these, ranging from severe trauma to acute pancreatitis, have in common a period of inadequate blood flow to the peripheral organs, usually accompanied by marked hypotension and shock. This pattern is called *ischemic ATI*. The second pattern, called *nephrotoxic ATI*, is caused by a multitude of drugs, such as gentamicin, radiographic contrast agents, poisons, including heavy metals (e.g., mercury), and organic solvents (e.g., carbon tetrachloride). Combinations of ischemic and nephrotoxic ATI also can occur, exemplified by mismatched blood transfusions and other hemolytic crises causing *hemoglobinuria* and skeletal muscle injuries causing *myoglobinuria*.

Pathogenesis

The critical events in both ischemic and nephrotoxic ATI are (1) tubular injury and (2) persistent and severe disturbances in blood flow (Fig. 20.28).

- **Tubular cell injury:** Tubular epithelial cells, especially those of proximal tubules, are particularly sensitive to ischemia because of the high rate of metabolism and oxygen consumption that are required to perform transport and reabsorption functions; these cells are also vulnerable to toxins. Several factors predispose the tubules to toxic injury, including an increased surface area for tubular reabsorption, active transport systems for ions and organic acids, and the ability to absorb and concentrate toxins. Ischemia causes numerous structural and functional alterations in epithelial cells, as discussed in Chapter 2. One early reversible result of ischemia is loss of cell polarity due to redistribution of membrane proteins (e.g., the enzyme $\text{Na,K}^+\text{-ATPase}$) from the basolateral to the luminal surface of the tubular cells, resulting in abnormal ion transport across the cells and increased sodium delivery to distal tubules, which incites vasoconstriction via *tubuloglomerular feedback*. Initially, this response to the increased sodium lowers the GFR to maintain distal blood flow. In addition, ischemic tubular

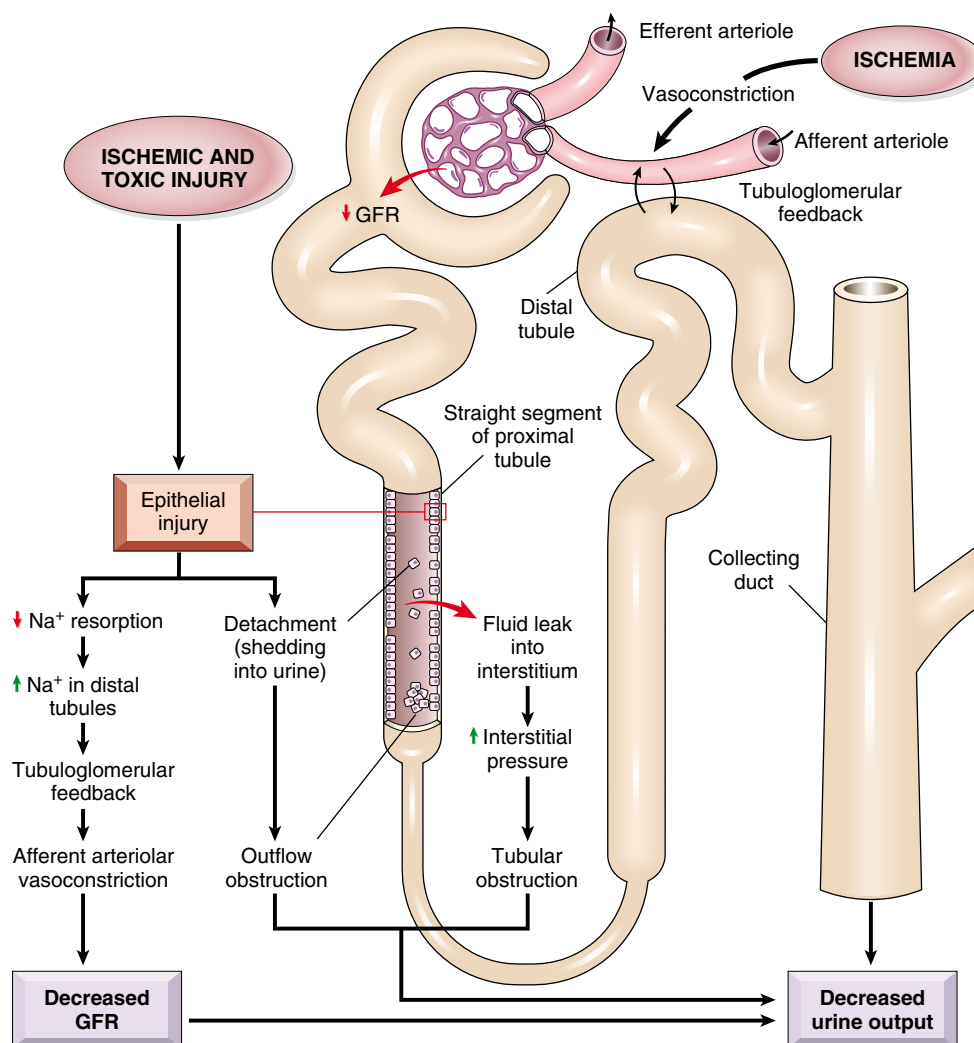


Fig. 20.28 Postulated sequence in ischemic or toxic acute tubular injury. Defective sodium resorption in injured tubules leads to increased concentration of Na^+ , which results in constriction of the afferent arterioles and decreased GFR. Sloughing of damaged epithelial cells causes obstruction, which, in combination with reduced GFR, results in decreased urine output. *GFR*, Glomerular filtration rate.

cells express cytokines and other chemotactic molecules, thus recruiting leukocytes that may contribute to the subsequent injury. In time, injured cells detach from the basement membranes and cause *luminal obstruction*, leading to increased intratubular pressure and further decrease in GFR. In addition, glomerular filtrate in the lumen of the damaged tubules can leak back into the interstitium, resulting in interstitial edema, increased interstitial pressure, and ischemic damage to the tubule. All of these effects, as shown in Fig. 20.29, contribute to the decreased GFR.

- **Disturbances in blood flow:** Ischemic renal injury is also characterized by *hemodynamic alterations* that cause reduced GFR. The major change is *intrarenal vasoconstriction*, which results in both reduced glomerular blood flow and reduced oxygen delivery to the functionally important tubules in the outer medulla (thick ascending limb and straight segment of the proximal tubule). Several vasoconstrictor pathways have been implicated, including the renin-angiotensin system, stimulated by decreased sodium in the tubules because of decreased blood pressure, and sublethal endothelial injury, leading to increased release of the vasoconstrictor endothelin and decreased production of the vasodilators nitric oxide and prostacyclin (prostaglandin I_2). There is also some evidence of a direct effect of ischemia or toxins on the glomerulus, causing a reduced glomerular ultrafiltration coefficient.

The patchiness of tubular necrosis and maintenance of the integrity of the basement membrane along many segments allow repair of the injured foci and recovery of function if the precipitating cause is

removed. This repair is dependent on the capacity of reversibly injured epithelial cells to proliferate and differentiate. Re-epithelialization is mediated by a variety of growth factors and cytokines produced locally by the tubular cells themselves or by inflammatory cells in the vicinity of necrotic foci.

MORPHOLOGY

ATI is characterized by **tubular epithelial injury** at multiple points along the nephron, with large skip areas in between, often accompanied by rupture of basement membranes (tubulorrhexis) and **occlusion of tubular lumens by casts** (Figs. 20.30A–B). The distinct patterns of tubular injury in ischemic and toxic ATI are shown in Fig. 20.29. The straight portion of the proximal tubule and the ascending thick limb in the renal medulla are especially vulnerable, but focal lesions may also occur in the distal tubule, often in conjunction with casts. It should be noted that the severity of the morphologic findings often does not correlate with the severity of clinical manifestations.

Other findings in ischemic ATI are interstitial edema and accumulations of leukocytes within dilated vasa recta. There is also evidence of epithelial regeneration in the form of flattened epithelial cells with hyperchromatic nuclei and mitotic figures. In time, this regeneration repopulates the tubules so that no residual evidence of damage is seen.

Toxic ATI is most obvious in the proximal convoluted tubules. On histologic examination, the tubular injury can be distinctive in poisoning with certain agents. With mercuric chloride, for example, severely injured cells may contain large acidophilic inclusions. Later, these cells become necrotic, are desquamated into the lumen, and may undergo calcification. Ethylene glycol poisoning produces marked ballooning and hydropic or vacuolar degeneration of proximal convoluted tubules, and calcium oxalate crystals are often also found in the tubular lumens.

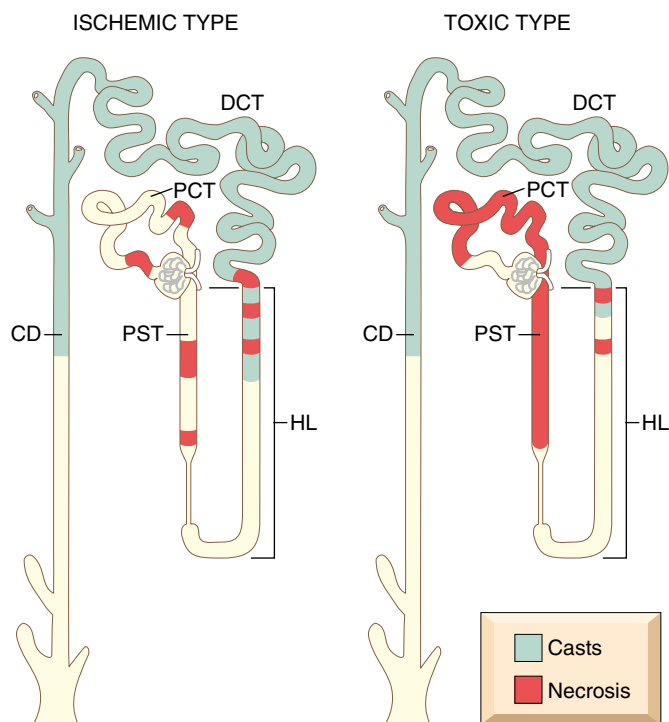


Fig. 20.29 Patterns of tubular damage in ischemic and toxic acute tubular injury. In the ischemic type, tubular necrosis is patchy, relatively short lengths of tubules are affected, and proximal straight tubule (PST) segments and ascending limbs of the Henle loop (HL) are most vulnerable. In toxic acute tubular injury, extensive necrosis is present along the proximal convoluted tubule (PCT) segments with many toxins (e.g., mercury), but necrosis of the distal tubule, particularly ascending HL, also occurs. In both types, lumens of the distal convoluted tubules (DCT) and collecting ducts (CD) contain casts.

Clinical Features

The clinical course of ATI is highly variable, but the classic case may be divided into three stages.

- **Initiation phase**, lasting about 36 hours, is dominated by the inciting medical, surgical, or obstetric event. The only indication of renal involvement is a modest decline in urine output and a rise in BUN. At this point, oliguria could be explained by a transient decrease in blood flow and declining GFR.
- **Maintenance phase** is characterized by sustained decreases in urine output to between 40 and 400 mL per day (oliguria), salt and water overload, rising BUN concentrations, hyperkalemia, metabolic acidosis, and other manifestations of uremia. With appropriate management, the patient can overcome this oliguric crisis.
- **Recovery phase** is ushered in by a steady increase in urine volume that may reach up to 3 L/day. The tubules are still damaged, so large amounts of water, sodium, and potassium are lost in the flood of urine. There is a peculiar increased vulnerability to infection at this stage. Eventually, renal tubular function is restored and concentrating ability improves. At the same time, BUN and creatinine levels begin to return to normal. Subtle tubular functional impairment may persist for months, but most patients who reach this phase eventually recover completely.

The prognosis of ATI depends on the magnitude and duration of injury. Recovery is expected with nephrotoxic ATI when the toxin has not caused serious damage to other organs, such as the liver or heart. With supportive care, 95% of those who do not succumb to the precipitating event recover. Conversely, in shock related to sepsis, extensive burns, or other causes of multiorgan failure, the mortality rate can exceed 50%.

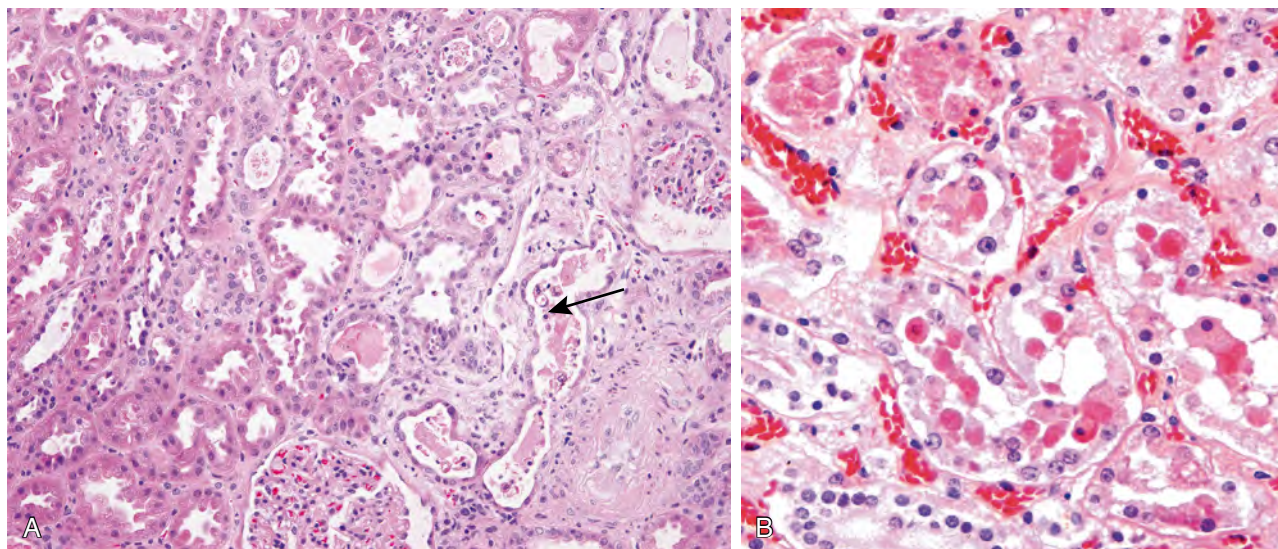


Fig. 20.30 Acute tubular injury. (A) Some tubular epithelial cells are necrotic, and many have detached (from their basement membranes) and been sloughed into the tubular lumens (arrow). (B) Necrotic tubular epithelial cells and cellular debris in tubular lumens. Congestion of peritubular capillaries is prominent.

KEY CONCEPTS

Acute Tubular Injury

- ATI is the most common cause of acute kidney injury and is attributable to ischemia and/or toxicity from an endogenous or exogenous substance.
- Tubular epithelial cell injury and altered intrarenal hemodynamics are the primary contributors to ATI.
- The clinical outcome is determined by the magnitude and duration of ATI.

Other Tubulointerstitial Diseases

Urate Nephropathy

Three types of nephropathy can occur in persons with hyperuricemic disorders:

- *Acute uric acid nephropathy* is caused by the precipitation of uric acid crystals in the renal tubules, principally in collecting ducts, leading to obstruction of nephrons and the development of acute renal failure. This is likely to occur in individuals with leukemias or lymphomas who are undergoing chemotherapy (tumor lysis syndrome); the drugs kill tumor cells, and uric acid is produced as released nucleic acids are broken down. Precipitation of uric acid is favored by the acidic pH in collecting tubules.
- *Chronic urate nephropathy*, or gouty nephropathy, occurs rarely in protracted forms of hyperuricemia. The monosodium urate crystals deposit in the acidic milieu of the distal tubules and collecting ducts, and a foreign-body giant cell reaction forms an outline of the needle-like crystals either in the tubular lumens or in the interstitium, but the crystals often dissolve during tissue processing (Fig. 20.31). This lesion is called a *tophus* (Chapter 26). Tubular obstruction by the urates causes cortical atrophy and scarring. Clinically, urate nephropathy is a subtle disease associated with tubular defects that may progress slowly.
- *Nephrolithiasis*: uric acid stones are present in 22% of individuals with gout and 42% of those with secondary hyperuricemia (see later discussion of renal stones).

Hypercalcemia and Nephrocalcinosis

Hypercalcemia, which is caused by several disorders such as hyperparathyroidism, multiple myeloma, vitamin D intoxication, metastatic

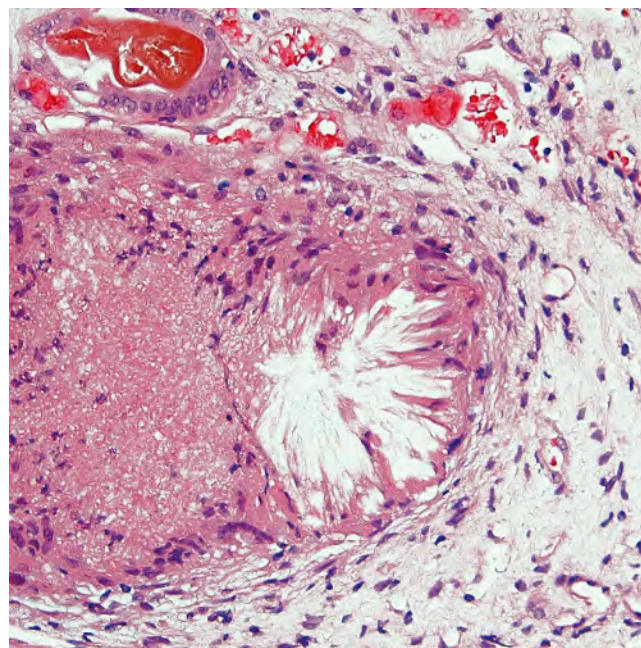


Fig. 20.31 Urate nephropathy. Granulomatous inflammation outlines slender urate crystals in the renal medulla.

cancer, or excess calcium intake (milk-alkali syndrome), may induce the formation of calcium stones and deposition of calcium in the kidney (nephrocalcinosis). Extensive degrees of calcinosis, under certain conditions, may lead to chronic tubulointerstitial disease and renal insufficiency.

The earliest functional defect is an inability to concentrate the urine. Other tubular defects, such as tubular acidosis and salt-losing nephritis, may also occur. With further damage, a slowly progressive chronic kidney disease develops. This is usually due to nephrocalcinosis, but many of these patients also have calcium stones and secondary pyelonephritis.

Autosomal Dominant Tubulointerstitial Kidney Disease

Autosomal dominant tubulointerstitial kidney disease (ADTKD) was previously known as medullary cystic kidney disease. Given the variable presence of cysts, which often are not located in the medulla, and the distinctive genetics, it is now called ADTKD. It is associated with mutations in the following genes and proteins:

MUC1 (encodes mucin-1, expressed in distal nephrons); *UMOD* (encodes uromodulin, expressed in thick ascending limb of loop of Henle); *REN* (encodes preprorenin, expressed by juxtaglomerular apparatus); and *HNF1B* (encodes hepatocyte nuclear factor 1 β , a transcription factor that regulates multiple genes, including *UMOD*). Almost half of the ADTKDs do not have a detectable mutation.

ADTKD produces nonspecific clinical and pathologic findings and causes progressive renal failure in adult life with an autosomal dominant pattern of transmission. The pathogenic mechanisms are unknown.

Light-Chain Cast Nephropathy (“Myeloma Kidney”)

Nonrenal malignant tumors, particularly those of hematopoietic origin, affect the kidneys in several ways (Table 20.9). The most common involvements are tubulointerstitial, caused by complications of the tumor (hypercalcemia, ureteral obstruction) or therapy (irradiation, hyperuricemia, chemotherapy, hematopoietic cell transplantation, infections in immunosuppressed patients). We limit the discussion here to the tubulointerstitial lesions in *multiple myeloma* patients.

Overt renal insufficiency occurs in half of those with multiple myeloma and related lymphoplasmacytic disorders. Several factors contribute to renal damage:

- *Bence-Jones proteinuria and cast nephropathy.* The main cause of renal dysfunction is related to Bence-Jones (light-chain) proteinuria and correlates with the degree of proteinuria. Two mechanisms seem to account for the renal toxicity of Bence-Jones proteins. First, some Ig light chains are directly toxic to epithelial cells, because of their intrinsic physicochemical properties. Second, Bence-Jones proteins combine with urinary glycoprotein (Tamm-Horsfall protein) under acidic conditions to form large, histologically distinct tubular casts that obstruct the tubular lumens and induce a characteristic inflammatory reaction (light-chain cast nephropathy).
- *Amyloidosis of AL type,* formed from free light chains (usually of λ type), occurs in 6% to 24% of individuals with myeloma.
- *Light-chain deposition disease.* In some patients, light chains (usually of κ type) deposit in the GBM and mesangium in nonfibrillar forms, causing a glomerulopathy (described earlier), and in tubular basement membranes, which may cause tubulointerstitial nephritis.
- *Hypercalcemia and hyperuricemia* are often present in these patients.

TABLE 20.9 Renal Disease Related to Nonrenal Neoplasms

Direct or metastatic tumor invasion of renal parenchyma
Ureters (obstruction)
Artery (renovascular hypertension)
Hypercalcemia
Hyperuricemia
Amyloidosis (AL, light-chain type)
Excretion of abnormal proteins (multiple myeloma)
Glomerulopathies
Membranous nephropathy, secondary (carcinomas)
Minimal change disease (Hodgkin disease)
Membranoproliferative glomerulonephritis (leukemias and lymphomas)
Monoclonal immunoglobulin/light-chain deposition disease (multiple myeloma)
Effects of radiation therapy, chemotherapy, hematopoietic cell transplantation, secondary infection

MORPHOLOGY

The tubulointerstitial changes in light-chain cast nephropathy are characteristic. The Bence-Jones protein **tubular casts** appear as pink to blue amorphous masses, sometimes concentrically laminated and often fractured, which fill and distend the tubular lumens. Some casts are surrounded by multinucleated giant cells that are derived from activated macrophages (Fig. 20.32). The adjacent interstitial tissue usually shows an inflammatory response and fibrosis. On occasion, the casts rupture the tubules, evoking a granulomatous inflammatory reaction. Amyloidosis, light-chain deposition disease, nephrocalcinosis, and infection may also be present.

Clinical Features

Clinically, the renal manifestations are of several types. In the most common form, *chronic kidney disease* develops insidiously and progresses slowly during a period of several months to years. Another form occurs suddenly and is manifested by *acute kidney injury* with oliguria. Precipitating factors include dehydration, hypercalcemia, acute infection, and treatment with nephrotoxic antibiotics. *Bence-Jones proteinuria* occurs in 70% of individuals with multiple myeloma; the presence of significant non-light-chain proteinuria (e.g., albuminuria) suggests AL amyloidosis or light-chain deposition disease.

Bile Cast Nephropathy

Impairment of renal function often occurs in patients with severe acute or advanced chronic liver disease. In this setting, serum bilirubin levels can be markedly elevated, particularly in jaundiced patients, with bile cast formation (also known as cholemic nephrosis) in distal nephron segments. The casts can extend to proximal tubules, resulting in both direct toxic effects on tubular epithelial cells and obstruction of the involved nephron. This mechanism of injury is analogous to that with monoclonal immunoglobulin and myoglobin casts. The tubular bile casts can range from yellowish-green to red-pink and contain variable degrees of sloughed cells or cellular debris. The reversibility of the renal injury depends on the severity and duration of the liver dysfunction.

VASCULAR DISEASES

Nearly all diseases of the kidney involve the renal blood vessels secondarily. Systemic vascular diseases, such as various forms of vasculitis, also affect renal vessels, and their effects on the kidney are clinically important. Hypertension, as discussed in Chapter 11, is intimately linked with the kidney, because kidney disease can be both a cause and consequence

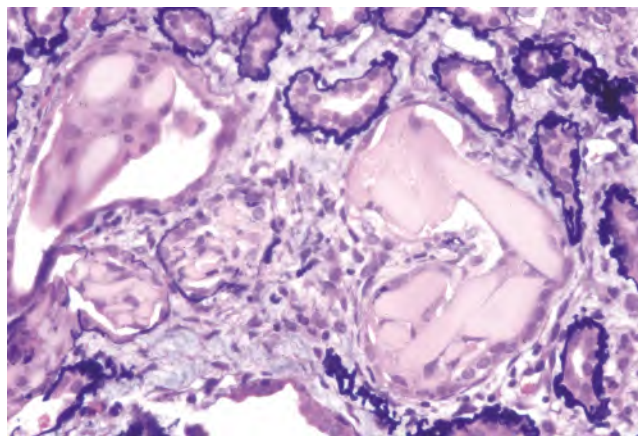


Fig. 20.32 Light-chain cast nephropathy. Note the angulated and tubular casts, surrounded by macrophages, including multinucleated cells.

of increased blood pressure. In this chapter, we discuss nephrosclerosis and renal artery stenosis, lesions associated with hypertension, and sundry lesions involving mostly smaller vessels of the kidney.

Nephrosclerosis

Nephrosclerosis is the term used for the constellation of non-specific pathologic features (glomerulosclerosis, tubulointerstitial scarring, arteriosclerosis, and arteriolosclerosis); it is strongly associated with hypertension. Nephrosclerosis is also associated with advanced age, is more frequent in people of African descent than in people of Northern European descent, and may be seen in the absence of hypertension.

Pathogenesis

Two processes participate in the arterial lesions:

- Medial and intimal thickening, a response to hemodynamic changes (e.g., hypertension), aging, genetic defects, or some combination of these
- Hyalinization of arteriolar walls, caused by extravasation of plasma proteins through injured endothelium, which may also be caused by hypertension

Because of thickened walls, the affected vessels have narrowed lumens, which results in focal parenchymal ischemia. Ischemia leads to glomerulosclerosis and chronic tubulointerstitial injury, and it produces a reduction in functional renal mass.

MORPHOLOGY

The kidneys are either normal or moderately reduced in size, with average weights between 110 and 130 g. The cortical surfaces have a fine, even granularity that resembles grain leather (Fig. 20.33). The loss of mass is due mainly to **cortical scarring and shrinking**.

On histologic examination, there is narrowing of the lumens of arterioles and small arteries, caused by thickening and hyalinization of the walls (**hyaline arteriolosclerosis**) (Fig. 20.34). Corresponding to the finely granular surface are microscopic subcapsular scars with sclerotic glomeruli and tubular drop-out, alternating with better preserved parenchyma. In addition, the interlobular and arcuate arteries show medial hypertrophy, replication of the internal elastic lamina, and increased myofibroblastic tissue in the intima, all of which narrow the lumen. This change, called **fibroelastic hyperplasia**, often accompanies hyaline arteriolosclerosis and increases in severity with age and in the presence of hypertension.

Consequent to the vascular narrowing, there is patchy ischemic atrophy, which consists of (1) foci of **tubular atrophy and interstitial fibrosis** and (2) a variety of **glomerular alterations**. The latter include collapse of the GBM, deposition of collagen within the Bowman space, periglomerular fibrosis, and total sclerosis of glomeruli. When the ischemic changes are pronounced and affect large areas of parenchyma, they can produce wedge-shaped infarcts or regional scars with histologic alterations that may resemble those in renal ablation injury, mentioned earlier.

Clinical Features

It is unusual for uncomplicated nephrosclerosis to cause uremia. However, two groups of hypertensive patients with nephrosclerosis are at increased risk of developing renal failure: people of African descent, and those with severe blood pressure elevations. In these groups, the development of the malignant or accelerated phase of hypertension often correlates with rapid renal failure.

Severe (Malignant) Hypertension. A small percentage of hypertensive persons (as many as 5%) show a rapidly rising blood pressure that, if untreated, leads to death within 1 to 2 years. This form of hypertension,

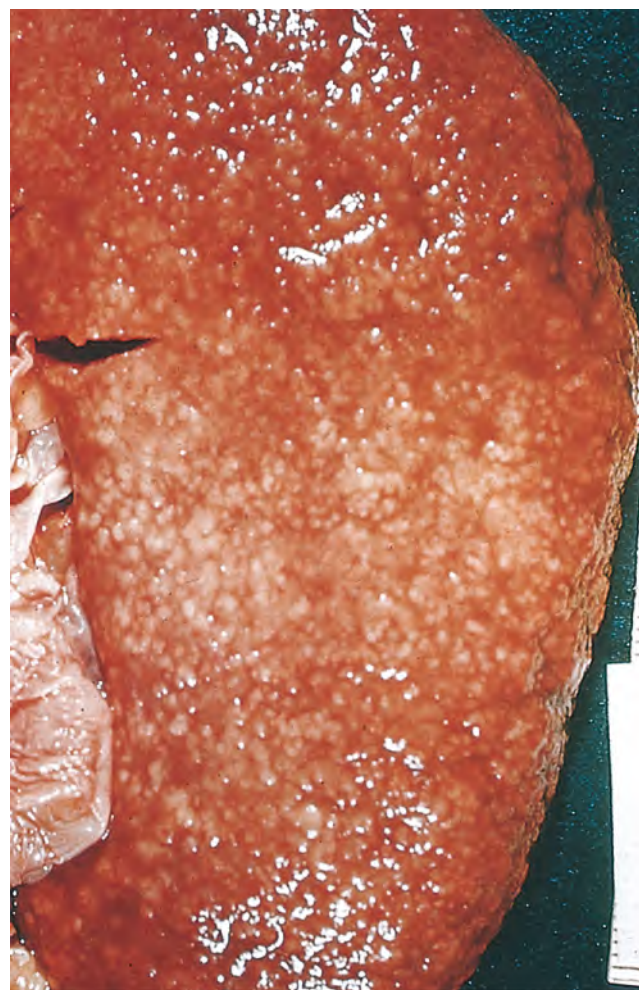


Fig. 20.33 Nephrosclerosis. Close-up of the cortical surface illustrating the fine, leathery granularity of the surface.

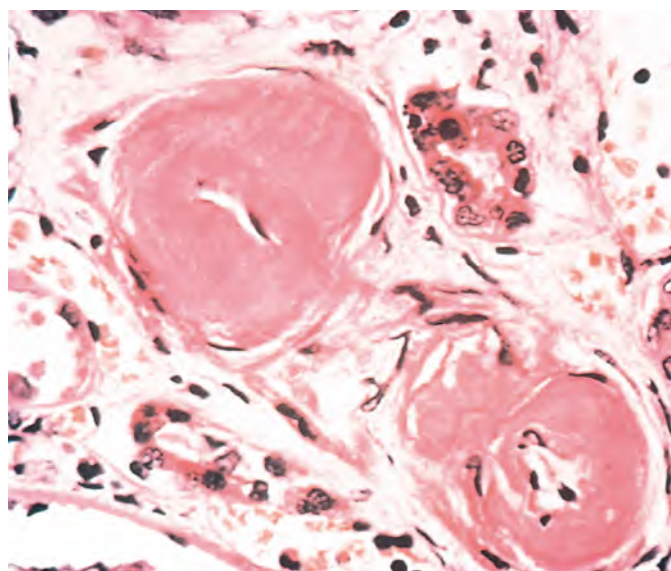


Fig. 20.34 Hyaline arteriolosclerosis. High-power view of two arterioles with hyaline deposition, marked thickening of the walls, and a narrowed lumen. (Courtesy Dr. MA Venkatachalam, Department of Pathology, University of Texas Health Sciences Center, San Antonio.)

called *severe (malignant) hypertension*, is characterized by severe pressure elevations (i.e., systolic pressure more than 180 mm Hg, diastolic pressure more than 120 mm Hg), renal failure, and retinal hemorrhages and exudates, with or without papilledema (swelling of the optic nerve that reflects increased intracranial pressures). There is clinical and morphologic overlap between the renal lesions of severe hypertension and thrombotic microangiopathy (TMA, discussed later). In both, endothelial injury and microvascular disease occur, which may result in microangiopathic hemolytic anemia (Chapter 14). However, pathogenetically, TMA and hypertension are distinct, as the molecular defects characteristic of TMA are not seen in hypertension.

MORPHOLOGY

In severe (malignant) hypertension, interlobular arteries and larger arterioles are marked by proliferation of intimal cells producing an “onion skin” appearance (Fig. 20.35). This lesion, called hyperplastic arteriolosclerosis, may cause obliteration of small arteries and arterioles. Fibrinoid necrosis may involve arterioles and glomeruli, sometimes with microthrombi within the glomeruli.

KEY CONCEPTS

Nephrosclerosis

- Nephrosclerosis, which is commonly associated with hypertension, is defined by the presence of varying degrees of glomerulosclerosis, interstitial fibrosis and tubular atrophy, arteriosclerosis, and arteriolosclerosis.
- Luminal reduction of the renal vasculature (arteries and arterioles) contributes to glomerulosclerosis (both global and segmental), which can subsequently cause interstitial fibrosis and tubular atrophy.

Renal Artery Stenosis

Unilateral renal artery stenosis is responsible for 2% to 5% of hypertension and is important to recognize because it is potentially curable by surgery. Furthermore, important insights into renal mechanisms of

hypertension came from studies of experimental and human renal artery stenosis.

Pathogenesis

Hypertension secondary to renal artery stenosis is caused by increased production of renin from the ischemic kidney. The classic experiments of Goldblatt and colleagues showed that constriction of one renal artery in dogs results in hypertension and that the magnitude of the effect is proportional to the amount of narrowing. Elevation in blood pressure, at least initially, is due to stimulation of renin secretion by the juxtaglomerular apparatus and the subsequent production of the vasoconstrictor angiotensin II. A large proportion of individuals with renovascular hypertension have elevated renin levels, and almost all show a reduction of blood pressure when given drugs that block angiotensin II activity. Other factors, however, may contribute to the maintenance of renovascular hypertension after the renin-angiotensin system has initiated it, including sodium retention.

MORPHOLOGY

The most common cause of renal artery stenosis (70% of cases) is narrowing at the origin of the renal artery by an **atheromatous plaque**. This occurs more frequently in males, and the incidence increases with advancing age and diabetes. The plaque is usually concentric, and superimposed thrombosis often occurs.

The second most frequent cause of stenosis is **fibromuscular dysplasia** of the renal artery. This heterogeneous entity is characterized by fibrous or fibromuscular thickening that may involve the intima, the media, or the adventitia of the artery (Fig. 20.36). Fibromuscular dysplasia is more common in females and tends to occur in younger age groups (i.e., in the third and fourth decades).

The ischemic kidney is reduced in size and shows diffuse atrophy, with crowded glomeruli, atrophic tubules, interstitial fibrosis, and focal inflammatory infiltrates. The arterioles in the ischemic kidney are usually protected from the effects of high pressure, thus showing only mild arteriolosclerosis. In contrast, the contralateral nonischemic kidney may show more severe arteriolosclerosis, depending on the severity of the hypertension.

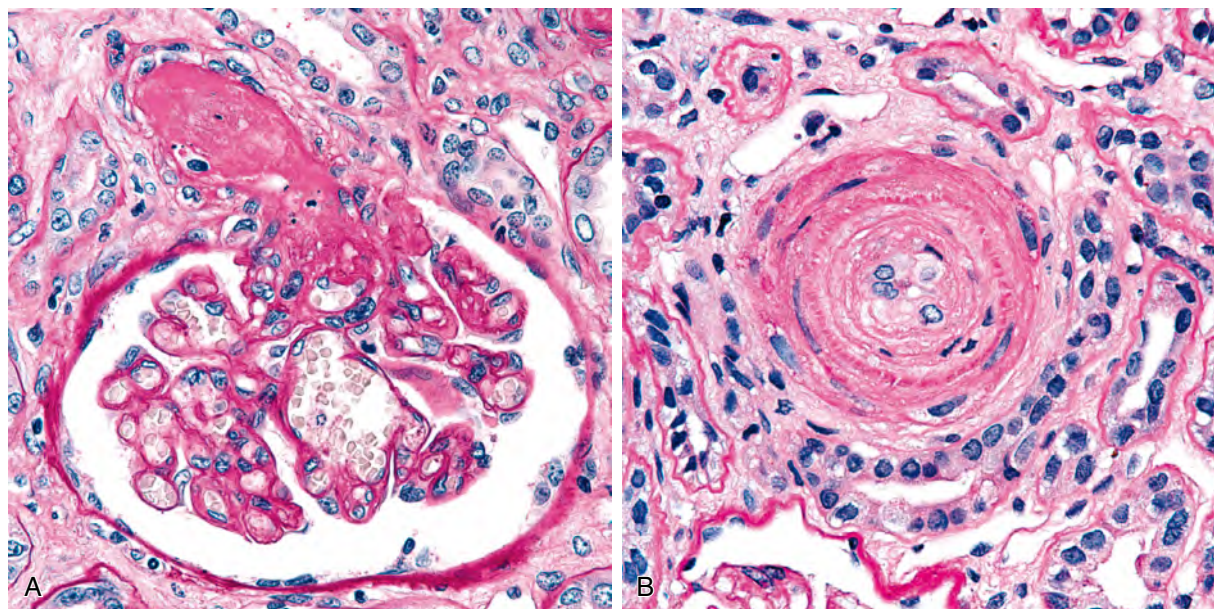


Fig. 20.35 Malignant hypertension. (A) Fibrinoid necrosis of afferent arteriole. (B) Hyperplastic arteriolosclerosis (onion-skin lesion), typically seen in cases with long-standing hypertension.

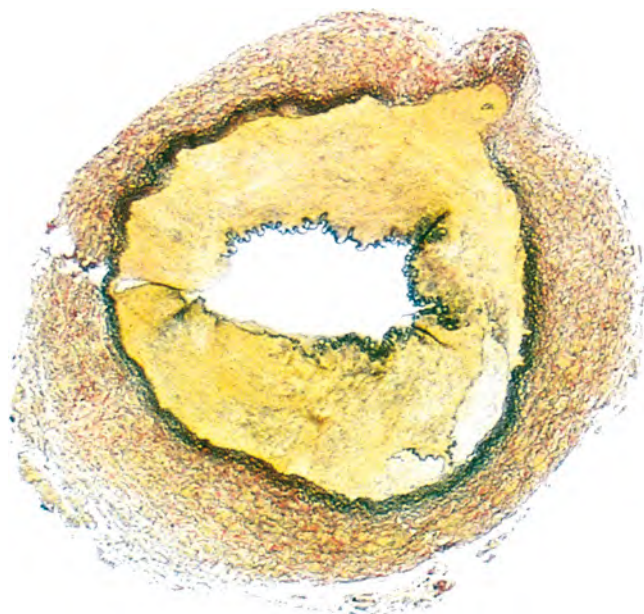


Fig. 20.36 Fibromuscular dysplasia of the renal artery, medial type (elastic tissue stain). The media shows marked fibrous thickening, and the lumen is stenotic. (Courtesy Dr. Seymour Rosen, Beth Israel Hospital, Boston.)

Clinical Features

Few distinctive features suggest the presence of renal artery stenosis, and in general, these patients resemble those with essential hypertension. On occasion, a bruit can be heard on auscultation of the affected kidneys. Elevated plasma or renal vein renin, response to angiotensin-converting enzyme inhibitor, renal scans, and intravenous pyelography may aid with diagnosis, but arteriography is required to localize the stenotic lesion. The cure rate after surgery is 70% to 80% in well-selected cases.

Thrombotic Microangiopathies

The term *thrombotic microangiopathy* encompasses a spectrum of clinical syndromes that include thrombotic thrombocytopenic purpura (TTP) and hemolytic uremic syndrome (HUS). As discussed in Chapter 14, **HUS and TTP are caused by diverse mechanisms that lead to the formation of thrombi in capillaries and/or arterioles in various tissue beds, including those of the kidney (Fig. 20.37).** The thrombi cause tissue ischemia and organ dysfunction. Thrombocytopenia is typically more pronounced in TTP than HUS.

This group of disorders is classified according to the current understanding of their causes or associations (Table 20.10):

- **Typical HUS (epidemic, classic, diarrhea-positive HUS)** is most frequently associated with consumption of food contaminated by bacteria producing Shiga-like toxins.
- **Atypical HUS (nonepidemic, diarrhea-negative HUS)** is associated with:
 - Inherited mutations of or autoantibodies targeting complement-regulatory proteins; now called *complement-mediated TMA*
 - Diverse acquired causes of endothelial injury, including antiphospholipid antibodies; complications of pregnancy; vascular renal diseases such as scleroderma and severe hypertension; chemotherapeutic and immunosuppressive drugs; and radiation
- **TTP** is characterized by inherited or acquired deficiencies of ADAMTS13, a plasma metalloprotease that cleaves large multimers of von Willebrand factor (vWF).

TABLE 20.10 Major Forms of Thrombotic Microangiopathy

	Forms	Etiology
Shiga toxin–mediated HUS	Acquired	Shiga-like toxin–producing <i>E. coli</i> , <i>Shigella dysenteriae</i> serotype 1
Atypical HUS (complement-mediated TMA)	Inherited	Complement dysregulation due to mutations in complement regulatory proteins (relatively common)
	Acquired	Complement dysregulation due to autoantibodies against complement regulatory proteins Various causes of endothelial injury
TTP	Inherited	Genetic ADAMTS13 deficiency (rare)
	Acquired	ADAMTS13 deficiency due to autoantibodies (relatively common)

ADAMTS13, A disintegrin and metalloproteinase with thrombospondin type 1 motif, member 13; *HUS*, hemolytic uremic syndrome; *TMA*, thrombotic microangiopathy; *TTP*, thrombotic thrombocytopenic purpura.

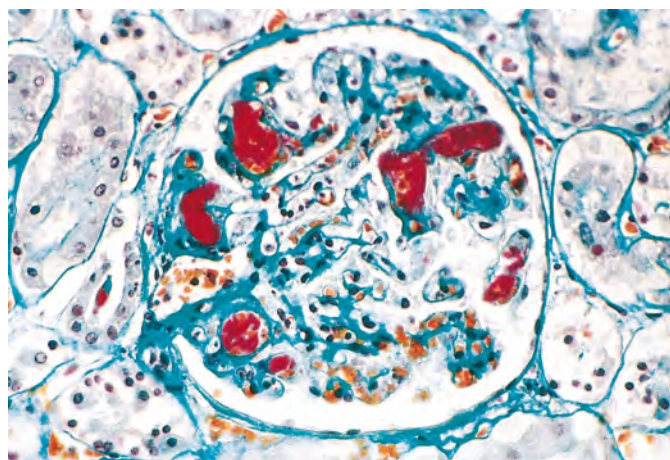


Fig. 20.37 Thrombotic microangiopathy. Masson trichrome stain showing thrombi (red) in the glomerular capillaries.

Pathogenesis

The pathogenesis of thrombotic microangiopathies (TMAs) is discussed in Chapter 14 and briefly described here. Within the TMAs, two pathogenetic triggers dominate: (1) *endothelial injury* and (2) *excessive platelet activation and aggregation*. As we will discuss, endothelial injury appears to be the primary mechanism of HUS, whereas platelet activation may be the inciting event in TTP.

- **Endothelial injury:** In typical HUS, the trigger for endothelial injury and activation is usually a Shiga-like toxin, while for inherited forms of atypical HUS (complement-mediated TMA), the cause of the endothelial injury is excessive activation of complement. The endothelial injury in HUS appears to cause platelet activation and thrombosis within microvascular beds.
- **Platelet aggregation:** In contrast to HUS, the initiating event in TTP appears to be platelet aggregation induced by very large multimers of vWF, which accumulate due to a deficiency of ADAMTS13, a plasma protease that cleaves vWF multimers into smaller sizes. The

deficiency of ADAMTS13 is most often caused by autoantibodies that inhibit ADAMTS13 function and less commonly, inherited deficiencies of ADAMTS13. Very large vWF multimers can bind platelet surface glycoproteins and activate platelets spontaneously, providing a pathophysiologic explanation for the microthrombi that are observed in vascular beds.

With this introduction, we now briefly delve into the various subtypes of HUS/TTP and then return to the morphologic features that are common to all.

Typical (Epidemic, Classic, Diarrhea-Positive) Hemolytic Uremic Syndrome

This is the best-characterized form of HUS. Most cases occur after intestinal infection with strains of *E. coli* (the most common being O157:H7) that produce Shiga-like toxins, so-called because they resemble those made by *Shigella dysenteriae* (Chapter 17). Less commonly, infections by other agents, including *S. dysenteriae*, can give rise to a similar disorder.

Precisely how Shiga-like toxin causes HUS is not well understood. According to one model, the toxin activates endothelial cells, which respond by increasing their expression of leukocyte adhesion molecules and endothelin and decreasing nitric oxide production. In the presence of cytokines such as TNF, Shiga-like toxin may cause endothelial apoptosis. These alterations lead to platelet activation and induce vasoconstriction, resulting in the characteristic microangiopathy, but other possibilities remain. For example, there is evidence that Shiga-like toxins may bind the regulatory complement protein Factor H and inhibit its activity, causing hyperactivation of complement, an intriguing idea given the clear role of complement activation in atypical HUS (described in the next section).

In typical HUS, if the renal failure is less severe, most patients recover renal function in a matter of weeks. However, due to underlying renal damage, the long-term (15- to 25-year) outlook is more guarded. In one study, only 10 of 25 patients with prior typical HUS had normal renal function.

Atypical (Nonepidemic, Diarrhea-Negative) Hemolytic Uremic Syndrome

Atypical HUS occurs in several different settings. More than half of those affected have identifiable mutations of complement-regulatory proteins, most commonly Factor H, which promotes breakdown of the alternative pathway C3 convertase and protects cells from damage by uncontrolled complement activation (Chapter 3). Other patients have mutations in other complement regulatory proteins, such as complement Factor I and CD46 (membrane cofactor protein). Having multiple mutations in complement-regulatory proteins may cause atypical HUS to manifest at a younger age. Autoantibodies that block the activity of complement-regulatory proteins are another cause of atypical HUS.

The remaining cases of atypical HUS arise in association with a variety of miscellaneous conditions or exposures. These include:

- The *antiphospholipid syndrome*, described in detail in Chapter 4.
- Complications of pregnancy or the postpartum period. So-called *postpartum renal failure* is a form of TMA that can occur after pregnancy, 1 day to several months after delivery. The underlying mechanism is unclear; some patients may have a mutation in a complement regulatory protein. The condition has a grave prognosis, although recovery can occur in milder cases.
- *Vascular diseases affecting the kidney*, such as systemic sclerosis.

Patients with atypical HUS do not fare as well as those with typical HUS, in large part because the underlying triggers may be genetic or chronic conditions that are difficult to treat. As in typical HUS, some patients have neurologic symptoms; the disease in these patients can be distinguished from TTP by normal ADAMTS13 levels in the plasma (see later).

Thrombotic Thrombocytopenic Purpura

TTP is classically manifested by the pentad of fever, neurologic symptoms, microangiopathic hemolytic anemia, thrombocytopenia, and renal dysfunction. The most common cause of deficient ADAMTS13 activity is inhibitory autoantibodies; less commonly, patients inherit an inactivating mutation in *ADAMTS13*.

For unknown reasons, in TTP, central nervous system involvement is the dominant feature, whereas severe renal injury is uncommon. The clinical findings are dictated by the distribution of the microthrombi, which are found in arterioles throughout the body. With plasma exchange, which removes autoantibodies and provides functional ADAMTS13, TTP (which once was uniformly fatal) can be treated successfully in more than 80% of patients.

MORPHOLOGY

The morphologic findings in the various forms of HUS/TTP are indistinguishable and vary mainly in chronicity. In acute, active disease, the kidney may show patchy or diffuse cortical necrosis and subcapsular petechiae. The glomerular capillaries are distended and occluded by thrombi. Disruption of the mesangial matrix and damage to the mesangial cells often results in destruction of the mesangium. Interlobular arteries and arterioles often show occlusive thrombi. Chronic disease is common in patients with atypical HUS and has features that stem from continued injury and attempts at healing. The renal cortex reveals various degrees of scarring. By light microscopy, the glomeruli are shrunken and have marked thickening of the capillary walls associated with duplication of the basement membrane (so-called double contours or tram tracks). The walls of arteries often exhibit increased layers of cells and connective tissue that narrow the vessel lumens. When chronic features predominate, the arteriolar vessel walls may be thickened, with prominent hyalinosis. These changes lead to persistent hypoperfusion and potentially diffuse cortical necrosis, which manifests clinically as renal failure. The gross alterations of massive ischemic necrosis are sharply limited to the cortex (Fig. 20.38). The histologic appearance is that of acute ischemic infarction. The lesions may be patchy, with areas of coagulative necrosis and apparently better preserved cortex.

KEY CONCEPTS

Thrombotic Microangiopathy

- Thrombotic microangiopathy encompasses a diverse set of conditions that all lead to deposition of thrombi in the microvasculature, accompanied by red cell hemolysis, tissue ischemia and organ dysfunction, and a consumptive thrombocytopenia.
- In typical HUS, Shiga-like toxin produced by bacteria, most commonly *E. coli* strain O157:H7, is responsible for producing endothelial damage and platelet activation and thrombosis.
- In most cases of atypical HUS, aberrant activation of complement due to inherited mutations or acquired autoantibodies is the key pathogenic abnormality.
- In TTP, deficiency of ADAMTS13, a negative regulator of vWF, caused by antibodies or mutations, permits the formation of abnormally large multimers of vWF that are capable of activating platelets.



Fig. 20.38 Diffuse cortical necrosis. The pale ischemic necrotic areas are confined to the cortex and columns of Bertin.



Fig. 20.39 Atheroembolus with typical cholesterol clefts in an interlobular artery.

Other Vascular Disorders

Atherosclerotic Ischemic Renal Disease

We have discussed earlier that unilateral atherosclerotic renal artery stenosis can lead to hypertension. *Bilateral renal artery disease*, usually diagnosed definitively by arteriography, is a fairly common cause of chronic ischemia with renal insufficiency in older individuals, sometimes in the absence of hypertension. The importance of recognizing this condition is that surgical revascularization can prevent further decline in renal function.

Atheroembolic Renal Disease

Embolization of fragments of atheromatous plaques from the aorta or renal artery into intrarenal vessels occurs in older adults with severe atherosclerosis, especially after surgery on the abdominal aorta, aortography, or intraaortic cannulization. These emboli can be recognized in the lumens of arcuate and interlobular arteries by their content of cholesterol crystals, which appear as rhomboid clefts (Fig. 20.39). The clinical consequences of atheroemboli vary according to the number of emboli and the preexisting state of renal function. Frequently they are of no significance. However, acute renal injury or failure may develop in older adults in whom renal function is already compromised.

Sickle Cell Nephropathy

Sickle cell disease (homozygous) or trait (heterozygous) may lead to a variety of alterations in renal morphology and function, some of which

produce clinically significant abnormalities. The various manifestations are grouped under *sickle cell nephropathy*.

The most common abnormalities are hematuria and a diminished concentrating ability (*hyposthenuria*). These are thought to be due to accelerated sickling in the hypertonic hypoxic milieu of the renal medulla; the hyperosmolarity dehydrates red cells and increases intracellular sickle hemoglobin (HbS) concentrations, which likely explains why even those with sickle cell trait are affected. Patchy papillary necrosis may occur in both homozygotes and heterozygotes; this is sometimes associated with cortical scarring. Proteinuria is also common in sickle cell disease, occurring in about 30% of patients. It is usually mild to moderate, but on occasion the overt nephrotic syndrome arises, associated with sclerosing glomerular lesions.

Renal Infarcts

The kidneys are common sites for the development of infarcts. Contributing to this predisposition is the extensive blood flow to the kidneys (one-quarter of the cardiac output), but probably more important is the limited collateral circulation from extrarenal sites (small blood vessels penetrating from the renal capsule supply only the very outer rim of the cortex). Although thrombosis in advanced atherosclerosis and the acute vasculitis of polyarteritis nodosa may occlude arteries, most infarcts are due to embolism. A major source of such emboli is mural thrombosis in the left atrium and ventricle because of myocardial infarction. Vegetative endocarditis, aortic aneurysms, and aortic atherosclerosis are less frequent sources of emboli.

MORPHOLOGY

Because of the lack of a collateral blood supply, most renal infarcts are of the “white” anemic variety. Within 24 hours, infarcts become sharply demarcated, pale, yellow-white areas that may contain small irregular foci of hemorrhagic discoloration. They are usually ringed by a zone of intense hyperemia.

The infarcts are wedge-shaped, with the base against the cortical surface and the apex pointing toward the medulla. In time, these acute areas of ischemic necrosis undergo progressive fibrous scarring, giving rise to depressed, pale, gray-white scars that assume a V-shape on section. The histologic changes in renal infarcts are those of ischemic coagulative necrosis, described in Chapter 2.

Many renal infarcts are clinically silent. Sometimes, pain with tenderness localized to the costovertebral angle occurs, associated with showers of red cells in the urine. Large infarcts of one kidney are probably associated with narrowing of the renal artery or one of its major branches, which, in turn, may cause hypertension.

Chronic Kidney Disease

Chronic kidney disease is a clinical term that evolved from chronic renal failure and represents the progression of any glomerular, tubulointerstitial, or vascular injury that results in nephron loss. Typically, there is a constellation of nonspecific findings, which include glomerulosclerosis, interstitial fibrosis, tubular atrophy, and arteriosclerosis. Hyperfiltration of surviving glomeruli occurs when there is a reduced number of functional nephrons, which can lead to further progression toward end-stage kidney disease. Chronic kidney disease is more common in those with African ancestry, possibly due, at least in part, to high-risk *APOL1* gene variants

CONGENITAL AND DEVELOPMENTAL ANOMALIES

About 10% of people are born with significant malformations of the urinary system. Renal dysplasias and hypoplasias account for 20% of chronic kidney disease in children.

Congenital renal disease can be hereditary but most often results from an acquired developmental defect during gestation. As discussed in Chapter 10, defects in genes involved in normal renal development, including the Wilms tumor–associated genes, understandably cause urogenital anomalies. As a rule, the resulting developmental abnormalities involve structural components of both the kidney and urinary tract. Other genetic defects primarily produce functional abnormalities in tubular transport, such as cystinuria and renal tubular acidosis. Here, we restrict the discussion to structural anomalies primarily involving the kidney. All except horseshoe kidney are uncommon. Anomalies of the lower urinary tract are discussed in Chapter 21.

Agenesis of the Kidney

Bilateral agenesis is incompatible with life and usually encountered in stillborn infants. It is often associated with other congenital disorders (e.g., limb defects, hypoplastic lungs). Unilateral agenesis is uncommon and compatible with normal life if no other abnormalities exist. The solitary kidney enlarges because of compensatory hypertrophy. Some patients eventually develop progressive glomerular sclerosis in the remaining kidney as a result of the adaptive changes in hypertrophied nephrons, discussed earlier in the chapter, and, in time, chronic kidney disease ensues.

Hypoplasia

Hypoplasia refers to failure of the kidneys to develop to a normal size. This anomaly may occur bilaterally, resulting in renal failure in early childhood, but it is more commonly encountered as a unilateral defect. True renal hypoplasia is observed in low-birth-weight infants and may contribute to their increased lifetime risk for chronic kidney disease.

Differentiation between congenital and acquired atrophic kidneys may be impossible, but a truly hypoplastic kidney shows no scars and has a reduced number of renal lobes and pyramids, usually six or fewer.

Ectopic Kidneys

The development of the metanephros into the kidneys may occur in ectopic foci. These kidneys lie either just above the pelvic brim or sometimes within the pelvis. They are usually normal or slightly small in size but otherwise are not remarkable. Because of their abnormal position, kinking or tortuosity of the ureters may cause obstruction to urinary flow, which predisposes to bacterial infections.

Horseshoe Kidneys

Fusion of the upper (10%) or lower poles (90%) of the kidneys produces a horseshoe-shaped structure that is continuous across the midline anterior to the great vessels. This anomaly is found in 1 in 500 to 1000 autopsies.

CYSTIC DISEASES OF THE KIDNEY

Cystic diseases of the kidney are heterogeneous, comprising hereditary, developmental, and acquired disorders. They are important for several reasons: (1) They are reasonably common and often represent diagnostic problems for clinicians, radiologists, and pathologists; (2) some forms, such as adult polycystic kidney disease, are major causes of chronic kidney disease; and (3) they can occasionally be confused with malignant tumors. A useful classification of renal cysts is in [Table 20.11](#).

Autosomal Dominant (Adult) Polycystic Kidney Disease

Autosomal dominant (adult) polycystic kidney disease is a hereditary disorder characterized by multiple expanding cysts of both kidneys that ultimately destroy the renal parenchyma and cause

TABLE 20.11 Summary of Renal Cystic Diseases

Disease	Inheritance	Pathologic Features	Clinical Features or Complications	Typical Outcome	Diagrammatic Representation
Adult polycystic kidney disease	Autosomal dominant	Large multicystic kidneys, liver cysts, berry aneurysms	Hematuria, flank pain, urinary tract infection, renal stones, hypertension	Chronic renal failure beginning at 40–60 years of age	
Childhood polycystic kidney disease	Autosomal recessive	Enlarged, cystic kidneys at birth	Hepatic fibrosis	Variable, death in infancy or childhood	
Medullary sponge kidney	None	Medullary cysts on excretory urography	Hematuria, urinary tract infection, recurrent renal stones	Benign	
Familial juvenile nephronophthisis	Autosomal recessive	Corticomedullary cysts, shrunken kidneys	Salt wasting, polyuria, growth retardation, anemia	Progressive renal failure beginning in childhood	
Multicystic renal dysplasia	None	Irregular kidneys with cysts of variable size	Association with other renal anomalies	Renal failure if bilateral, surgically curable if unilateral	
Acquired renal cystic disease	None	Cystic degeneration in end-stage kidney disease	Hemorrhage, erythrocytosis, neoplasia	Dependence on dialysis	
Simple cysts	None	Single or multiple cysts in normal-sized kidneys	Microscopic hematuria	Benign	

renal failure. It is a common condition affecting roughly 1 in 400 to 1000 live births and accounting for about 5% to 10% of cases of ESKD requiring transplantation or dialysis. Although susceptibility to this disease is inherited as an autosomal dominant trait, both alleles of the involved genes must be nonfunctional for cyst development to occur. The cysts are clonal, each cell having one inherited mutation and acquired somatic mutations that interfere with the function of the second allele. The disease is always bilateral; reported unilateral cases probably represent multicystic dysplasia. The cysts initially involve a minority of the nephrons, so renal function is retained until about the fourth or fifth decade of life.

Genetics and Pathogenesis

Almost all cases of adult polycystic kidney disease are caused by mutations in one of two genes, *PKD1* or *PKD2*.

- The *PKD1* gene encodes a large (460-kD) integral membrane protein named *polycystin-1*, which has a large extracellular region, multiple transmembrane domains, and a short cytoplasmic tail. Polycystin-1 is expressed in tubular epithelial cells, particularly those of the distal nephron. Its precise function is not known, but it contains domains that are usually involved in cell-cell and cell-matrix interactions. Mutations in *PKD1* account for about 85% of cases. In individuals with these mutations, the likelihood of developing renal failure is less than 5% by 40 years of age but rises to more than 35% by 50 years of age, exceeds 70% by 60 years of age, and is more than 95% by 70 years of age.
- The *PKD2* gene accounts for most of the remaining cases of polycystic disease. Its product, *polycystin-2*, is an integral membrane protein that is expressed in all segments of the renal tubules and in many extrarenal tissues. Polycystin-2 functions as a Ca^{2+} -permeable cation channel. Overall, the disease is less severe than that associated with *PKD1* mutations. Renal failure occurs in less than 5% of patients with *PKD2* mutations at 50 years of age, but this rises to 15% at 60 years of age and 45% at 70 years of age.

The pathogenesis of polycystic disease is not established, but the currently favored hypothesis places the cilia-centrosome complex of tubular epithelial cells at the center of the disorder (Fig. 20.40). The tubular epithelial cells of the kidney contain a single nonmotile primary cilium, a 2- to 3- μm -long hairlike organelle that projects into the tubular lumen from the apical surface of the cells. The cilium is made up of microtubules, and it arises from and is attached to a basal body derived from the centriole. The cilia are part of a system of organelles and cellular structures that sense mechanical signals. The apical cilia in the kidney tubule function as a mechanosensor to monitor changes in fluid flow and shear stress, while intercellular junctional complexes monitor forces between cells, and focal adhesions sense attachment to extracellular matrices. In response to external signals, these sensors regulate ion flux (cilia can induce Ca^{2+} flux in cultured kidney epithelial cells) and cellular behavior, including cell polarity and proliferation. The idea that defects in mechanosensing, Ca^{2+} flux, and signal transduction underlie cyst formation is supported by several observations.

- Both polycystin-1 and polycystin-2 are localized to the primary cilium.
- Other genes that are mutated in cystic diseases (e.g., the nephrocystin genes, described later) encode proteins that also localize to cilia and/or basal bodies.
- Knockout of the *PKD1* gene in one model organism (the worm *Caenorhabditis elegans*) results in ciliary abnormalities and cyst formation.
- Tubular cells obtained from mice with a deletion of the *PKD1* gene (which causes embryonic lethality in this species) retain normal

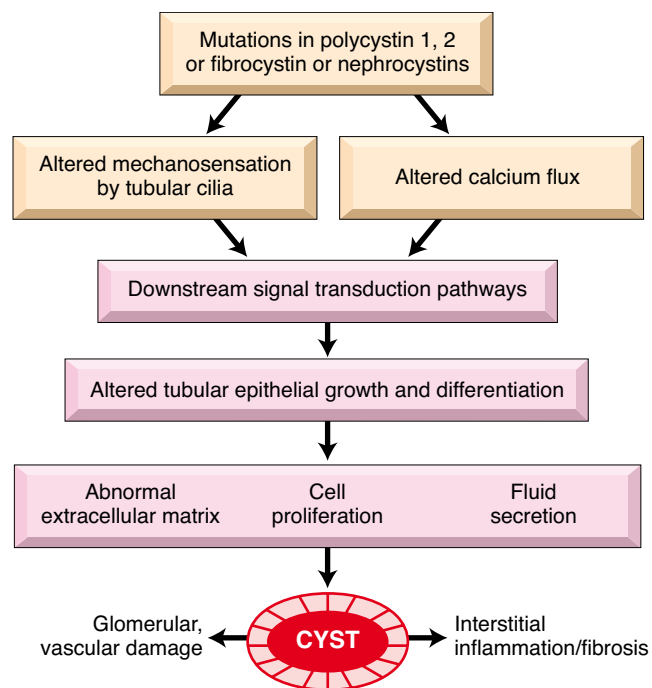


Fig. 20.40 Possible mechanisms of cyst formation in cystic kidney diseases (see text).

ciliary architecture but lack the flow-induced Ca^{2+} flux that occurs in normal tubular cells.

Polycystin-1 and polycystin-2 appear to form a complex that regulates intracellular Ca^{2+} in response to fluid flow, perhaps because fluid moving through the kidney tubules causes ciliary bending that opens Ca^{2+} channels. Mutation of either *PKD* gene leads to loss of the polycystin complex or formation of an aberrant complex. The consequent disruption of normal polycystin activity results in alterations of intracellular Ca^{2+} , which (you will recall) regulates many downstream signaling events, including pathways that directly or indirectly impact cellular proliferation, apoptosis, and secretory functions. The increase in calcium is thought to stimulate proliferation and secretion from epithelial cells lining the cysts, which together result in progressive cyst formation and enlargement. The calcium-induced signals also alter the interaction of epithelial cells with ECM, and this too is thought to contribute to the cyst formation and interstitial fibrosis that are characteristic of progressive polycystic kidney disease.

MORPHOLOGY

In gross appearance, the kidneys are bilaterally enlarged and may achieve enormous sizes; weights of as much as 4 kg for each kidney have been reported. The external surface appears to be composed solely of a mass of cysts, up to 3 to 4 cm in diameter, with no intervening parenchyma (Fig. 20.41A and B). However, microscopic examination reveals functioning nephrons dispersed between the cysts. The cysts may be filled with a clear, serous fluid or with turbid, red to brown, sometimes hemorrhagic fluid. As these cysts enlarge, they may encroach on the calyces and pelvis to produce pressure defects. The cysts arise from the tubules throughout the nephron and therefore have variable lining epithelia. On occasion, papillary epithelial formations and polyps project into the lumen.

Clinical Features

Many patients remain asymptomatic until renal insufficiency announces the presence of the disease. In others, hemorrhage or progressive dilation of cysts may produce pain. Excretion of blood clots causes renal colic. The enlarged kidneys, usually apparent on abdominal palpation, may induce a dragging sensation. The disease occasionally begins with the insidious onset of hematuria, followed by other features of progressive chronic kidney disease, such as proteinuria (rarely more than 2 g/day), polyuria, and hypertension. Patients with *PKD2* mutations tend to develop symptoms and renal failure at older ages. Both genetic and environmental factors influence disease severity. Progression is accelerated in people of African descent (particularly in those with sickle cell trait), in males, and in the presence of hypertension.

Individuals with polycystic kidney disease also frequently have extrarenal anomalies. About 40% have one to several cysts in the liver (*polycystic liver disease*) that are usually asymptomatic. The cysts are derived from biliary epithelium. Cysts occur much less frequently in the spleen, pancreas, and lungs. *Intracranial berry aneurysms*, presumably from altered expression of polycystin in vascular smooth muscle, arise in the circle of Willis, and subarachnoid hemorrhages from these account for death in 4% to 10% of individuals. *Mitral valve prolapse* and other cardiac valvular anomalies occur in 20% to 25% of patients, but most are asymptomatic. The diagnosis is made by radiologic imaging techniques.

This form of chronic kidney disease is remarkable in that patients may survive for many years with azotemia slowly progressing to uremia. Ultimately, about 40% of adult patients die of coronary or hypertensive heart disease, 25% of infection, 15% of a ruptured berry aneurysm or hypertensive intracerebral hemorrhage, and the rest of other causes.

Autosomal Recessive (Childhood) Polycystic Kidney Disease

Autosomal recessive (childhood) polycystic kidney disease is genetically distinct from adult polycystic kidney disease. *Perinatal*, *neonatal*, *infantile*, and *juvenile* subcategories have been defined, depending on the time of presentation and presence of associated hepatic lesions. The

first two are the most common; serious manifestations are usually present at birth, and affected infants may succumb to renal failure.

Genetics and Pathogenesis

In most cases, the disease is caused by mutations in the *PKHD1* gene, which is highly expressed in adult and fetal kidney, liver, and pancreas. The *PKHD1* gene encodes *fibrocystin*, a 447-kD integral membrane protein with a large extracellular region, a single transmembrane component, and a short cytoplasmic tail. Like polycystins 1 and 2, fibrocystin localizes to the primary cilium of tubular cells. The function of fibrocystin is unknown, but its structure indicates it may be a cell surface receptor with a role in collecting duct and biliary differentiation.

Analysis of autosomal recessive polycystic disease patients has revealed a wide range of different mutations. Most cases are compound heterozygotes (i.e., inherit a different mutant allele from each of the two parents), complicating molecular diagnosis of the disorder.

MORPHOLOGY

The kidneys are enlarged and have a smooth external appearance. On cut section, numerous small cysts in the cortex and medulla give the kidney a spongelike appearance. Dilated elongated channels are present at right angles to the cortical surface, completely replacing the medulla and cortex (Fig. 20.41C). On microscopic examination, there is cylindrical or, less commonly, saccular dilation of all collecting tubules. The cysts have a uniform lining of cuboidal cells, reflecting their origin from the collecting ducts. In almost all cases, the liver has cysts (Fig. 20.41D) associated with portal fibrosis and proliferation of portal bile ducts.

Patients who survive infancy (infantile and juvenile forms) may develop a peculiar hepatic injury characterized by bland periportal fibrosis and proliferation of well-differentiated biliary ductules, termed *congenital hepatic fibrosis*. In older children, hepatic disease is the predominant clinical concern. Such patients may develop portal hypertension with splenomegaly. Curiously, congenital hepatic fibrosis

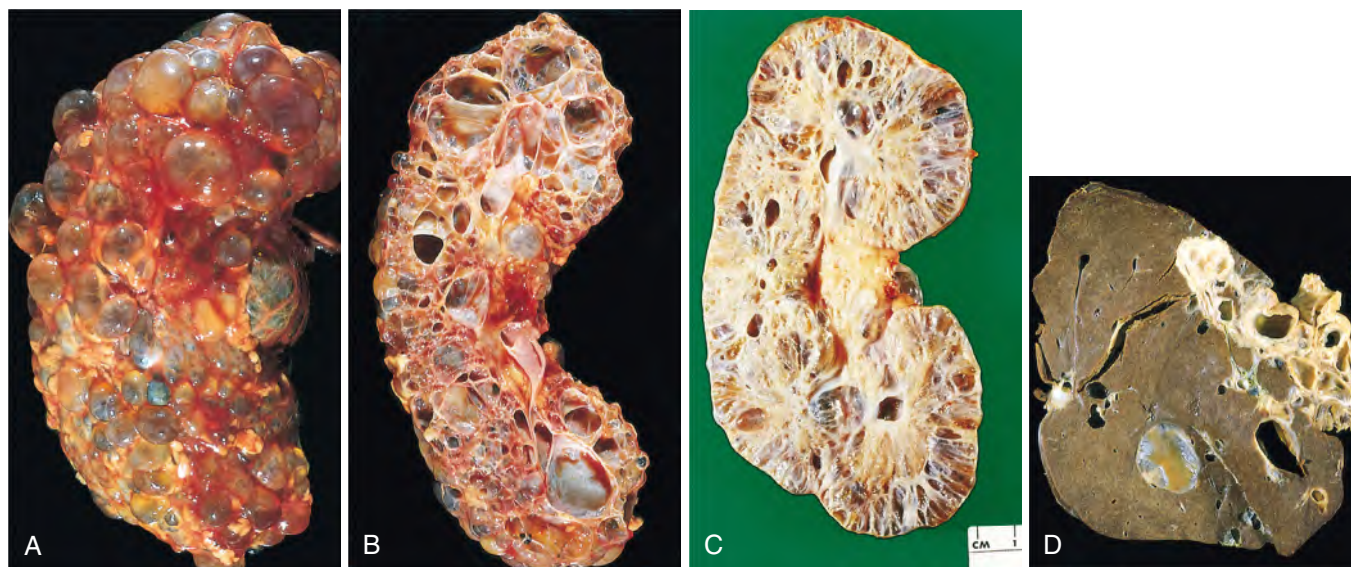


Fig. 20.41 (A–B) Autosomal dominant adult polycystic kidney disease (ADPKD) viewed from the external surface and bisected. The kidney is markedly enlarged and contains numerous dilated cysts. (C) Autosomal recessive childhood PKD, showing smaller cysts and dilated channels at right angles to the cortical surface. (D) Liver cysts in adult PKD.

sometimes occurs in the absence of polycystic kidneys or with adult-onset polycystic kidney disease.

Cystic Diseases of the Renal Medulla

The two major types of medullary cystic disease are *medullary sponge kidney*, a relatively common and usually innocuous structural change, and *nephronophthisis*, which is almost always associated with renal dysfunction.

Medullary Sponge Kidney

The term *medullary sponge kidney* is restricted to multiple cystic dilations of the collecting ducts in the medulla. The condition occurs in adults and is usually discovered radiographically. Renal function is usually normal. On gross inspection, the papillary ducts in the medulla are dilated, and small cysts may be present. The cysts are lined by cuboidal epithelium or occasionally by transitional epithelium. Unless there is superimposed pyelonephritis, cortical scarring is absent. The pathogenesis is unknown.

Nephronophthisis

This group of progressive renal disorders is characterized by variable number of cysts in the medulla, usually concentrated at the corticomedullary junction. Initial injury probably involves the distal tubules with tubular basement membrane disruption, followed by chronic and progressive tubular atrophy involving both medulla and cortex and interstitial fibrosis. Although the medullary cysts are important, the **cortical tubulointerstitial damage is the cause of the eventual renal insufficiency.**

Three variants of the nephronophthisis disease complex are recognized: (1) sporadic, nonfamilial; (2) familial juvenile nephronophthisis (most common); and (3) renal-retinal dysplasia (15%) in which the kidney disease is accompanied by ocular lesions. The familial forms are inherited as autosomal recessive traits and usually become manifest in childhood or adolescence. As a group, the nephronophthisis complex is the most common genetic cause of ESKD in children and young adults.

Children affected with nephronophthisis present first with polyuria and polydipsia, which reflect a marked defect in the concentrating ability of renal tubules. Sodium wasting and tubular acidosis are also prominent. The expected course is progression to ESKD in 5 to 10 years.

Genetics and Pathogenesis

Mutations in sixteen genes loci, *NPHP1* to *NPHP11* (that encode proteins called nephrocystins), *JBTS2*, *JBTS3*, *JBTS9*, and *JBTS11*, are implicated in the various juvenile forms of nephronophthisis, and the list continually expands as additional loci that contribute to this ciliopathy are identified. These proteins are present in the primary cilia, basal bodies attached to these cilia, or the centrosome organelle from which the basal bodies originate. The *NPHP2* gene product has been identified as *inversin* (or nephrocystin-2), which mediates left-right patterning during embryogenesis.

MORPHOLOGY

In nephronophthisis, the kidneys are small, have contracted granular surfaces, and show cysts in the medulla, most prominently at the corticomedullary junction. Small cysts are also seen in the cortex. The cysts are lined by flattened or cuboidal epithelium and are usually surrounded by either inflammatory cells or fibrous tissue. In the cortex, there is widespread atrophy and thickening of the tubular basement membranes, together with interstitial fibrosis. In general, glomerular structure is preserved.

There are few specific clues to diagnosis, because the medullary cysts might be too small to be visualized radiographically. The disease should be strongly considered in children or adolescents with otherwise unexplained chronic renal failure, a positive family history, and chronic tubulointerstitial nephritis on biopsy.

Multicystic Renal Dysplasia

Renal dysplasia is a sporadic disorder that can be unilateral or bilateral and is often cystic. The kidney is usually enlarged, extremely irregular, and multicystic (Fig. 20.42A). The cysts vary in size from several millimeters to centimeters in diameter. On histologic examination, they are lined by flattened epithelium. Although normal nephrons are present, the characteristic histologic feature is the presence of islands of undifferentiated mesenchyme, often with cartilage, and immature collecting ducts (Fig. 20.42B). Most cases are associated with ureteropelvic obstruction, ureteral agenesis or atresia, and other anomalies of the lower urinary tract.

When unilateral, the dysplasia may mimic a neoplasm and lead to surgical exploration and nephrectomy. The opposite kidney functions normally, and such patients have an excellent prognosis after surgical removal of the affected kidney. In bilateral multicystic renal dysplasia, renal failure may ultimately result.

Acquired Cystic Disease

Patients with ESKD who have undergone prolonged dialysis develop numerous cortical and medullary renal cysts. The cysts measure 0.1 to

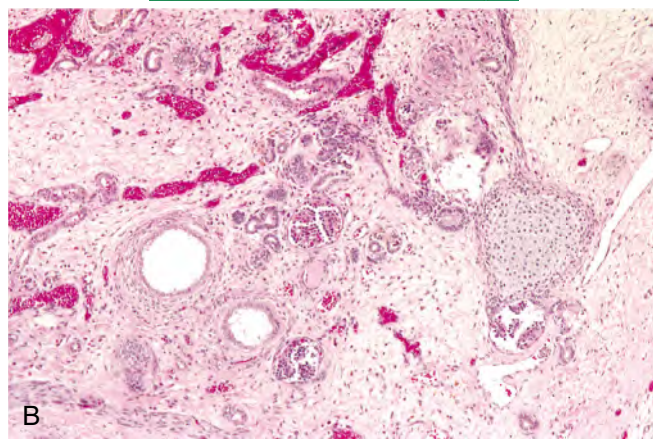


Fig. 20.42 Multicystic renal dysplasia. (A) Gross appearance. (B) Histologic section showing disorganized architecture, dilated tubules with cuffs of primitive stroma, and an island of cartilage (hematoxylin and eosin stain). (A, Courtesy Dr. D Schofield, Children's Hospital, Los Angeles. B, courtesy Dr. Laura Finn, Children's Hospital, Seattle.)

4 cm in diameter, contain clear fluid, are lined by either hyperplastic or flattened tubular epithelium, and often contain calcium oxalate crystals. They probably form as a result of obstruction of tubules by interstitial fibrosis or by oxalate crystals. Most are asymptomatic, but sometimes the cysts bleed, causing hematuria. There is up to 100-fold increased risk of renal cell carcinoma, which develops in 7% of patients observed for 10 years.

Simple Cysts

Simple cysts may be single or multiple and usually involve the cortex. They are commonly 1 to 5 cm but may reach 10 cm or more in size. They are translucent, lined by a gray, glistening, smooth membrane, and filled with clear fluid. On microscopic examination, these membranes are composed of a single layer of cuboidal or flattened cuboidal epithelium, which in many instances may be completely atrophic.

Simple cysts are common postmortem findings without clinical significance. On occasion, hemorrhage into them may cause sudden distention and pain, and calcification of the hemorrhage may give rise to bizarre radiographic shadows. The main importance of cysts lies in their differentiation from kidney tumors. Radiologic studies show that in contrast to renal tumors, renal cysts have smooth contours, are almost always avascular, and give fluid rather than solid signals on ultrasonography.

KEY CONCEPTS

Cystic Diseases

- Autosomal dominant polycystic kidney disease accounts for a small yet significant subset of ESRD.
- Ciliopathies or abnormalities of the cilium-centrosome complex underlie the major cystic kidney diseases, including polycystic kidney disease (both autosomal dominant and autosomal recessive forms) and nephronophthisis.
- Dysfunction of the primary cilium of tubular epithelial cells results in alterations in ion flux and changes in cell proliferation and function, culminating in renal cyst formation.

URINARY TRACT OBSTRUCTION (OBSTRUCTIVE UROPATHY)

Obstructive lesions of the urinary tract increase susceptibility to infection and to stone formation, and unrelieved obstruction almost always leads to permanent renal atrophy, termed *hydronephrosis* or *obstructive uropathy*. Fortunately, many causes of obstruction are surgically correctable or medically treatable.

Obstruction may be sudden or insidious, partial or complete, unilateral or bilateral; it may occur at any level of the urinary tract from the urethra to the renal pelvis. It can be caused by *intrinsic* lesions of the urinary tract or *extrinsic* lesions that compress the ureter. The common causes are (Fig. 20.43):

- **Congenital anomalies:** posterior urethral valves and urethral strictures, meatal stenosis, bladder neck obstruction; ureteropelvic junction narrowing or obstruction; severe vesicoureteral reflux
- **Urinary calculi**
- **Benign prostatic hypertrophy**
- **Tumors:** carcinoma of the prostate, bladder tumors, contiguous malignant disease (retroperitoneal lymphoma), carcinoma of the cervix or uterus
- **Inflammation:** prostatitis, ureteritis, urethritis, retroperitoneal fibrosis
- **Sloughed papillae or blood clots**

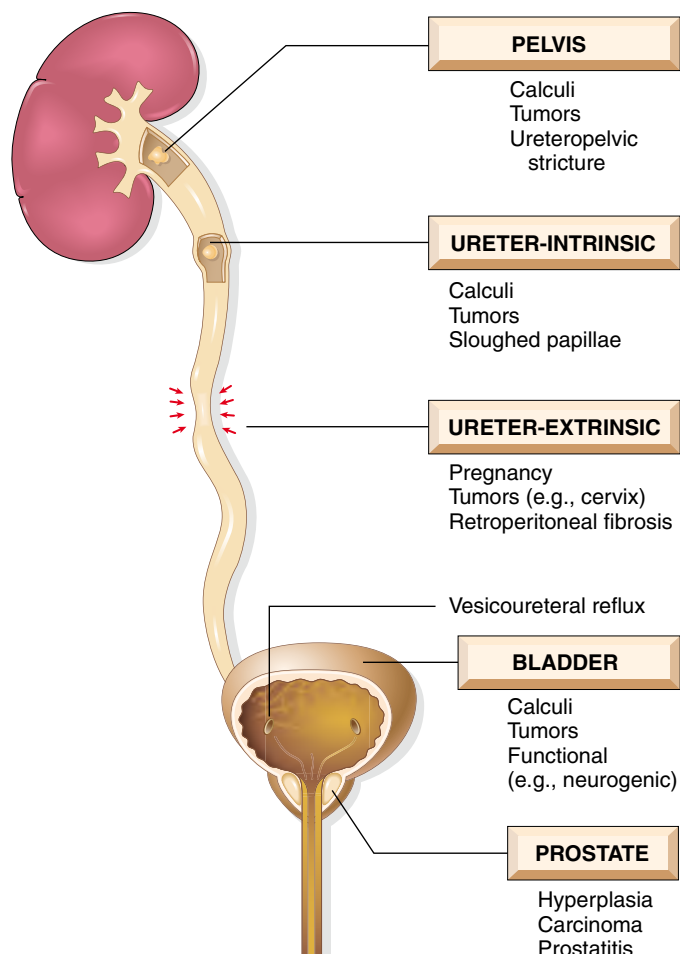


Fig. 20.43 Obstructive lesions of the urinary tract.

- **Pregnancy**
- **Uterine prolapse and cystocele**
- **Functional disorders:** neurogenic (spinal cord damage or diabetic nephropathy) and other functional abnormalities of the ureter or bladder (often termed *dysfunctional obstruction*)

Hydronephrosis is the term used to describe dilation of the renal pelvis and calyces associated with progressive atrophy of the kidney due to obstruction to the outflow of urine. Even with complete obstruction, glomerular filtration persists for some time because the filtrate subsequently diffuses back into the renal interstitium and perirenal spaces, from where it ultimately returns to the lymphatic and venous systems. Because of this continued filtration, the affected calyces and pelvis become dilated, often markedly so. The high pressure in the pelvis is transmitted back through the collecting ducts into the cortex, causing renal atrophy, but it also compresses the renal vasculature of the medulla, causing a diminution in inner medullary blood flow. The medullary vascular defects are initially reversible but lead to medullary functional disturbances. Accordingly, the initial functional alterations caused by obstruction are largely tubular, manifested primarily by impaired concentrating ability. Only later does the GFR begin to fall. Obstruction also triggers an interstitial inflammatory reaction, leading eventually to interstitial fibrosis.

MORPHOLOGY

When the obstruction is sudden and complete, it leads to mild dilation of the pelvis and calyces and sometimes to atrophy of the renal parenchyma. When the obstruction is subtotal or intermittent, progressive dilation ensues, giving rise to hydronephrosis (Fig. 20.44). Depending on the level of urinary block, the dilation may affect the bladder first, or the ureter and then the kidney.

The kidney may be slightly to massively enlarged, depending on the degree and the duration of the obstruction. The earlier features are those of simple dilation of the pelvis and calyces, but in addition there is often significant interstitial inflammation, even in the absence of infection. In chronic cases, the picture is one of cortical tubular atrophy with marked diffuse interstitial fibrosis. Progressive blunting of the apices of the pyramids occurs, and these eventually become cupped. In advanced cases, the kidney may become transformed into a thin-walled cystic structure having a diameter of up to 15 to 20 cm with striking parenchymal atrophy, total obliteration of the pyramids, and thinning of the cortex.

Clinical Features

Acute obstruction may provoke pain attributed to distention of the collecting system or renal capsule. Most of the early symptoms are produced by the underlying cause of the hydronephrosis. Thus, calculi lodged in the ureters may give rise to renal colic, and prostatic enlargements may give rise to bladder symptoms.

Unilateral complete or partial hydronephrosis may remain silent for long periods because the unaffected kidney can maintain adequate renal function. Sometimes its existence first becomes apparent during imaging studies. In its early stages, perhaps the first few weeks, relief of obstruction leads to reversion to normal function. Ultrasonography is a useful noninvasive technique in the diagnosis of obstructive uropathy.

In *bilateral partial obstruction*, the earliest manifestation is inability to concentrate urine, reflected by polyuria and nocturia. Some patients develop distal tubular acidosis, renal salt wasting, secondary renal calculi, and chronic tubulointerstitial nephritis with scarring and atrophy of the papilla and medulla. Hypertension is common.

Complete bilateral obstruction of rapid onset results in oliguria or anuria and is incompatible with survival unless the obstruction is

relieved. Curiously, diuresis frequently occurs after relief of complete urinary tract obstruction. This can often be massive, with the kidney excreting large amounts of urine that is rich in sodium chloride.

UROLITHIASIS (RENAL CALCULI, STONES)

Urolithiasis affects 5% to 10% of people in the United States in their lifetime. Stones may form anywhere in the urinary tract, but most arise in the kidney. Males are affected more often than females, and the peak age at onset is between 20 and 30 years. Familial and hereditary predisposition to stone formation has long been known. Many inborn errors of metabolism, such as cystinuria and primary hyperoxaluria, provide examples of hereditary disease characterized by excessive production and excretion of stone-forming substances.

Etiology and Pathogenesis

There are four main types of calculi (Table 20.12): (1) *calcium stones* (about 70%), composed largely of calcium oxalate or calcium oxalate mixed with calcium phosphate; (2) another 5% to 10% are so-called *triple stones* or *struvite stones*, composed of magnesium ammonium phosphate; (3) 5% to 10% are *uric acid stones*; and (4) 1% to 2% are made up of *cystine*. An organic mucoprotein matrix, making up 1% to 5% of the stone by weight, is present in all calculi. **Although there are many causes for the initiation and propagation of stones, the most important determinant is an increased urinary concentration of the stones' constituents, such that it exceeds their solubility (supersaturation).** A low urine volume in some metabolically normal patients may also favor supersaturation.

- Calcium oxalate stones* are associated in about 5% of patients with hypercalcemia and hypercalciuria, such as occurs with hyperparathyroidism, diffuse bone disease, sarcoidosis, and other hypercalcemic states. About 55% have hypercalciuria without hypercalcemia, which is caused by several factors, including excessive absorption of calcium from the intestine (absorptive hypercalciuria), an intrinsic impairment in renal tubular reabsorption of calcium (renal hypercalciuria), or idiopathic fasting hypercalciuria with normal parathyroid



Fig. 20.44 Hydronephrosis of the kidney, with marked dilation of the pelvis and calyces and thinning of the renal parenchyma.

TABLE 20.12 Prevalence of Various Types of Renal Stones

Stone Type	Percentage of All Stones
Calcium oxalate and phosphate	70
Idiopathic hypercalciuria (50%)	
Hypercalciuria and hypercalcemia (10%)	
Hyperoxaluria (5%)	
Enteric (4.5%)	
Primary (0.5%)	
Hyperuricosuria (20%)	
Hypocitraturia	
No known metabolic abnormality (15% to 20%)	
Magnesium ammonium phosphate (struvite)	5–10
Uric acid	5–10
Associated with hyperuricemia	
Associated with hyperuricosuria	
Idiopathic (50% of uric stones)	
Cystine	1–2
Others or unknown	±5

function. As many as 20% of calcium oxalate stones are associated with increased uric acid secretion (*hyperuricosuric calcium nephrolithiasis*), with or without hypercalciuria. The mechanism of stone formation in this setting involves “nucleation” of calcium oxalate by uric acid crystals in the collecting ducts. Five percent are associated with *hyperoxaluria*, either hereditary (primary oxaluria) or, more commonly, acquired by intestinal overabsorption in patients with enteric diseases. *Hypocitraturia*, which can be idiopathic or associated with acidosis and chronic diarrhea of unknown cause, may produce calcium stones. In a variable proportion of individuals with calcium stones, no cause can be found (idiopathic calcium stone disease).

- *Magnesium ammonium phosphate stones* are formed largely after infections by urea-splitting bacteria (e.g., *Proteus* and some staphylococci) that convert urea to ammonia. The resultant alkaline urine causes the precipitation of magnesium ammonium phosphate salts. These form some of the largest stones, as the amount of urea excreted normally is very large. Indeed, so-called *staghorn calculi* occupying large portions of the renal pelvis are frequently a consequence of infection.
- *Uric acid stones* are common in individuals with hyperuricemia, such as patients with gout, and diseases involving rapid cell turnover, such as leukemia. However, **more than half of all patients with uric acid calculi have neither hyperuricemia nor increased urinary excretion of uric acid.** In this group, it is thought that a tendency to excrete urine of pH below 5.5 may predispose to uric acid stones, because uric acid is insoluble in acidic urine. In contrast to the radiopaque calcium stones, uric acid stones are radiolucent.
- *Cystine stones* are caused by genetic defects in the renal reabsorption of amino acids, including cystine, leading to cystinuria. These stones also form at low urinary pH.

It can therefore be appreciated that **increased concentration of stone constituents, changes in urinary pH, decreased urine volume, and the presence of bacteria influence the formation of calculi.** However, many calculi occur in the absence of these factors; conversely, individuals with hypercalciuria, hyperoxaluria, and hyperuricosuria often do not form stones. It has been postulated that stone formation is enhanced by a deficiency in inhibitors of crystal formation in urine. The list of such inhibitors is long, including pyrophosphate, diphosphonate, citrate, glycosaminoglycans, osteopontin, and a glycoprotein called nephrocalcin.

MORPHOLOGY

Stones are unilateral in about 80% of patients. The favored sites for their formation are within the renal calyces and pelves (Fig. 20.45) and in the bladder. If formed in the renal pelvis, they tend to remain small, having an average diameter of 2 to 3 mm. These may have smooth contours or may take the form of an irregular, jagged mass of spicules. Often, many stones are found within one kidney. On occasion, progressive accretion of salts leads to the development of branching structures known as **staghorn calculi**, which create a cast of the pelvic and calyceal system.

Clinical Features

Urolithiasis may be asymptomatic, produce severe renal colic and abdominal pain, or may cause significant renal damage. Larger stones often manifest themselves by hematuria. Stones also predispose to superimposed infection, both by their obstructive nature and by the trauma they produce.



Fig. 20.45 Nephrolithiasis. A large stone is impacted in the renal pelvis. (Courtesy Dr. E Mosher, Brigham and Women's Hospital, Boston.)

NEOPLASMS OF THE KIDNEY

Both benign and malignant neoplasms occur in the kidney. Malignant neoplasms are of great importance clinically. By far the most common malignant tumor is renal cell carcinoma, followed by Wilms tumor, which is found in children and described in Chapter 10, and finally urothelial carcinoma of the calyces and pelves.

Benign Neoplasms

Renal Papillary Adenoma

Small, discrete adenomas arising from the renal tubular epithelium are found commonly (7% to 22%) at autopsy. They are most frequently papillary and therefore called *papillary adenomas*.

MORPHOLOGY

These tumors are typically less than 1.5 cm in diameter. They are present in the cortex and appear grossly as pale yellow-gray, discrete, well-circumscribed nodules. On microscopic examination, they are composed of complex, branching, papillomatous structures with numerous complex fronds or tubules forming solid cell nests. The cells are cuboidal to polygonal in shape and have regular, small central nuclei, scanty cytoplasm, and no atypia.

By histologic criteria, these tumors do not differ from low-grade papillary renal cell carcinoma and share some immunohistochemical and cytogenetic features (trisomies 7 and 17) with papillary cancers, to be discussed later.

Angiomyolipoma

This benign neoplasm consists of vessels, smooth muscle, and fat. **Angiomyolipomas are present in 25% to 50% of patients with tuberous sclerosis**, a disease caused by loss-of-function mutations in the *TSC1* or *TSC2* tumor suppressor genes. Tuberous sclerosis is characterized by lesions of the cerebral cortex that produce epilepsy and intellectual disability, a variety of skin abnormalities, and unusual benign tumors at other sites, such as the heart. The clinical importance of angiomyolipomas is due largely to their susceptibility to undergo spontaneous hemorrhage.

Renal Oncocytoma

This is an epithelial neoplasm composed of eosinophilic cells having small, round, benign-appearing nuclei with small but conspicuous nucleoli. It is thought to arise from the intercalated cells of collecting

ducts, and accounts for approximately 5% of renal neoplasms. Ultrastructurally, the eosinophilic cells have numerous mitochondria. In gross appearance, the tumors are tan or mahogany brown, relatively homogeneous, and usually well circumscribed with a central scar in one-third of cases. However, they may achieve a large size (up to 12 cm in diameter). The term *renal oncocytosis* is used for some familial cases when the tumors are multifocal.

Malignant Neoplasms

Renal Cell Carcinoma

Renal cell carcinomas represent about 3% of all newly diagnosed cancers in the United States and account for 85% of renal cancers in adults. There are approximately 65,000 new cases per year and 13,000 deaths from the disease. The tumors occur most often in the sixth and seventh decades of life, and show a 2:1 male preponderance.

Epidemiology. Tobacco use is the most significant risk factor. Cigarette smoking doubles the incidence of renal cell carcinoma, and pipe and cigar smokers are also more susceptible. An international study has identified additional risk factors, including obesity (particularly in females); hypertension; unopposed estrogen therapy; and exposure to asbestos, petroleum products, and heavy metals. There is also an increased risk in patients with ESKD, chronic kidney disease, acquired cystic disease, and tuberous sclerosis.

Most renal cancers are sporadic, but autosomal dominant familial cancers also occur, usually in younger individuals. Although they account for only 4% of renal cancers, familial variants have been instructive in understanding renal carcinogenesis.

- *von Hippel-Lindau (VHL) syndrome.* One-half to two-thirds of individuals with VHL (nearly all, if they live long enough; Chapter 28) develop renal cysts and bilateral, often multiple, renal cell carcinomas. As will be described, loss of function mutations in the *VHL* gene also have a central role in sporadic renal cell carcinoma.
- *Hereditary leiomyomatosis and renal cell carcinoma syndrome.* This autosomal dominant disease is caused by loss of function mutations of the *FH* gene, which encodes fumarate hydratase, and is characterized

by cutaneous and uterine leiomyomata and an aggressive type of high-grade papillary carcinoma with increased propensity for metastatic spread. Hereditary leiomyomatosis and renal cell carcinoma syndrome-associated RCCs are now called *fumarate hydratase-deficient RCCs* since a subset of patients do not have a family history, that is, they are not hereditary. Of interest, fumarate hydratase is an enzyme that participates in the Krebs cycle, another example of an alteration in metabolism that promotes carcinogenesis (discussed in Chapter 7).

- *Hereditary papillary carcinoma.* This autosomal dominant form is manifested by multiple bilateral tumors with papillary histology and usually low nuclear grade. These tumors show numerous cytogenetic abnormalities and, as we will describe, mutations in the *MET* proto-oncogene.
- *Birt-Hogg-Dubé syndrome.* The autosomal dominant inheritance pattern of this disease is due to loss of function mutations involving the *BHD* gene, which encodes the tumor suppressor folliculin. The syndrome features a constellation of skin (fibrofolliculomas, trichodiscomas, and acrochordons), pulmonary (cysts or blebs), and renal tumors with a wide range of morphologic subtypes.

Classification of Renal Cell Carcinoma: Histology, Cytogenetics, and Genetics. The classification of renal cell carcinoma is based on cytogenetic, genetic, and histologic features. The major types of tumor are (Fig. 20.46):

- *Clear cell renal cell carcinoma.* This is the most common type, accounting for 70% to 80% of renal cell cancers. The tumors are made up of cells with clear or granular cytoplasm arranged in variably sized nests. Most cases (95%) are sporadic. In 98% of these tumors, whether sporadic, familial, or associated with VHL syndrome, there is a deletion in one copy of chromosome 3 at band p25.3, the site of the *VHL* gene. In up to 80% of clear cell cancers, the second copy of the *VHL* gene is disabled by acquired somatic loss-of-function mutations or is silenced by hypermethylation, consistent with *VHL* acting as a tumor suppressor gene (Chapter 7). The *VHL* gene encodes a protein that is part of a ubiquitin ligase complex that targets the transcription factor hypoxia-inducible factor-1 (HIF-1) for degradation in an O_2 -dependent fashion. When

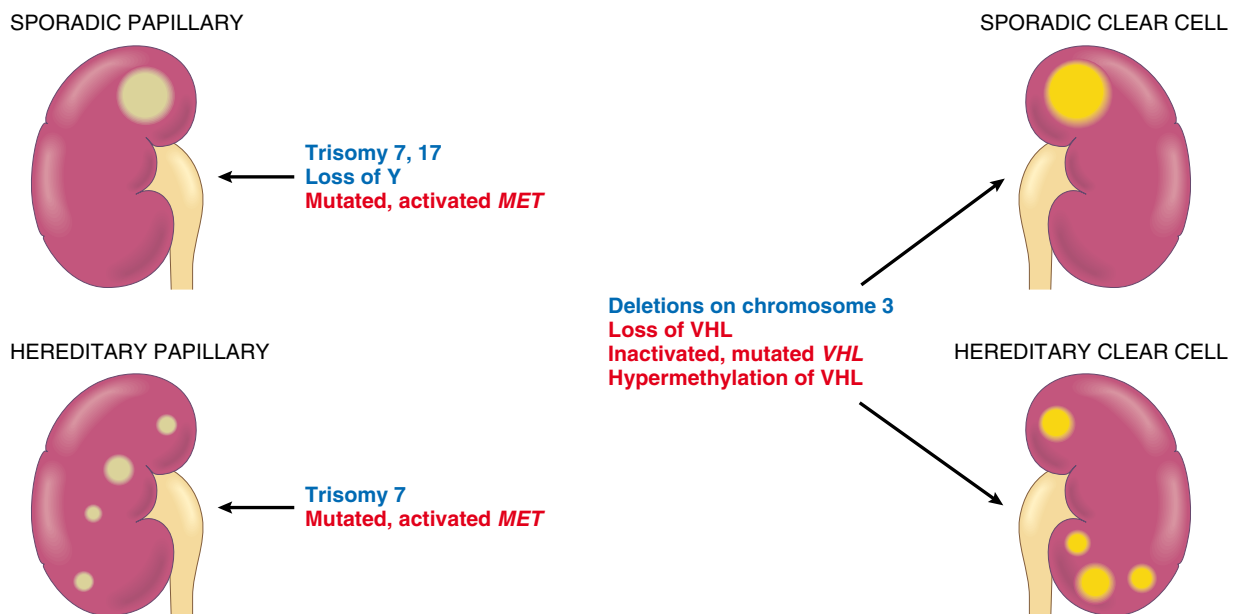


Fig. 20.46 Cytogenetics (blue) and genetics (red) of clear cell versus papillary renal cell carcinoma. (Courtesy Dr. Keith Ligon, Brigham and Women's Hospital, Boston.)

VHL is inactive, HIF-1 levels remain high, even under normoxic conditions, causing inappropriate expression of genes that are turned on by HIF. These include genes that promote angiogenesis, such as VEGF, and genes that stimulate cell growth, such as insulin-like growth factor-1 (IGF-1). In addition, HIF collaborates in complex ways with the oncogenic protein MYC to “reprogram” cellular metabolism in a way that favors growth. Deep sequencing of renal carcinoma genomes has revealed frequent mutations in several genes that regulate histone modifications, indicating that dysregulation of the epigenome also has an important role in clear cell carcinoma.

- **Papillary renal cell carcinoma** accounts for 10% to 15% of renal cancers. It is characterized by papillary, tubulopapillary, or solid growth patterns, and occurs in both familial and sporadic forms. These tumors are not associated with chromosome 3p deletions. The most common cytogenetic abnormalities are trisomies 7 and 17 and loss of the Y chromosome in male patients in the sporadic form, and trisomy 7 in the familial form. The familial form is caused by gain-of-function mutations in *MET*, a proto-oncogene located on chromosome 7 that encodes the tyrosine kinase receptor for *hepatocyte growth factor*. *MET* is also mutated in a small proportion of sporadic papillary carcinomas. Hepatocyte growth factor (also called *scatter factor*) stimulates cell growth, mobility, invasion, and morphogenetic differentiation. Unlike clear cell carcinomas, papillary carcinomas are frequently multifocal in origin.
- **Chromophobe renal cell carcinoma** represents 5% of renal cell cancers and is composed of cells with prominent cell membranes and pale eosinophilic cytoplasm, usually with a halo around the nucleus. On cytogenetic examination, these tumors show multiple chromosome losses and extreme hypodiploidy. They are thought to originate from intercalated cells of collecting ducts and have an excellent prognosis compared to clear cell and papillary cancers. Histologic distinction from oncocytoma can be difficult.
- **Xp11 translocation renal cell carcinoma** (or *TFE3-rearranged renal cell carcinoma*) is a genetically distinct subtype of renal cell carcinoma. It often occurs in young patients and is defined by translocations of the *TFE3* gene located at Xp11.2 with several partner genes, all of which result in overexpression of the TFE3 transcription factor. The morphology of TFE3 translocation renal cell carcinoma depends on the partner gene and includes nested, solid, and tubulopapillary.
- **Collecting duct carcinoma** represents approximately 1% or less of renal epithelial neoplasms. They arise from collecting duct cells in the medulla. Several chromosomal losses and deletions have been described, but a distinct pattern has not been identified. Histologically, these tumors are characterized by malignant cells forming glands enmeshed within a prominent desmoplastic stroma, typically in a medullary location. *Medullary carcinoma* is a morphologically similar neoplasm that is seen in patients with sickle cell trait.

MORPHOLOGY

Renal cell carcinomas may arise in any portion of the kidney, but more commonly affects the poles. **Clear cell renal cell carcinomas** most likely arise from proximal tubular epithelium, and usually occur as solitary unilateral lesions. They are bright yellow-gray-white spherical masses of variable size that distort the kidney. The yellow color is a consequence of prominent lipid accumulations in tumor cells. There are commonly large areas of gray-white necrosis and foci of hemorrhagic discoloration. The margins are usually sharply defined and confined within the renal capsule (Fig. 20.47). The growth pattern varies from nested, tubular, solid, to pseudopapillary. The tumor cells have a rounded or polygonal

shape and abundant clear or granular cytoplasm, which contains glycogen and lipids (Fig. 20.48A). The tumors have delicate branching vasculature and may show cystic areas. Larger tumors often show hyalinization. Most tumors are well differentiated, but some show atypia including bizarre nuclei and giant cells.

As tumors enlarge, they may bulge into the calyces and pelvis and eventually fungate through the walls of the collecting system to extend into the ureter.

One of the striking characteristics of clear cell renal cell carcinoma is its tendency to invade the renal vein (see Fig. 20.47), in which it may grow as a solid column of cells that extends up the inferior vena cava, sometimes as far as the right side of the heart.

Papillary renal cell carcinomas, thought to arise from distal convoluted tubules, can be multifocal and bilateral. They are typically hemorrhagic and cystic, especially when large. The tumor is composed of cuboidal or low columnar cells arranged in papillary formations. Interstitial foam histiocytes are common in the papillary cores (Fig. 20.48B). Psammoma bodies may be present. **Chromophobe renal cell carcinoma** is made up of pale eosinophilic cells, often with a perinuclear halo, arranged in solid sheets with a concentration of the largest cells around blood vessels (Fig. 20.48C). **Collecting duct carcinoma** is a rare variant showing irregular channels lined by highly atypical epithelium with a hobnail pattern. Sarcomatoid changes arise infrequently in all types of renal cell carcinoma and are associated with a very poor prognosis.

Clinical Features

The classic clinical features of renal cell carcinoma are *costovertebral pain*, *palpable mass*, and *hematuria*, but all three are seen in only 10% of cases. The most reliable clue is hematuria, but it is usually intermittent and may be microscopic; thus, the tumor may remain silent until it attains a large size, often greater than 10 cm. At this time, it is often associated with generalized constitutional symptoms, such as fever, malaise, weakness, and weight loss. Currently, an increasing number of tumors are being discovered in the asymptomatic state by incidental radiologic studies (e.g., computed tomography or magnetic resonance imaging) performed for other indications.

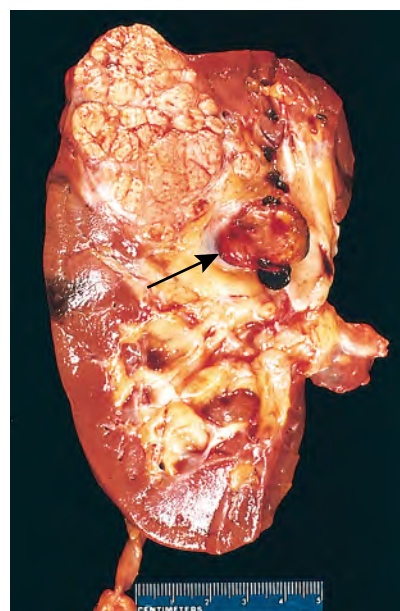


Fig. 20.47 Renal cell carcinoma. Typical Cross-section of a typical yellowish, spherical neoplasm in one pole of the kidney. Note the tumor in the dilated thrombosed renal vein (arrow).

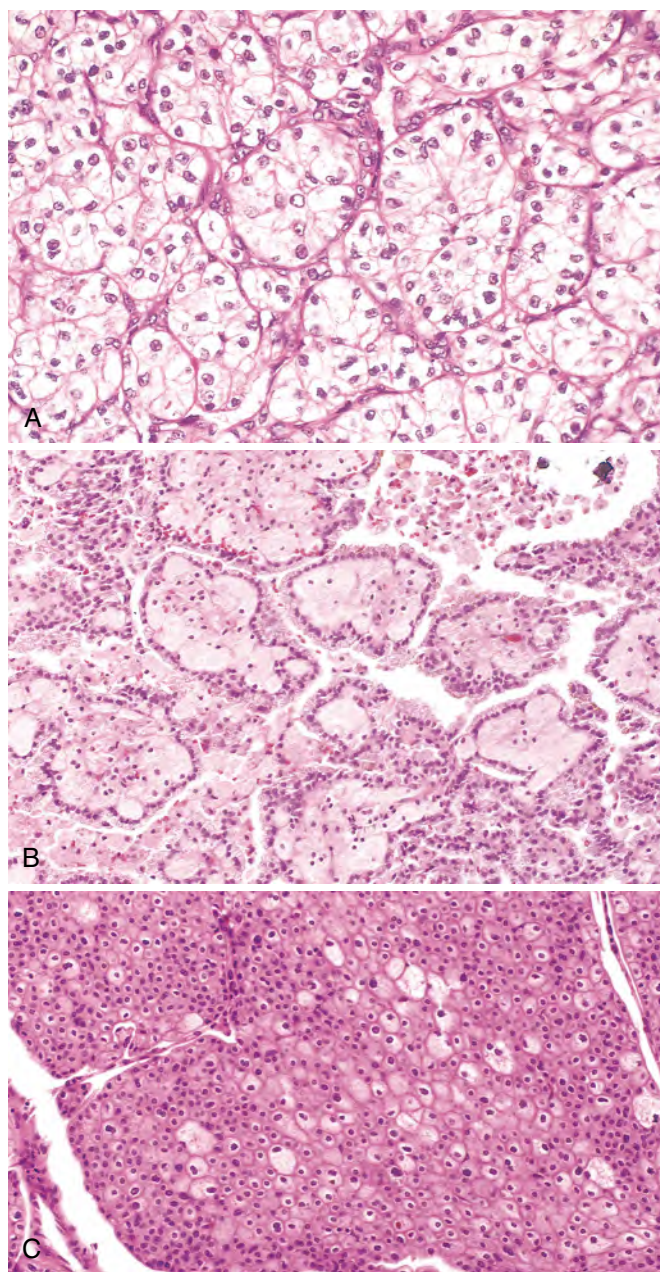


Fig. 20.48 Renal cell carcinoma. (A) Clear cell type. (B) Papillary type. Note the papillae and foamy macrophages in the stalk. (C) Chromophobe type. (Courtesy Dr. A. Renshaw, Baptist Hospital, Miami.)

Renal cell carcinoma is considered one of the great mimics in medicine, because it tends to produce a diversity of systemic symptoms not related to the kidney. In addition to fever and constitutional symptoms mentioned earlier, renal cell carcinomas produce several syndromes ascribed to abnormal hormone production, including polycythemia, hypercalcemia, hypertension, hepatic dysfunction, feminization or masculinization, Cushing syndrome, eosinophilia, leukemoid reactions, and amyloidosis.

A particularly troublesome feature of renal cell carcinoma is its tendency to metastasize widely before giving rise to any local symptoms or signs. In 15% of new patients with renal cell carcinoma, there is radiologic evidence of metastases at the time of presentation. Overall, the most common locations of metastasis are the lungs (more than

50%) and bones (33%), followed in frequency by the regional lymph nodes, liver, adrenal glands, and brain.

The average 5-year survival rate is about 70% and close to 100% in the absence of distant metastases. With renal vein invasion or extension into the perinephric fat, 5-year survival is reduced to approximately 60%. Radical nephrectomy has been the treatment of choice, but nephron-sparing surgery to preserve renal function is recommended for T1a tumors (<4 cm) as well as larger tumors when technically feasible. Drugs that inhibit VEGF and various tyrosine kinases as well as immunotherapy are used as an adjunct to chemotherapy for metastatic disease.

Urothelial Carcinoma of the Renal Pelvis

Approximately 5% to 10% of primary renal tumors originate from the urothelium of the renal pelvis (Fig. 20.49). These tumors range from apparently benign papillomas to invasive urothelial (transitional cell) carcinomas.

Renal pelvic tumors usually become clinically apparent within a relatively short time, because they lie within the pelvis and produce noticeable hematuria because of fragmentation. They are almost invariably small when discovered. These tumors may block urinary outflow and lead to palpable hydronephrosis and flank pain. On histologic examination, pelvic tumors are the exact counterpart of those found in the urinary bladder; further details are in Chapter 21.

Urothelial tumors may occasionally be multiple, involving the pelvis, ureters, and bladder. In 50% of renal pelvic tumors, there is a preexisting or concomitant bladder urothelial tumor. On histologic examination, there are also foci of atypia or carcinoma in situ in grossly normal urothelium remote from the pelvic tumor. There is an increased incidence of urothelial carcinomas of the renal pelvis in individuals with Lynch syndrome.

Infiltration of the wall of the pelvis and calyces is common. For this reason, despite their apparently small, deceptively benign appearance, the prognosis for these tumors is guarded. Reported 5-year survival rates vary from 50% to 100% for low-grade noninvasive lesions to 10% with high-grade infiltrating tumors.

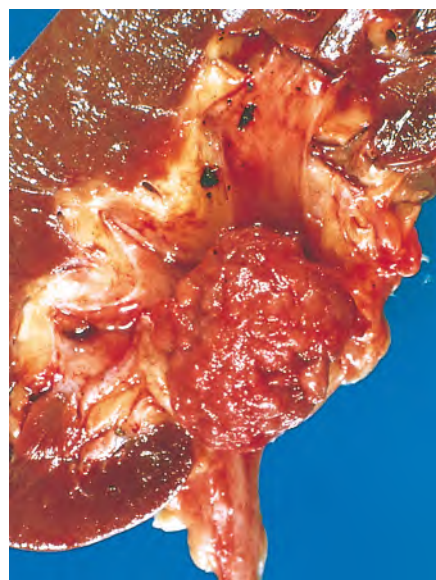


Fig. 20.49 Urothelial carcinoma of the renal pelvis. The pelvis has been opened to expose the nodular irregular neoplasm, just proximal to the ureter.

KEY CONCEPTS

Kidney Neoplasms

- Clear cell renal cell carcinoma is the most common subtype of malignant renal neoplasms and often is associated with loss of function mutation in *VHL*, a tumor suppressor gene.
- Papillary renal cell carcinoma is the second most common subtype of malignant renal neoplasms, and are often associated with gain of function mutations in *MET* proto-oncogene.
- Hereditary forms of renal cell carcinoma have led to the discovery of important genes (e.g., *VHL*, *BHD*) in renal carcinogenesis.
- Urothelial tumors resembling tumors in the urinary bladder also originate in the renal pelvis. These tumors have a poor prognosis.

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